DBMS ASSIGNMENT PGCP-AC-

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Assignment 1 (DAY 1)

Question 1.

1.To list all records with sal > 2000 and comm>200

```
mysql> select* from emp where sal>2000 and comm>200;
Empty set (0.00 sec)
```

2.To list all record with job=’Clerk’ or sal>2000

```
mysql> select* from emp where job='Clerk' or sal>2000;
+-------+--------+-----------+------+------------+---------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+--------+-----------+------+------------+---------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  800.00 | NULL |     20 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 2975.00 | NULL |     20 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 2850.00 | NULL |     30 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000.00 | NULL |     20 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL |     10 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1100.00 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950.00 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3000.00 | NULL |     20 |
|  7934 | MILLER | CLERK     | 7782 | 1982-01-23 | 1300.00 | NULL |     10 |
+-------+--------+-----------+------+------------+---------+------+--------+
9 rows in set (0.00 sec)
```

3.To list all the record with sal=1250 or 1100 or 2850

```
mysql> select* from emp where sal in(1250,1100,2850);
+-------+-------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME | JOB      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+-------+----------+------+------------+---------+--------+--------+
|  7521 | WARD  | SALESMAN | 7698 | 1981-02-22 | 1250.00 | 500.00 |     30 |
|  7698 | BLAKE | MANAGER  | 7839 | 1981-05-01 | 2850.00 |   NULL |     30 |
|  7876 | ADAMS | CLERK    | 7788 | 1983-01-12 | 1100.00 |   NULL |     20 |
+-------+-------+----------+------+------------+---------+--------+--------+
3 rows in set (0.00 sec)
```

4. To list all employees with sal>1250 and <2850

```
mysql> select* from emp where sal between 1251 and 2849;
+-------+--------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+--------+----------+------+------------+---------+--------+--------+
|  7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00 |     30 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00 |   0.00 |     30 |
|  7934 | MILLER | CLERK    | 7782 | 1982-01-23 | 1300.00 |   NULL |     10 |
+-------+--------+----------+------+------------+---------+--------+--------+
3 rows in set (0.00 sec)
```

5. To list all employees with name ends with AS

```
mysql> select* from emp where ename regexp 'AS$';
Empty set (0.01 sec)
```

6. To list all employees with job starts with C and ends with K

```
mysql> SELECT * FROM emp WHERE job LIKE 'C%K';
+-------+--------+-------+------+------------+---------+------+--------+
| EMPNO | ENAME  | JOB   | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+--------+-------+------+------------+---------+------+--------+
|  7369 | SMITH  | CLERK | 7902 | 1980-12-17 |  800.00 | NULL |     20 |
|  7876 | ADAMS  | CLERK | 7788 | 1983-01-12 | 1100.00 | NULL |     20 |
|  7900 | JAMES  | CLERK | 7698 | 1981-12-03 |  950.00 | NULL |     30 |
|  7934 | MILLER | CLERK | 7782 | 1982-01-23 | 1300.00 | NULL |     10 |
+-------+--------+-------+------+------------+---------+------+--------+
4 rows in set (0.00 sec)
```

7. To list all employees with job contains L at third position and M at third last position

```
mysql> SELECT * FROM emp WHERE job LIKE '__L%M__';
+-------+--------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+--------+----------+------+------------+---------+--------+--------+
|  7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00 |     30 |
|  7521 | WARD   | SALESMAN | 7698 | 1981-02-22 | 1250.00 | 500.00 |     30 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00 |   0.00 |     30 |
+-------+--------+----------+------+------------+---------+--------+--------+
3 rows in set (0.00 sec)
```

8. To list all the record with sal not equal to 1250 or 1100 or 2850

```
mysql> SELECT * FROM emp WHERE sal NOT IN (1250, 1100, 2850);
+-------+--------+-----------+------+------------+---------+---------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
+-------+--------+-----------+------+------------+---------+---------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  800.00 |    NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600.00 |  300.00 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 2975.00 |    NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000.00 |    NULL |     20 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5000.00 |    NULL |     10 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500.00 |    0.00 |     30 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950.00 |    NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3000.00 |    NULL |     20 |
|  7934 | MILLER | CLERK     | 7782 | 1982-01-23 | 1300.00 |    NULL |     10 |
+-------+--------+-----------+------+------------+---------+---------+--------+
9 rows in set (0.00 sec)
```

9. To list all employees with sal not >1250 and <2850

```
mysql> SELECT * FROM emp WHERE NOT (sal > 1250 AND sal < 2850);
+-------+-------+-----------+------+------------+---------+--------+--------+
| EMPNO | ENAME | JOB       | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+-------+-----------+------+------------+---------+--------+--------+
|  7369 | SMITH | CLERK     | 7902 | 1980-12-17 |  800.00 |   NULL |     20 |
|  7521 | WARD  | SALESMAN  | 7698 | 1981-02-22 | 1250.00 | 500.00 |     30 |
|  7566 | JONES | MANAGER   | 7839 | 1981-04-02 | 2975.00 |   NULL |     20 |
|  7698 | BLAKE | MANAGER   | 7839 | 1981-05-01 | 2850.00 |   NULL |     30 |
|  7788 | SCOTT | ANALYST   | 7566 | 1982-12-09 | 3000.00 |   NULL |     20 |
|  7839 | KING  | PRESIDENT | NULL | 1981-11-17 | 5000.00 |   NULL |     10 |
|  7876 | ADAMS | CLERK     | 7788 | 1983-01-12 | 1100.00 |   NULL |     20 |
|  7900 | JAMES | CLERK     | 7698 | 1981-12-03 |  950.00 |   NULL |     30 |
|  7902 | FORD  | ANALYST   | 7566 | 1981-12-03 | 3000.00 |   NULL |     20 |
+-------+-------+-----------+------+------------+---------+--------+--------+
9 rows in set (0.00 sec)
```

10. To list all employees with job starts with C , E at 3rd position and ends with K

```
mysql> SELECT* FROM emp WHERE JOB LIKE 'C_E%K';
+-------+--------+-------+------+------------+---------+------+--------+
| EMPNO | ENAME  | JOB   | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+--------+-------+------+------------+---------+------+--------+
|  7369 | SMITH  | CLERK | 7902 | 1980-12-17 |  800.00 | NULL |     20 |
|  7876 | ADAMS  | CLERK | 7788 | 1983-01-12 | 1100.00 | NULL |     20 |
|  7900 | JAMES  | CLERK | 7698 | 1981-12-03 |  950.00 | NULL |     30 |
|  7934 | MILLER | CLERK | 7782 | 1982-01-23 | 1300.00 | NULL |     10 |
+-------+--------+-------+------+------------+---------+------+--------+
4 rows in set (0.00 sec)
```

11. To list all rows with comm is null

```
mysql> SELECT * FROM emp WHERE comm IS NULL;
+-------+--------+-----------+------+------------+---------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+--------+-----------+------+------------+---------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  800.00 | NULL |     20 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 2975.00 | NULL |     20 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 2850.00 | NULL |     30 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000.00 | NULL |     20 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL |     10 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1100.00 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950.00 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3000.00 | NULL |     20 |
|  7934 | MILLER | CLERK     | 7782 | 1982-01-23 | 1300.00 | NULL |     10 |
+-------+--------+-----------+------+------------+---------+------+--------+
9 rows in set (0.00 sec)
```

12. To list all employees with sal is null and name starts with ‘S’

```
mysql> SELECT* FROM emp WHERE SAL IS NULL AND ENAME LIKE 'S%';
Empty set (0.00 sec)
```

13. To list all employees with job contains 5 characters

```
mysql> SELECT* FROM emp WHERE JOB LIKE '_____';
+-------+--------+-------+------+------------+---------+------+--------+
| EMPNO | ENAME  | JOB   | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+--------+-------+------+------------+---------+------+--------+
|  7369 | SMITH  | CLERK | 7902 | 1980-12-17 |  800.00 | NULL |     20 |
|  7876 | ADAMS  | CLERK | 7788 | 1983-01-12 | 1100.00 | NULL |     20 |
|  7900 | JAMES  | CLERK | 7698 | 1981-12-03 |  950.00 | NULL |     30 |
|  7934 | MILLER | CLERK | 7782 | 1982-01-23 | 1300.00 | NULL |     10 |
+-------+--------+-------+------+------------+---------+------+--------+
4 rows in set (0.00 sec)
```

14. To list all employees with name contain ‘A’ at 1 position and job Contains 5 characters

```
mysql> SELECT* FROM emp WHERE ENAME LIKE 'A%' AND JOB LIKE '_____';
+-------+-------+-------+------+------------+---------+------+--------+
| EMPNO | ENAME | JOB   | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+-------+-------+------+------------+---------+------+--------+
|  7876 | ADAMS | CLERK | 7788 | 1983-01-12 | 1100.00 | NULL |     20 |
+-------+-------+-------+------+------------+---------+------+--------+
1 row in set (0.00 sec)
```

15. Retrieve the details (Name, Salary and dept no) of the emp who are working in department code 20, 30 and 40.

```
mysql> select ename,sal,deptno from emp where deptno in (20,30,40);
+--------+---------+--------+
| ename  | sal     | deptno |
+--------+---------+--------+
| SMITH  |  800.00 |     20 |
| ALLEN  | 1600.00 |     30 |
| WARD   | 1250.00 |     30 |
| JONES  | 2975.00 |     20 |
| BLAKE  | 2850.00 |     30 |
| SCOTT  | 3000.00 |     20 |
| TURNER | 1500.00 |     30 |
| ADAMS  | 1100.00 |     20 |
| JAMES  |  950.00 |     30 |
| FORD   | 3000.00 |     20 |
+--------+---------+--------+
10 rows in set (0.00 sec)
```

16. Display the total salary of all employees . Total salary will be calculated as sal+comm+sal*0.10

```
mysql> select ename, (sal+ifnull(comm,0)+sal*0.10)Total_Sal from emp;
+--------+-----------+
| ename  | Total_Sal |
+--------+-----------+
| SMITH  |  880.0000 |
| ALLEN  | 2060.0000 |
| WARD   | 1875.0000 |
| JONES  | 3272.5000 |
| BLAKE  | 3135.0000 |
| SCOTT  | 3300.0000 |
| KING   | 5500.0000 |
| TURNER | 1650.0000 |
| ADAMS  | 1210.0000 |
| JAMES  | 1045.0000 |
| FORD   | 3300.0000 |
| MILLER | 1430.0000 |
+--------+-----------+
12 rows in set (0.00 sec)
```

17. List the Name and job of the emp who have joined before 1 jan 1986 and whose salary range is between 1200and 2500. Display the columns with user defined Column headers.

```
mysql> SELECT ename AS 'Employee Name', job AS 'Job Title' FROM
emp WHERE hiredate < '1986-01-01' AND sal BETWEEN 1200 AND 2500;

+---------------+-----------+
| Employee Name | Job Title |
+---------------+-----------+
| ALLEN         | SALESMAN  |
| WARD          | SALESMAN  |
| TURNER        | SALESMAN  |
| MILLER        | CLERK     |
+---------------+-----------+
4 rows in set (0.00 sec)
```

18. List the empno, name, and department number of the emp works under manager with id 7698

```
mysql> select empno, ename, deptno from emp where mgr=7698;
+-------+--------+--------+
| empno | ename  | deptno |
+-------+--------+--------+
|  7499 | ALLEN  |     30 |
|  7521 | WARD   |     30 |
|  7844 | TURNER |     30 |
|  7900 | JAMES  |     30 |
+-------+--------+--------+
4 rows in set (0.00 sec)
```

19. List the name, job, and salary of the emp who are working in departments 10 and 30.

```
mysql> select ename, job, sal from emp where deptno in (10,30);
+--------+----------+---------+
| ename  | job      | sal     |
+--------+----------+---------+
| ALLEN  | SALESMAN | 1600.00 |
| WARD   | SALESMAN | 1250.00 |
| BLAKE  | MANAGER  | 2850.00 |
| KING   | PRESIDENT| 5000.00 |
| TURNER | SALESMAN | 1500.00 |
| JAMES  | CLERK    |  950.00 |
| MILLER | CLERK    | 1300.00 |
+--------+----------+---------+
7 rows in set (0.00 sec)
```

20. Display the emp details who do not have manager.

```
mysql> select* from emp where mgr is null;
+-------+-------+----------+------+------------+---------+------+--------+
| EMPNO | ENAME | JOB      | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+-------+----------+------+------------+---------+------+--------+
|  7839 | KING  | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL |     10 |
+-------+-------+----------+------+------------+---------+------+--------+
1 row in set (0.00 sec)
```

21. Write a query which will display name, department no and date of joining of all employee who were joined January 1, 1981 and March 31, 1983.

```
mysql> select ename, deptno, hiredate from emp
    -> where hiredate between '1981-01-01' and '1983-03-31'
    -> order by hiredate;
+--------+--------+------------+
| ename  | deptno | hiredate   |
+--------+--------+------------+
| ALLEN  |     30 | 1981-02-20 |
| WARD   |     30 | 1981-02-22 |
| JONES  |     20 | 1981-04-02 |
| BLAKE  |     30 | 1981-05-01 |
| TURNER |     30 | 1981-09-08 |
| KING   |     10 | 1981-11-17 |
| JAMES  |     30 | 1981-12-03 |
| FORD   |     20 | 1981-12-03 |
| MILLER |     10 | 1982-01-23 |
| SCOTT  |     20 | 1982-12-09 |
| ADAMS  |     20 | 1983-01-12 |
+--------+--------+------------+
11 rows in set (0.00 sec)
```

22. Display the employee details where the job contains word ‘AGE’ anywhere in the Job

```
mysql> select* from emp where job like '%AGE%';
+-------+-------+---------+------+------------+---------+------+--------+
| EMPNO | ENAME | JOB     | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+-------+---------+------+------------+---------+------+--------+
|  7566 | JONES | MANAGER | 7839 | 1981-04-02 | 2975.00 | NULL |     20 |
|  7698 | BLAKE | MANAGER | 7839 | 1981-05-01 | 2850.00 | NULL |     30 |
+-------+-------+---------+------+------------+---------+------+--------+
2 rows in set (0.00 sec)
```

23. List the details of the employee , whose names start with ‘A’ and end with ‘S’ or whose names contains N as the second or third character, and ending with either ‘N’ or ‘S’.

```
mysql> select* from emp
    -> where ename regexp '^A.*S$|^..?N.*[NS]$';
+-------+-------+---------+------+------------+---------+------+--------+
| EMPNO | ENAME | JOB     | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+-------+---------+------+------------+---------+------+--------+
|  7566 | JONES | MANAGER | 7839 | 1981-04-02 | 2975.00 | NULL |     20 |
|  7876 | ADAMS | CLERK   | 7788 | 1983-01-12 | 1100.00 | NULL |     20 |
+-------+-------+---------+------+------------+---------+------+--------+
2 rows in set (0.01 sec)

mysql>
```

24. List the names of the emp having ‘_’ character in their name.

```
mysql> select* from emp where ename like '_';
Empty set (0.00 sec)
```

25.List all emplyees who joined in september.(without using functions)

```
mysql> select *
    -> from emp
    -> where hiredate like '____-09-%';
+-------+--------+----------+------+------------+---------+---------+--------+---------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO | netsal  |
+-------+--------+----------+------+------------+---------+---------+--------+---------+
|  7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1437.50 | 1400.00 |     30 | 1000.00 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00 |    0.00 |     30 | 1000.00 |
+-------+--------+----------+------+------------+---------+---------+--------+---------+
2 rows in set (0.01 sec)
```

Single Row Functions

1. To list all employees and their email, to generate email use 2 to 5 characters from ename
Concat it with 2 to 4 characters in job and then concat it with '@mycompany.com'

```
mysql> select ename, concat(substr(ename,2,4), substr(job,2,3),
'@mycompany.com') as EMAILS from emp;
+--------+----------------------+
| ename  | EMAILS               |
+--------+----------------------+
| SMITH  | MITHLER@mycompany.com |
| ALLEN  | LLENALE@mycompany.com |
| WARD   | ARDALE@mycompany.com  |
| JONES  | ONESANA@mycompany.com |
| BLAKE  | LAKEANA@mycompany.com |
| SCOTT  | COTTNAL@mycompany.com |
| KING   | INGRES@mycompany.com  |
| TURNER | URNEALE@mycompany.com |
| ADAMS  | DAMSLER@mycompany.com |
| JAMES  | AMESLER@mycompany.com |
| FORD   | ORDNAL@mycompany.com  |
| MILLER | ILLELER@mycompany.com |
+--------+----------------------+
12 rows in set (0.00 sec)
```

2. List the empno, name, and department number of the emp who have experience of 18 or
more years and sort them based on their experience.

```
mysql> select empno,ename,deptno from emp
    -> where hiredate <= '2008-04-06'
    -> order by hiredate;
+-------+--------+--------+
| empno | ename  | deptno |
+-------+--------+--------+
|  7369 | SMITH  |     20 |
|  7499 | ALLEN  |     30 |
|  7521 | WARD   |     30 |
|  7566 | JONES  |     20 |
|  7698 | BLAKE  |     30 |
|  7844 | TURNER |     30 |
|  7839 | KING   |     10 |
|  7900 | JAMES  |     30 |
|  7902 | FORD   |     20 |
|  7934 | MILLER |     10 |
|  7788 | SCOTT  |     20 |
|  7876 | ADAMS  |     20 |
+-------+--------+--------+
12 rows in set (0.00 sec)
```

3. Display the employee details who joined on 3rd of any month or any year

```
mysql> select* from emp where hiredate regexp '-03$';
+-------+-------+---------+------+------------+---------+------+--------+
| EMPNO | ENAME | JOB     | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+-------+---------+------+------------+---------+------+--------+
|  7900 | JAMES | CLERK   | 7698 | 1981-12-03 |  950.00 | NULL |     30 |
|  7902 | FORD  | ANALYST | 7566 | 1981-12-03 | 3000.00 | NULL |     20 |
+-------+-------+---------+------+------------+---------+------+--------+
2 rows in set (0.00 sec)
```

4. display all employees who joined between years 1981 to 1983.

```
mysql> select* from emp
    -> where hiredate between '1981-01-01' and '1983-12-31';
+-------+--------+-----------+------+------------+---------+--------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+--------+-----------+------+------------+---------+--------+--------+
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600.00 | 300.00 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250.00 | 500.00 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 2975.00 |   NULL |     20 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 2850.00 |   NULL |     30 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000.00 |   NULL |     20 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5000.00 |   NULL |     10 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500.00 |   0.00 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1100.00 |   NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950.00 |   NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3000.00 |   NULL |     20 |
|  7934 | MILLER | CLERK     | 7782 | 1982-01-23 | 1300.00 |   NULL |     10 |
+-------+--------+-----------+------+------------+---------+--------+--------+
11 rows in set (0.00 sec)
```

5. Display name concatenated with dept code separated by comma and space. Name the column as ‘Emp info’.

```
mysql> SELECT CONCAT(ename, ', ', deptno) AS 'Emp info'
    -> from emp;
+------------+
| Emp info   |
+------------+
| SMITH, 20  |
| ALLEN, 30  |
| WARD, 30   |
| JONES, 20  |
| MARTIN, 30 |
| BLAKE, 30  |
| CLARK, 10  |
| SCOTT, 20  |
| KING, 10   |
| TURNER, 30 |
| ADAMS, 20  |
| JAMES, 30  |
| FORD, 20   |
| MILLER, 10 |
+------------+
14 rows in set (0.01 sec)
```

Group functions

6. Display the Highest, Lowest, Total & Average salary of all employee. Label the columns Maximum, Minimum, Total and Average respectively for each Department. Also round the result to the nearest whole number.

```
mysql> select deptno,
    -> round(max(sal))Maximum, round(min(sal))Minimum,
    -> round(sum(sal))Total, round(avg(sal))Average
    -> from emp group by deptno;
+--------+---------+---------+-------+----------+
| deptno | Maximum | Minimum | Total | Average |
+--------+---------+---------+-------+----------+
|     20 |    3000 |     800 | 10875 |    2175 |
|     30 |    2850 |     950 |  8150 |    1630 |
|     10 |    5000 |    1300 |  6300 |    3150 |
+--------+---------+---------+-------+----------+
3 rows in set (0.00 sec)
```

7. Display Department no and number of managers working in that department. Label the column as 'Total Number of Managers' for each department.

```
mysql> select deptno, count(*)'Total Number of Managers'
    -> from emp where job='Manager' group by deptno;
+--------+--------------------------+
| deptno | Total Number of Managers |
+--------+--------------------------+
|     20 |                        1 |
|     30 |                        1 |
+--------+--------------------------+
2 rows in set (0.00 sec)
```

8. Get the Department number, and sum of Salary of all non managers where the sum is greater than 20000.

```
mysql> select deptno, sum(sal) from emp
    -> where job!='MANAGER'
    -> group by deptno
    -> having sum(sal>20000);
Empty set (0.00 sec)
```

Product assignment

1. Display the product name and quantity of all products whose quantity is greater than 20.

```
mysql> SELECT pname,qty
    -> from product
    -> where qty>20;
+----------------+-----+
| pname          | qty |
+----------------+-----+
| Wireless Mouse |  25 |
| USB Cable      |  40 |
| Notebook       | 100 |
| Pen Pack       |  60 |
| Basmati Rice   |  50 |
| Cooking Oil    |  35 |
| Power Bank     |  22 |
| Marker Set     |  45 |
| Toothpaste     |  30 |
| Green Tea      |  25 |
+----------------+-----+
10 rows in set (0.00 sec)
```

2. Display pid, pname and qty for products whose quantity is less than or equal to 15.

```
mysql> SELECT pid,pname,qty
    -> from product
    -> where qty<=15;
+-----+----------------+-----+
| pid | pname          | qty |
+-----+----------------+-----+
| 102 | Keyboard       |  15 |
| 110 | Football       |  12 |
| 111 | Cricket Bat    |   8 |
| 112 | Mixer Grinder  |  10 |
| 113 | Electric Kettle |  14 |
| 119 | Air Fryer      |   7 |
| 120 | Webcam         |  11 |
+-----+----------------+-----+
7 rows in set (0.00 sec)
```

3. Display pname and a calculated column showing the quantity after adding 10 units to the current quantity.

```
mysql> select pname,qty+10
    -> from product;
+----------------+--------+
| pname          | qty+10 |
+----------------+--------+
| Wireless Mouse |     35 |
| Keyboard       |     25 |
| USB Cable      |     50 |
| Notebook       |    110 |
| Pen Pack       |     70 |
| Basmati Rice   |     60 |
| Cooking Oil    |     45 |
| Shampoo        |     30 |
| Face Wash      |     28 |
| Football       |     22 |
| Cricket Bat    |     18 |
| Mixer Grinder  |     20 |
| Electric Kettle |     24 |
| Power Bank     |     32 |
| Marker Set     |     55 |
| Toothpaste     |     40 |
| Green Tea      |     35 |
| Yoga Mat       |     26 |
| Air Fryer      |     17 |
| Webcam         |     21 |
+----------------+--------+
20 rows in set (0.00 sec)
```

4. Display pname and a calculated column showing the quantity after removing

5 units from the current quantity.

```
mysql> select pname,qty-5
    -> from product;
+-----------------+-------+
| pname           | qty-5 |
+-----------------+-------+
| Wireless Mouse  |    20 |
| Keyboard        |    10 |
| USB Cable       |    35 |
| Notebook        |    95 |
| Pen Pack        |    55 |
| Basmati Rice    |    45 |
| Cooking Oil     |    30 |
| Shampoo         |    15 |
| Face Wash       |    13 |
| Football        |     7 |
| Cricket Bat     |     3 |
| Mixer Grinder   |     5 |
| Electric Kettle |     9 |
| Power Bank      |    17 |
| Marker Set      |    40 |
| Toothpaste      |    25 |
| Green Tea       |    20 |
| Yoga Mat        |    11 |
| Air Fryer       |     2 |
| Webcam          |     6 |
+-----------------+-------+
20 rows in set (0.00 sec)
```

5. Display pname and a calculated column showing double the current quantity.

```
mysql> select pname,2*qty
    -> from product;
+-----------------+-------+
| pname           | 2*qty |
+-----------------+-------+
| Wireless Mouse  |    50 |
| Keyboard        |    30 |
| USB Cable       |    80 |
| Notebook        |   200 |
| Pen Pack        |   120 |
| Basmati Rice    |   100 |
| Cooking Oil     |    70 |
| Shampoo         |    40 |
| Face Wash       |    36 |
| Football        |    24 |
| Cricket Bat     |    16 |
| Mixer Grinder   |    20 |
| Electric Kettle |    28 |
| Power Bank      |    44 |
| Marker Set      |    90 |
| Toothpaste      |    60 |
| Green Tea       |    50 |
| Yoga Mat        |    32 |
| Air Fryer       |    14 |
| Webcam          |    22 |
+-----------------+-------+
20 rows in set (0.00 sec)
```

6. Display pname and a calculated column showing half of the current quantity.

```
mysql> select pname,0.5*qty
    -> from product;
+-----------------+---------+
| pname           | 0.5*qty |
+-----------------+---------+
| Wireless Mouse  |    12.5 |
| Keyboard        |     7.5 |
| USB Cable       |    20.0 |
| Notebook        |    50.0 |
| Pen Pack        |    30.0 |
| Basmati Rice    |    25.0 |
| Cooking Oil     |    17.5 |
| Shampoo         |    10.0 |
| Face Wash       |     9.0 |
| Football        |     6.0 |
| Cricket Bat     |     4.0 |
| Mixer Grinder   |     5.0 |
| Electric Kettle |     7.0 |
| Power Bank      |    11.0 |
| Marker Set      |    22.5 |
| Toothpaste      |    15.0 |
| Green Tea       |    12.5 |
| Yoga Mat        |     8.0 |
| Air Fryer       |     3.5 |
| Webcam          |     5.5 |
+-----------------+---------+
20 rows in set (0.01 sec)
```

7. Display pid, pname and qty for products whose quantity is between 10 and 30.

```
mysql> select pid,pname,qty
    -> from product
    -> where qty BETWEEN 10 and 30;
+-----+-----------------+-----+
| pid | pname           | qty |
+-----+-----------------+-----+
| 101 | Wireless Mouse  |  25 |
| 102 | Keyboard        |  15 |
| 108 | Shampoo         |  20 |
| 109 | Face Wash       |  18 |
| 110 | Football        |  12 |
| 112 | Mixer Grinder   |  10 |
| 113 | Electric Kettle |  14 |
| 114 | Power Bank      |  22 |
| 116 | Toothpaste      |  30 |
| 117 | Green Tea       |  25 |
| 118 | Yoga Mat        |  16 |
| 120 | Webcam          |  11 |
+-----+-----------------+-----+
12 rows in set (0.01 sec)
```

8. Display all products whose quantity is NOT between 10 and 30.

```
mysql> select pid,pname,qty
    -> from product
    -> where qty NOT BETWEEN 10 and 30;
+-----+--------------+-----+
| pid | pname        | qty |
+-----+--------------+-----+
| 103 | USB Cable    |  40 |
| 104 | Notebook     | 100 |
| 105 | Pen Pack     |  60 |
| 106 | Basmati Rice |  50 |
| 107 | Cooking Oil  |  35 |
| 111 | Cricket Bat  |   8 |
| 115 | Marker Set   |  45 |
| 119 | Air Fryer    |   7 |
+-----+--------------+-----+
8 rows in set (0.00 sec)
```

9. Display products belonging to category IDs 1, 3, or 5 using the IN operator.

```
mysql> select *
   -> from product
   -> where  cid in (1, 3,5);
+-----+----------------+-----+------------+------------+------+
| pid | pname          | qty | mfgdate    | expdate    | cid  |
+-----+----------------+-----+------------+------------+------+
| 101 | Wireless Mouse |  25 | 2025-01-10 | 2028-01-09 |    1 |
| 102 | Keyboard       |  15 | 2025-02-15 | 2028-02-14 |    1 |
| 103 | USB Cable      |  40 | 2025-03-01 | 2028-02-28 |    1 |
| 104 | Notebook       | 100 | 2025-01-05 | 2030-01-04 |    3 |
| 105 | Pen Pack       |  60 | 2025-04-10 | 2028-04-09 |    3 |
| 110 | Football       |  12 | 2025-01-20 | 2029-01-19 |    5 |
| 111 | Cricket Bat    |   8 | 2025-02-08 | 2029-02-07 |    5 |
| 114 | Power Bank     |  22 | 2025-06-10 | 2028-06-09 |    1 |
| 115 | Marker Set     |  45 | 2025-07-01 | 2028-06-30 |    3 |
| 118 | Yoga Mat       |  16 | 2025-08-20 | 2029-08-19 |    5 |
| 120 | Webcam         |  11 | 2025-09-15 | 2028-09-14 |    1 |
+-----+----------------+-----+------------+------------+------+
11 rows in set (0.00 sec)
```

10. Display products whose category ID is NOT IN 2, 4, and 6.

```
mysql> select *
    -> from product
    -> where  cid not in (2,4,5);
+-----+----------------+-----+------------+------------+------+
| pid | pname          | qty | mfgdate    | expdate    | cid  |
+-----+----------------+-----+------------+------------+------+
| 101 | Wireless Mouse |  25 | 2025-01-10 | 2028-01-09 |    1 |
| 102 | Keyboard       |  15 | 2025-02-15 | 2028-02-14 |    1 |
| 103 | USB Cable      |  40 | 2025-03-01 | 2028-02-28 |    1 |
| 104 | Notebook       | 100 | 2025-01-05 | 2030-01-04 |    3 |
| 105 | Pen Pack       |  60 | 2025-04-10 | 2028-04-09 |    3 |
| 112 | Mixer Grinder  |  10 | 2025-04-15 | 2029-04-14 |    6 |
| 113 | Electric Kettle |  14 | 2025-05-05 | 2029-05-04 |    6 |
| 114 | Power Bank     |  22 | 2025-06-10 | 2028-06-09 |    1 |
| 115 | Marker Set     |  45 | 2025-07-01 | 2028-06-30 |    3 |
| 119 | Air Fryer      |   7 | 2025-09-01 | 2029-08-31 |    6 |
| 120 | Webcam         |  11 | 2025-09-15 | 2028-09-14 |    1 |
+-----+----------------+-----+------------+------------+------+
11 rows in set (0.00 sec)
```

11. Display products whose quantity is greater than 20 AND category ID is 1.

```
mysql> select *
    -> from product
    -> where qty>20 and cid=1;
+-----+----------------+-----+------------+------------+------+
| pid | pname          | qty | mfgdate    | expdate    | cid  |
+-----+----------------+-----+------------+------------+------+
| 101 | Wireless Mouse |  25 | 2025-01-10 | 2028-01-09 |    1 |
| 103 | USB Cable      |  40 | 2025-03-01 | 2028-02-28 |    1 |
| 114 | Power Bank     |  22 | 2025-06-10 | 2028-06-09 |    1 |
+-----+----------------+-----+------------+------------+------+
3 rows in set (0.01 sec)
```

12. Display products whose quantity is less than 20 OR category ID is 5.

```
mysql> select *
    -> from product
    -> where qty<20 and cid=5;
+-----+-------------+-----+------------+------------+------+
| pid | pname       | qty | mfgdate    | expdate    | cid  |
+-----+-------------+-----+------------+------------+------+
| 110 | Football    |  12 | 2025-01-20 | 2029-01-19 |    5 |
| 111 | Cricket Bat |   8 | 2025-02-08 | 2029-02-07 |    5 |
| 118 | Yoga Mat    |  16 | 2025-08-20 | 2029-08-19 |    5 |
+-----+-------------+-----+------------+------------+------+
3 rows in set (0.00 sec)
```

13. Display products whose quantity is between 10 and 30 AND category ID is either 1 or 3.

```
mysql> select *
    -> from product
    -> where qty between 10 and 30 and cid in (1,3);
+-----+----------------+-----+------------+------------+------+
| pid | pname          | qty | mfgdate    | expdate    | cid  |
+-----+----------------+-----+------------+------------+------+
| 101 | Wireless Mouse |  25 | 2025-01-10 | 2028-01-09 |    1 |
| 102 | Keyboard       |  15 | 2025-02-15 | 2028-02-14 |    1 |
| 114 | Power Bank     |  22 | 2025-06-10 | 2028-06-09 |    1 |
| 120 | Webcam         |  11 | 2025-09-15 | 2028-09-14 |    1 |
+-----+----------------+-----+------------+------------+------+
4 rows in set (0.00 sec)
```

14. Display products whose quantity is greater than 40 OR quantity is less than 10.

```
mysql> select * from product where qty>40 or qty<10;
+-----+--------------+-----+------------+------------+------+
| pid | pname        | qty | mfgdate    | expdate    | cid  |
+-----+--------------+-----+------------+------------+------+
| 104 | Notebook     | 100 | 2025-01-05 | 2030-01-04 |    3 |
| 105 | Pen Pack     |  60 | 2025-04-10 | 2028-04-09 |    3 |
| 106 | Basmati Rice |  50 | 2025-05-12 | 2027-05-11 |    2 |
| 111 | Cricket Bat  |   8 | 2025-02-08 | 2029-02-07 |    5 |
| 115 | Marker Set   |  45 | 2025-07-01 | 2028-06-30 |    3 |
| 119 | Air Fryer    |   7 | 2025-09-01 | 2029-08-31 |    6 |
+-----+--------------+-----+------------+------------+------+
6 rows in set (0.00 sec)
```

15. Display products whose quantity is greater than 15 AND quantity is less than 50 AND category ID is 1.

```
mysql> select * from product where qty>15 and qty<50 and cid=1;
+-----+----------------+-----+------------+------------+------+
| pid | pname          | qty | mfgdate    | expdate    | cid  |
+-----+----------------+-----+------------+------------+------+
| 101 | Wireless Mouse |  25 | 2025-01-10 | 2028-01-09 |    1 |
| 103 | USB Cable      |  40 | 2025-03-01 | 2028-02-28 |    1 |
| 114 | Power Bank     |  22 | 2025-06-10 | 2028-06-09 |    1 |
+-----+----------------+-----+------------+------------+------+
3 rows in set (0.00 sec)
```

16. Display products manufactured between '2025-02-01' and '2025-06-30' using the BETWEEN operator.

```
mysql> select * from product where mfgdate between '2025-02-01' and '2025-06-30';
+-----+----------------+-----+------------+------------+------+
| pid | pname          | qty | mfgdate    | expdate    | cid  |
+-----+----------------+-----+------------+------------+------+
| 102 | Keyboard       |  15 | 2025-02-15 | 2028-02-14 |    1 |
| 103 | USB Cable      |  40 | 2025-03-01 | 2028-02-28 |    1 |
| 105 | Pen Pack       |  60 | 2025-04-10 | 2028-04-09 |    3 |
| 106 | Basmati Rice   |  50 | 2025-05-12 | 2027-05-11 |    2 |
| 107 | Cooking Oil    |  35 | 2025-06-20 | 2027-06-19 |    2 |
| 108 | Shampoo        |  20 | 2025-02-25 | 2027-02-24 |    4 |
| 109 | Face Wash      |  18 | 2025-03-18 | 2027-03-17 |    4 |
| 111 | Cricket Bat    |   8 | 2025-02-08 | 2029-02-07 |    5 |
| 112 | Mixer Grinder  |  10 | 2025-04-15 | 2029-04-14 |    6 |
| 113 | Electric Kettle |  14 | 2025-05-05 | 2029-05-04 |    6 |
| 114 | Power Bank     |  22 | 2025-06-10 | 2028-06-09 |    1 |
+-----+----------------+-----+------------+------------+------+
11 rows in set (0.01 sec)
```

17. Display products whose expiry date is NOT between '2027-01-01' and '2028-12-31'.

```
''2028-12-31'.' at line 1
mysql> select * from product where expdate NOT between '2027-01-01' and '2028-12-31';
+-----+----------------+-----+------------+------------+------+
| pid | pname          | qty | mfgdate    | expdate    | cid  |
+-----+----------------+-----+------------+------------+------+
| 104 | Notebook       | 100 | 2025-01-05 | 2030-01-04 |    3 |
| 110 | Football       |  12 | 2025-01-20 | 2029-01-19 |    5 |
| 111 | Cricket Bat    |   8 | 2025-02-08 | 2029-02-07 |    5 |
| 112 | Mixer Grinder  |  10 | 2025-04-15 | 2029-04-14 |    6 |
| 113 | Electric Kettle |  14 | 2025-05-05 | 2029-05-04 |    6 |
| 118 | Yoga Mat       |  16 | 2025-08-20 | 2029-08-19 |    5 |
| 119 | Air Fryer      |   7 | 2025-09-01 | 2029-08-31 |    6 |
+-----+----------------+-----+------------+------------+------+
7 rows in set (0.00 sec)
```

18. Display products whose quantity, when divided by 5, has remainder 0. Use the modulo (%) operator.

```
mysql> select * from product where qty%5=0;
+-----+----------------+-----+------------+------------+------+
| pid | pname          | qty | mfgdate    | expdate    | cid  |
+-----+----------------+-----+------------+------------+------+
| 101 | Wireless Mouse |  25 | 2025-01-10 | 2028-01-09 |    1 |
| 102 | Keyboard       |  15 | 2025-02-15 | 2028-02-14 |    1 |
| 103 | USB Cable      |  40 | 2025-03-01 | 2028-02-28 |    1 |
| 104 | Notebook       | 100 | 2025-01-05 | 2030-01-04 |    3 |
| 105 | Pen Pack       |  60 | 2025-04-10 | 2028-04-09 |    3 |
| 106 | Basmati Rice   |  50 | 2025-05-12 | 2027-05-11 |    2 |
| 107 | Cooking Oil    |  35 | 2025-06-20 | 2027-06-19 |    2 |
| 108 | Shampoo        |  20 | 2025-02-25 | 2027-02-24 |    4 |
| 112 | Mixer Grinder  |  10 | 2025-04-15 | 2029-04-14 |    6 |
| 115 | Marker Set     |  45 | 2025-07-01 | 2028-06-30 |    3 |
| 116 | Toothpaste     |  30 | 2025-07-15 | 2027-07-14 |    4 |
| 117 | Green Tea      |  25 | 2025-08-05 | 2027-08-04 |    2 |
+-----+----------------+-----+------------+------------+------+
12 rows in set (0.00 sec)
```

19. Display products whose quantity is greater than 20 AND whose category ID is NOT 2.

```
mysql> select * from product where qty>20 and cid!=2;
+-----+----------------+-----+------------+------------+------+
| pid | pname          | qty | mfgdate    | expdate    | cid  |
+-----+----------------+-----+------------+------------+------+
| 101 | Wireless Mouse |  25 | 2025-01-10 | 2028-01-09 |    1 |
| 103 | USB Cable      |  40 | 2025-03-01 | 2028-02-28 |    1 |
| 104 | Notebook       | 100 | 2025-01-05 | 2030-01-04 |    3 |
| 105 | Pen Pack       |  60 | 2025-04-10 | 2028-04-09 |    3 |
| 114 | Power Bank     |  22 | 2025-06-10 | 2028-06-09 |    1 |
| 115 | Marker Set     |  45 | 2025-07-01 | 2028-06-30 |    3 |
| 116 | Toothpaste     |  30 | 2025-07-15 | 2027-07-14 |    4 |
+-----+----------------+-----+------------+------------+------+
7 rows in set (0.00 sec)
```

20. Display products where either:
    - quantity is between 10 and 20 AND category ID is 1,
    OR
    - quantity is between 30 and 50 AND category ID is 2.

```
mysql> select * from product where qty between 10 and 20 AND cid=1 or qty between 30 and 50 AND cid=2;
+-----+--------------+-----+------------+------------+------+
| pid | pname        | qty | mfgdate    | expdate    | cid  |
+-----+--------------+-----+------------+------------+------+
| 102 | Keyboard     |  15 | 2025-02-15 | 2028-02-14 |    1 |
| 106 | Basmati Rice |  50 | 2025-05-12 | 2027-05-11 |    2 |
| 107 | Cooking Oil  |  35 | 2025-06-20 | 2027-06-19 |    2 |
| 120 | Webcam       |  11 | 2025-09-15 | 2028-09-14 |    1 |
+-----+--------------+-----+------------+------------+------+
4 rows in set (0.00 sec)

mysql>
```

Day(3)

Date and Time functions

1. Write a query to display the first day of the month (in datetime format) three months before the current month.

Sample current date : 2014-09-03

Expected result : 2014-06-01

SELECT DATE_FORMAT(DATE_SUB('2014-09-03', INTERVAL 3 MONTH), '%Y-%m-01');

```
+-----------------------------------------------------------------------+
| DATE_FORMAT(DATE_SUB('2014-09-03', INTERVAL 3 MONTH), '%Y-%m-01') |
+-----------------------------------------------------------------------+
| 2014-06-01                                                            |
+-----------------------------------------------------------------------+
1 row in set (0.00 sec)
```

2. Write a query to display the last day of the month (in datetime format) three months before the current month.

Select * from emp where last_day(date_sub(curdate(),interval 3 month));

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

3. Write a query to get the distinct Mondays from hiredate in emp tables.

```
mysql> SELECT DISTINCT hiredate
    -> FROM emp
    -> WHERE DAYNAME(hiredate) = 'Monday';
+------------+
| hiredate   |
+------------+
| 1981-09-28 |
+------------+
1 row in set (0.01 sec)
```

4. Write a query to get the first day of the current year.

```
mysql> SELECT DATE_FORMAT(CURRENT_DATE(), '%Y-01-01') AS first_day_of_year;
+--------------------+
| first_day_of_year |
+--------------------+
| 2026-01-01         |
+--------------------+
1 row in set (0.00 sec)
```

5. Write a query to get the last day of the current year.

```
mysql> SELECT DATE_FORMAT(CURRENT_DATE(), '%Y-12-31') AS last_day_of_year;
+------------------+
| last_day_of_year |
+------------------+
| 2026-12-31       |
+------------------+
1 row in set (0.00 sec)
```

6. Write a query to calculate your age in year.

```
mysql> SELECT TIMESTAMPDIFF(YEAR, '2004-07-04', CURDATE()) AS age;
+------+
| age  |
+------+
|   22 |
+------+
1 row in set (0.00 sec)
```

7. Write a query to get the current date in the following format.

Sample date : 04-sep-2014

Output : September 4, 2014

```
mysql> SELECT DATE_FORMAT(CURDATE(), '%M %e, %Y') AS formatted_date;
+--------------------+
| formatted_date     |
+--------------------+
| September 7, 2026 |
+--------------------+
1 row in set (0.00 sec)
```

8. Write a query to get the current date in Thursday September 2014 format.

Thursday September 2014

```
mysql> SELECT DATE_FORMAT(CURDATE(), '%w, %M, %Y');
+--------------------------------------+
| DATE_FORMAT(CURDATE(), '%w, %M, %Y') |
+--------------------------------------+
| 1, September, 2026                   |
+--------------------------------------+
1 row in set (0.00 sec)
```

9. Write a query to extract the year from the current date.

```
mysql>  Select extract(year from curdate());
+------------------------------+
| extract(year from curdate()) |
+------------------------------+
|                         2026 |
+------------------------------+
1 row in set (0.00 sec)
```

10. Write a query to get the first name and hire date from employees table where hire date between '1987-06-01' and '1987-07-30'

```
mysql> Select ename as first_name,hiredate from emp where hiredate between  '1987-06-01' and '1987-07-30' ;
Empty set (0.00 sec)
```

11. Write a query to display the current date in the following format.
Sample output: Thursday 4th September 2014 00:00:00

```
mysql> SELECT DATE_FORMAT(CURDATE(), '%W, %d, %M, %Y, %r');
+-----------------------------------------------+
| DATE_FORMAT(CURDATE(), '%W, %d, %M, %Y, %r') |
+-----------------------------------------------+
| Monday, 07, September, 2026, 12:00:00 AM      |
+-----------------------------------------------+
1 row in set (0.00 sec)
```

12. Write a query to display the current date in the following format.
Sample output: 05/09/2014

```
mysql> SELECT DATE_FORMAT(CURDATE(), '%d/%m/%Y') AS formatted_date;
+----------------+
| formatted_date |
+----------------+
| 07/09/2026     |
+----------------+
1 row in set (0.00 sec)
```

13. Write a query to display the current date in the following format.
Sample output: 12:00 AM Sep 5, 2014

```
mysql> SELECT DATE_FORMAT(NOW(), '%r %b %m, %Y');
+-----------------------------------+
| DATE_FORMAT(NOW(), '%r %b %m, %Y') |
+-----------------------------------+
| 04:58:13 PM Sep 09, 2026          |
+-----------------------------------+
1 row in set (0.00 sec)
```

14. Write a query to get the employees who joined in the month of June.

```
mysql> Select * from emp where monthname(hiredate)= 'June' ;
+-------+-------+---------+------+------------+---------+------+--------+
| EMPNO | ENAME | JOB     | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+-------+---------+------+------------+---------+------+--------+
|  7782 | CLARK | MANAGER | 7839 | 1981-06-09 | 2450.00 | NULL |     10 |
+-------+-------+---------+------+------------+---------+------+--------+
1 row in set (0.00 sec)
```

15. Write a query to get the years in which more than 10 employees joined.

```
mysql> Select year(hiredate),count(*)
    -> From emp
    -> Group by hiredate
    -> Having count(*)>10;
Empty set (0.00 sec)
```

16. Write a query to get first name of employees who joined in 1987. Select ename from emp where hiredate between '1987-01-01' and1987-12-31';

```
mysql> Select ename
    -> From emp
    -> Where year(hiredate)=1987;
Empty set (0.00 sec)
```

17. Write a query to get employees whose experience is more than 5 years.

```
mysql> select empno,ename,hiredate,timestampdiff(YEAR,hiredate,curdate()) experience
    ->  from emp
    -> where timestampdiff(YEAR,hiredate,curdate())>5;
+-------+--------+------------+------------+
| empno | ename  | hiredate   | experience |
+-------+--------+------------+------------+
|  7369 | SMITH  | 1980-12-17 |         45 |
|  7499 | ALLEN  | 1981-02-20 |         45 |
|  7521 | WARD   | 1981-02-22 |         45 |
|  7566 | JONES  | 1981-04-02 |         45 |
|  7654 | MARTIN | 1981-09-28 |         44 |
|  7698 | BLAKE  | 1981-05-01 |         45 |
|  7782 | CLARK  | 1981-06-09 |         45 |
|  7788 | SCOTT  | 1982-12-09 |         43 |
|  7839 | KING   | 1981-11-17 |         44 |
|  7844 | TURNER | 1981-09-08 |         44 |
|  7876 | ADAMS  | 1983-01-12 |         43 |
|  7900 | JAMES  | 1981-12-03 |         44 |
|  7902 | FORD   | 1981-12-03 |         44 |
|  7934 | MILLER | 1982-01-23 |         44 |
+-------+--------+------------+------------+
14 rows in set (0.00 sec)
```

18. Write a query to get employee ID, last name, and date of first salary of the employees.

```
mysql> Select empno, ename as last_name, DATE_ADD(LAST_DAY(hiredate), INTERVAL 1 DAY) from emp;
+-------+-----------+-----------------------------------------------+
| empno | last_name | DATE_ADD(LAST_DAY(hiredate), INTERVAL 1 DAY) |
+-------+-----------+-----------------------------------------------+
|  7369 | SMITH     | 1981-01-01                                    |
|  7499 | ALLEN     | 1981-03-01                                    |
|  7521 | WARD      | 1981-03-01                                    |
|  7566 | JONES     | 1981-05-01                                    |
|  7654 | MARTIN    | 1981-10-01                                    |
|  7698 | BLAKE     | 1981-06-01                                    |
|  7782 | CLARK     | 1981-07-01                                    |
|  7788 | SCOTT     | 1983-01-01                                    |
|  7839 | KING      | 1981-12-01                                    |
|  7844 | TURNER    | 1981-10-01                                    |
|  7876 | ADAMS     | 1983-02-01                                    |
|  7900 | JAMES     | 1982-01-01                                    |
|  7902 | FORD      | 1982-01-01                                    |
|  7934 | MILLER    | 1982-02-01                                    |
+-------+-----------+-----------------------------------------------+
14 rows in set (0.00 sec)
```

19. Write a query to get first name, hire date and experience of the employees.

Sample table: employees

```
mysql> Select ename as first_name, hiredate,timestampdiff(year,hiredate,curdate()) as exp from emp ;
+------------+------------+------+
| first_name | hiredate   | exp  |
+------------+------------+------+
| SMITH      | 1980-12-17 |   45 |
| ALLEN      | 1981-02-20 |   45 |
| WARD       | 1981-02-22 |   45 |
| JONES      | 1981-04-02 |   45 |
| MARTIN     | 1981-09-28 |   44 |
| BLAKE      | 1981-05-01 |   45 |
| CLARK      | 1981-06-09 |   45 |
| SCOTT      | 1982-12-09 |   43 |
| KING       | 1981-11-17 |   44 |
| TURNER     | 1981-09-08 |   44 |
| ADAMS      | 1983-01-12 |   43 |
| JAMES      | 1981-12-03 |   44 |
| FORD       | 1981-12-03 |   44 |
| MILLER     | 1982-01-23 |   44 |
+------------+------------+------+
14 rows in set (0.00 sec)
```

20. Write a query to get the department ID, year, and number of employees joined.

```
mysql>  Select deptno,year(hiredate),count(*)
    -> From emp
    -> Group by deptno,year(hiredate)
    -> Order by deptno;
+--------+----------------+----------+
| deptno | year(hiredate) | count(*) |
+--------+----------------+----------+
|     10 |           1981 |        2 |
|     10 |           1982 |        1 |
|     20 |           1980 |        1 |
|     20 |           1981 |        2 |
|     20 |           1982 |        1 |
|     20 |           1983 |        1 |
|     30 |           1981 |        6 |
+--------+----------------+----------+
7 rows in set (0.00 sec)
```

(Day 4)

Question 1 – Online Shopping System

Create the following tables:

Customer

`customerid, customername, email, phone`

Orders

`orderid, orderdate, amount, status, customerid`

A. CREATE TABLE

1. Create `Customer` with `customerid` as the **PRIMARY KEY**.
2. `customername` should be **NOT NULL**.
3. `email` should be **NOT NULL** and **UNIQUE**.
4. `phone` should be **UNIQUE**.
5. Create `Orders` with `orderid` as the **PRIMARY KEY**.

```
mysql> CREATE TABLE Customer (
    ->     customerid INT AUTO_INCREMENT PRIMARY KEY,
    ->     customername VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) NOT NULL UNIQUE,
    ->     phone VARCHAR(15) UNIQUE
    -> );
Query OK, 0 rows affected (1.32 sec)
```

6. `orderdate` should be **NOT NULL**.
7. Set the default value of `amount` to `0`.
8. Set the default value of `status` to `'Pending'`.
9. `customerid` should be a **FOREIGN KEY** referencing `Customer(customerid)`.

```
mysql> CREATE TABLE Orders (
    ->     orderid INT AUTO_INCREMENT PRIMARY KEY,
    ->     orderdate DATE NOT NULL,
    ->     amount DECIMAL(10, 2) DEFAULT 0,
    ->     status VARCHAR(20) DEFAULT 'Pending',
    ->     customerid INT,
    ->     FOREIGN KEY (customerid) REFERENCES Customer(customerid)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

B. ALTER TABLE

10. Add a new column `city` to the `Customer` table.

```
mysql> ALTER TABLE Customer
    -> ADD city VARCHAR(50);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

11. Add a new column `shippingaddress` to the `Orders` table.
12. Add a `NOT NULL` constraint to `shippingaddress`.

```
mysql> ALTER TABLE Orders
    -> ADD shippingaddress VARCHAR(255) NOT NULL;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

13. Add a new column `paymentmode` to the `Orders` table.
14. Set the default value of `paymentmode` to `'Cash'`.

```
mysql> ALTER TABLE Orders
    -> ADD paymentmode VARCHAR(20) DEFAULT 'Cash';
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

15. Rename the `phone` column of `Customer` to `mobileno` if your database supports column renaming.

```
mysql> ALTER TABLE Customer
    -> RENAME COLUMN phone TO mobileno;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

C. INSERT

16. Insert **5 customers** into the `Customer` table.

```
mysql> INSERT INTO Customer (customername, email, mobileno, city) VALUES
    -> ('Alice Smith', 'alice@example.com', '9876543210', 'New York'),
    -> ('Bob Johnson', 'bob@example.com', '9876543211', 'Los Angeles'),
    -> ('Charlie Brown', 'charlie@example.com', '9876543212', 'Chicago'),
    -> ('Diana Prince', 'diana@example.com', '9876543213', 'Houston'),
    -> ('Evan Wright', 'evan@example.com', '9876543214', 'Phoenix');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

17. Insert **7 orders** into the `Orders` table.

```
mysql> INSERT INTO Orders (orderdate, amount, status, customerid, shippingad
dress, paymentmode) VALUES
    -> ('2026-01-10', 150.00, 'Pending', 1, '123 Main St, NY', 'Credit Card'
),
    -> ('2026-01-12', 250.50, 'Confirmed', 1, '123 Main St, NY', 'PayPal'),
    -> ('2026-01-15', 75.00, 'Pending', 2, '456 Elm St, LA', 'Cash'),
    -> ('2026-01-18', 500.00, 'Shipped', 3, '789 Oak St, Chicago', 'Credit C
ard'),
    -> ('2026-01-20', 120.00, 'Pending', 3, '789 Oak St, Chicago', 'Debit Ca
rd'),
    -> ('2026-01-22', 310.00, 'Delivered', 4, '101 Pine St, Houston', 'Cash'
),
    -> ('2026-01-25', 95.25, 'Pending', 4, '101 Pine St, Houston', 'PayPal')
;
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0
```

18. Insert one order without specifying `status` and verify that the default value is

inserted.

```
mysql> INSERT INTO Orders (orderdate, amount, customerid, shippingaddress, p
aymentmode)
    -> VALUES ('2026-02-01', 200.00, 2, '456 Elm St, LA', 'Credit Card');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM Orders WHERE orderid = LAST_INSERT_ID();
+---------+------------+--------+---------+------------+-----------------+--
-----------+
| orderid | orderdate  | amount | status  | customerid | shippingaddress | p
aymentmode |
+---------+------------+--------+---------+------------+-----------------+--
-----------+
|       8 | 2026-02-01 | 200.00 | Pending |          2 | 456 Elm St, LA  | C
redit Card |
+---------+------------+--------+---------+------------+-----------------+--
-----------+
1 row in set (0.01 sec)
```

19. Insert one order without specifying `amount` and verify the default value.

```
mysql> INSERT INTO Orders (orderdate, status, customerid, shippingaddress, p
aymentmode)
    -> VALUES ('2026-02-02', 'Pending', 1, '123 Main St, NY', 'Cash');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Orders WHERE orderid = LAST_INSERT_ID();
+---------+------------+--------+---------+------------+-----------------+--
-----------+
| orderid | orderdate  | amount | status  | customerid | shippingaddress | p
aymentmode |
+---------+------------+--------+---------+------------+-----------------+--
-----------+
|       9 | 2026-02-02 |   0.00 | Pending |          1 | 123 Main St, NY | C
ash        |
+---------+------------+--------+---------+------------+-----------------+--
-----------+
1 row in set (0.00 sec)
```

D. UPDATE

20. Update the city of a particular customer.

```
mysql> UPDATE Customer
    -> SET city = 'San Francisco'
    -> WHERE customerid = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

21. Increase the amount of a particular order.

```
mysql> UPDATE Orders
    -> SET amount = amount + 50.00
    -> WHERE orderid = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

22. Change the status of an order from `'Pending'` to `'Confirmed'`. 23. Change the payment mode of a particular order.

```
mysql> UPDATE Orders
    -> SET paymentmode = 'Credit Card'
    -> WHERE orderid = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

E. DELETE

24. Delete a particular order.

```
mysql> DELETE FROM Orders
    -> WHERE orderid = 7;
Query OK, 1 row affected (0.01 sec)
```

25. Delete a customer who does not have any orders.

```
mysql> DELETE FROM Customer
    -> WHERE customerid = 5
    ->   AND customerid NOT IN (SELECT DISTINCT customerid FROM Orders WHERE
 customerid IS NOT NULL);
Query OK, 1 row affected (0.01 sec)
```

26. Try deleting a customer who has orders and observe the result because of the **FOREIGN KEY** constraint.

```
mysql> DELETE FROM Customer WHERE customerid = 1;
```

Question 2 – Movie Ticket Booking System

Create the following tables:

Movie

`movieid, moviename, language, duration`

Booking

`bookingid, customername, ticketcount, bookingdate, movieid`

A. CREATE TABLE

1. Create `Movie` with `movieid` as the **PRIMARY KEY**.
2. `moviename` should be **NOT NULL** and **UNIQUE**.
3. `language` should be **NOT NULL**.

```
mysql> CREATE TABLE Movie (
    ->     movieid INT AUTO_INCREMENT PRIMARY KEY,
    ->     moviename VARCHAR(100) NOT NULL UNIQUE,
    ->     language VARCHAR(50) NOT NULL,
    ->     duration INT
    -> );
Query OK, 0 rows affected (0.04 sec)
```

4. Create `Booking` with `bookingid` as the **PRIMARY KEY**.
5. `customername` should be **NOT NULL**.
6. Set the default value of `ticketcount` to `1`.
7. `bookingdate` should be **NOT NULL**.
8. `movieid` should be a **FOREIGN KEY** referencing `Movie(movieid)`.

```
ysql> CREATE TABLE Booking (
   ->     bookingid INT AUTO_INCREMENT PRIMARY KEY,
   ->     customername VARCHAR(100) NOT NULL,
   ->     ticketcount INT DEFAULT 1,
   ->     bookingdate DATE NOT NULL,
   ->     movieid INT,
   ->     FOREIGN KEY (movieid) REFERENCES Movie(movieid)
   -> );
uery OK, 0 rows affected (0.04 sec)
```

B. ALTER TABLE

9. Add `price` to the `Booking` table.
10. Set the default value of `price` to `200`.

```
mysql> ALTER TABLE Booking
    -> ADD price DECIMAL(10, 2) DEFAULT 200;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

11. Add `theatre` to the `Booking` table.
12. Add a `NOT NULL` constraint to `theatre`.

```
mysql> ALTER TABLE Booking
    -> ADD theatre VARCHAR(100) NOT NULL;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

13. Add `genre` to the `Movie` table.

```
mysql> ALTER TABLE Movie
    -> ADD genre VARCHAR(50);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

14. Add a `UNIQUE` constraint on `theatre` **only if appropriate for your design**. Explain why it may or may not be suitable.

C. INSERT

15. Insert **5 movies**.

```
mysql> INSERT INTO Movie (moviename, language, duration, genre) VALUES
    -> ('Inception', 'English', 148, 'Sci-Fi'),
    -> ('RRR', 'Telugu', 187, 'Action'),
    -> ('Parasite', 'Korean', 132, 'Thriller'),
    -> ('Spirited Away', 'Japanese', 125, 'Animation'),
    -> ('Dangal', 'Hindi', 161, 'Biography');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

16. Insert **8 bookings**.

```
ysql> INSERT INTO Booking (customername, bookingdate, movieid, price, theat
e)
   -> VALUES ('Stanley Hudson', '2026-03-06', 1, 250.00, 'INOX IMAX');
uery OK, 1 row affected (0.01 sec)

ysql> SELECT * FROM Booking WHERE bookingid = LAST_INSERT_ID();
-----------+----------------+-------------+-------------+---------+--------
-----------+
 bookingid | customername   | ticketcount | bookingdate | movieid | price
 theatre   |
-----------+----------------+-------------+-------------+---------+--------
-----------+
         1 | Stanley Hudson |           1 | 2026-03-06  |       1 | 250.00
 INOX IMAX |
-----------+----------------+-------------+-------------+---------+--------
-----------+
 row in set (0.00 sec)
```

17. Insert a booking without specifying `ticketcount` and verify the default value. 18. Insert a booking without specifying `price` and verify the default value.

```
mysql> INSERT INTO Booking (customername, ticketcount, bookingdate, movieid,
 theatre)
    -> VALUES ('Kevin Malone', 2, '2026-03-06', 2, 'PVR Cinemas');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Booking WHERE bookingid = LAST_INSERT_ID();
+-----------+--------------+-------------+-------------+---------+--------+-
------------+
| bookingid | customername | ticketcount | bookingdate | movieid | price  |
theatre     |
+-----------+--------------+-------------+-------------+---------+--------+-
------------+
|         2 | Kevin Malone |           2 | 2026-03-06  |       2 | 200.00 |
PVR Cinemas |
+-----------+--------------+-------------+-------------+---------+--------+-
------------+
1 row in set (0.00 sec)
```

D. UPDATE

19. Change the language of a movie.

```
mysql> UPDATE Movie
    -> SET language = 'English Dubbed'
    -> WHERE movieid = 4;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

20. Update the duration of a movie.

```
mysql> UPDATE Movie
    -> SET duration = 150
    -> WHERE movieid = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

21. Change the number of tickets for a particular booking.

```
mysql> UPDATE Booking
    -> SET ticketcount = 4
    -> WHERE bookingid = 3;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

22. Change the price of a booking.

```
mysql> UPDATE Booking
    -> SET price = 220.00
    -> WHERE bookingid = 5;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

23. Change the theatre of a booking.

```
mysql> UPDATE Booking
    -> SET theatre = 'Regal Cinema'
    -> WHERE bookingid = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

E. DELETE

24. Delete a particular booking.

```
mysql> DELETE FROM Booking
    -> WHERE bookingid = 8;
Query OK, 0 rows affected (0.00 sec)
```

25. Delete a movie that has no bookings.

```
mysql> DELETE FROM Booking WHERE movieid = 5;
st
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DELETE FROM Movie WHERE movieid = 5;
Query OK, 1 row affected (0.00 sec)
```

26. Try deleting a movie that is referenced by a booking and observe the result.

```
mysql> DELETE FROM Movie WHERE movieid = 1;
```

Question 3 – Vehicle Service Center

Create the following tables:

Vehicle

`vehicleid, vehicleno, model, ownername, phone`

Service

`serviceid, servicedate, servicetype, charges,vehicleid` **A.**

CREATE TABLE

1. Create `Vehicle` with `vehicleid` as the **PRIMARY KEY**.
2. `vehicleno` should be **NOT NULL** and **UNIQUE**.
3. `ownername` should be **NOT NULL**.
4. `phone` should be **UNIQUE**.

```
ysql> CREATE TABLE Vehicle (
    ->     vehicleid INT AUTO_INCREMENT PRIMARY KEY,
    ->     vehicleno VARCHAR(20) NOT NULL UNIQUE,
    ->     model VARCHAR(50),
    ->     ownername VARCHAR(100) NOT NULL,
    ->     phone VARCHAR(15) UNIQUE
    -> );
uery OK, 0 rows affected (0.04 sec)
```

5. Create `Service` with `serviceid` as the **PRIMARY KEY**.
6. `servicedate` and `servicetype` should be **NOT NULL**.
7. Set the default value of `charges` to `0`.
8. Create a **FOREIGN KEY** on `vehicleid` referencing `Vehicle(vehicleid)`.

```
ysql> CREATE TABLE Service (
    ->     serviceid INT AUTO_INCREMENT PRIMARY KEY,
    ->     servicedate DATE NOT NULL,
    ->     servicetype VARCHAR(50) NOT NULL,
    ->     charges DECIMAL(10, 2) DEFAULT 0,
    ->     vehicleid INT,
    ->     FOREIGN KEY (vehicleid) REFERENCES Vehicle(vehicleid)
    -> );
uery OK, 0 rows affected (0.04 sec)
```

B. ALTER TABLE

9. Add `color` to the `Vehicle` table.

```
ysql> ALTER TABLE Vehicle
    -> ADD color VARCHAR(30);
uery OK, 0 rows affected (0.06 sec)
ecords: 0  Duplicates: 0  Warnings: 0
```

10. Add `address` to the `Vehicle` table.

```
mysql> ALTER TABLE Vehicle
    -> ADD address VARCHAR(255);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

11. Add `servicestatus` to the `Service` table.
12. Set the default value of `servicestatus` to `'Pending'`.

```
mysql> ALTER TABLE Service
    -> ADD servicestatus VARCHAR(20) DEFAULT 'Pending';
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

13. Add `mechanicname` to the `Service` table.

```
mysql> ALTER TABLE Service
    -> ADD mechanicname VARCHAR(100);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

14. Modify the size of `ownername` to allow longer names.

```
mysql> ALTER TABLE Vehicle
    -> MODIFY ownername VARCHAR(250) NOT NULL;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

15. Add a `UNIQUE` constraint on `phone` if it was not defined while creating the table.

```
mysql> ALTER TABLE Vehicle
    -> ADD CONSTRAINT unique_vehicle_phone UNIQUE (phone);
Query OK, 0 rows affected, 1 warning (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 1
```

C. INSERT

16. Insert **5 vehicles**.

```
mysql> INSERT INTO Vehicle (vehicleno, model, ownername, phone, color, addre
ss) VALUES
    -> ('KA-01-AB-1234', 'Honda City', 'Bruce Wayne', '9900011122', 'Black',
 'Gotham City'),
    -> ('KA-02-CD-5678', 'Hyundai Creta', 'Clark Kent', '9900011123', 'Blue'
, 'Metropolis'),
    -> ('KA-03-EF-9012', 'Toyota Fortuner', 'Tony Stark', '9900011124', 'Red
', 'Malibu'),
    -> ('KA-04-GH-3456', 'Maruti Swift', 'Peter Parker', '9900011125', 'Whit
e', 'Queens'),
    -> ('KA-05-IJ-7890', 'Ford Mustang', 'Steve Rogers', '9900011126', 'Silv
er', 'Brooklyn');
Query OK, 5 rows affected (0.01 sec)
```

17. Insert **8 service records**.

```
ysql> INSERT INTO Service (servicedate, servicetype, vehicleid, servicestat
s, mechanicname)
    -> VALUES ('2026-02-22', 'General Inspection', 1, 'Pending', 'Alex Wrenc
');
uery OK, 1 row affected (0.01 sec)

ysql> SELECT * FROM Service WHERE serviceid = LAST_INSERT_ID();
-----------+-------------+--------------------+---------+-----------+------
--------+--------------+
 serviceid | servicedate | servicetype        | charges | vehicleid | servi
estatus | mechanicname |
-----------+-------------+--------------------+---------+-----------+------
--------+--------------+
         9 | 2026-02-22  | General Inspection |    0.00 |         1 | Pendi
g       | Alex Wrench  |
-----------+-------------+--------------------+---------+-----------+------
--------+--------------+
 row in set (0.00 sec)
```

18. Insert a service record without specifying `charges` and verify the default value.
19. Insert a service record without specifying `servicestatus` and verify the default value.

```
mysql> INSERT INTO Service (servicedate, servicetype, charges, vehicleid, me
chanicname)
    -> VALUES ('2026-02-23', 'Tire Rotation', 600.00, 2, 'John Garage');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Service WHERE serviceid = LAST_INSERT_ID();
+-----------+-------------+---------------+---------+-----------+-----------
----+---------------+
| serviceid | servicedate | servicetype   | charges | vehicleid | servicesta
tus | mechanicname |
+-----------+-------------+---------------+---------+-----------+-----------
----+---------------+
|        10 | 2026-02-23  | Tire Rotation |  600.00 |         2 | Pending
    | John Garage  |
+-----------+-------------+---------------+---------+-----------+-----------
----+---------------+
1 row in set (0.00 sec)
```

D. UPDATE

20. Change the color of a particular vehicle.

```
mysql> UPDATE Vehicle
    -> SET color = 'Matte Black'
    -> WHERE vehicleid = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

21. Change the owner's phone number.

```
mysql> UPDATE Vehicle
    -> SET phone = '9900099999'
    -> WHERE vehicleid = 4;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

22. Increase the service charges for a particular service.

```
mysql> UPDATE Service
    -> SET charges = charges + 300.00
    -> WHERE serviceid = 3;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

23. Change `servicestatus` from `'Pending'` to `'Completed'`.

```
ysql> UPDATE Service
    -> SET servicestatus = 'Completed'
    -> WHERE serviceid = 4;
uery OK, 1 row affected (0.01 sec)
ows matched: 1  Changed: 1  Warnings: 0
```

24. Change the mechanic assigned to a service.

```
mysql> UPDATE Service
    -> SET mechanicname = 'Dave Tools'
    -> WHERE serviceid = 7;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

E. DELETE

25. Delete a particular service record.

```
mysql> DELETE FROM Service
    -> WHERE serviceid = 8;
Query OK, 1 row affected (0.01 sec)
```

26. Delete a vehicle that has no service records.

```
mysql> DELETE FROM Vehicle
    -> WHERE vehicleid = 5
    ->   AND vehicleid NOT IN (SELECT DISTINCT vehicleid FROM Service WHERE
vehicleid IS NOT NULL);
Query OK, 1 row affected (0.01 sec)
```

27. Try deleting a vehicle that has service records and observe the FOREIGN Keyconstraint.

```
mysql> DELETE FROM Vehicle WHERE vehicleid = 1;
```

(Day5)

Q1. Write SQL statement for the following

1.To find all managers with salary >1500

```
mysql> select * from emp where sal>1500;
+-------+-------+-----------+------+------------+---------+--------+--------+
| EMPNO | ENAME | JOB       | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+-------+-----------+------+------------+---------+--------+--------+
|  7499 | ALLEN | SALESMAN  | 7698 | 1981-02-20 | 1600.00 | 300.00 |     30 |
|  7566 | JONES | MANAGER   | 7839 | 1981-04-02 | 2975.00 |   NULL |     20 |
|  7698 | BLAKE | MANAGER   | 7839 | 1981-05-01 | 2850.00 |   NULL |     30 |
|  7782 | CLARK | MANAGER   | 7839 | 1981-06-09 | 2450.00 |   NULL |     10 |
|  7788 | SCOTT | ANALYST   | 7566 | 1982-12-09 | 3000.00 |   NULL |     20 |
|  7839 | KING  | PRESIDENT | NULL | 1981-11-17 | 5000.00 |   NULL |     10 |
|  7902 | FORD  | ANALYST   | 7566 | 1981-12-03 | 3000.00 |   NULL |     20 |
+-------+-------+-----------+------+------------+---------+--------+--------+
7 rows in set (0.00 sec)
```

2. list all employees with sal >1200 and < 2000

```
mysql> select * from emp where sal between 1201 and 1999;
+-------+--------+----------+------+------------+---------+---------+--------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
+-------+--------+----------+------+------------+---------+---------+--------+
|  7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1600.00 |  300.00 |     30 |
|  7521 | WARD   | SALESMAN | 7698 | 1981-02-22 | 1250.00 |  500.00 |     30 |
|  7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250.00 | 1400.00 |     30 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00 |    0.00 |     30 |
|  7934 | MILLER | CLERK    | 7782 | 1982-01-23 | 1300.00 |    NULL |     10 |
+-------+--------+----------+------+------------+---------+---------+--------+
5 rows in set (0.00 sec)
```

3. list all employees with sal is 1600 or sal is 800 or sal is 1900

```
mysql> select * from emp where sal in(1600,800,1900);
+-------+-------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME | JOB      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+-------+----------+------+------------+---------+--------+--------+
|  7369 | SMITH | CLERK    | 7902 | 1980-12-17 |  800.00 |   NULL |     20 |
|  7499 | ALLEN | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00 |     30 |
+-------+-------+----------+------+------------+---------+--------+--------+
2 rows in set (0.00 sec)
```

4. list all employees with ‘R’ at second last position in name

```
mysql> select * from emp where ename like '%R_';
+-------+-------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME | JOB      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+-------+----------+------+------------+---------+--------+--------+
|  7521 | WARD  | SALESMAN | 7698 | 1981-02-22 | 1250.00 | 500.00 |     30 |
|  7782 | CLARK | MANAGER  | 7839 | 1981-06-09 | 2450.00 |   NULL |     10 |
|  7902 | FORD  | ANALYST  | 7566 | 1981-12-03 | 3000.00 |   NULL |     20 |
+-------+-------+----------+------+------------+---------+--------+--------+
3 rows in set (0.00 sec)
```

5. List all employees with name starts with A and ends with N

```
mysql> select * from emp where ename regexp'^[Aa].*[N]$';
+-------+-------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME | JOB      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+-------+----------+------+------------+---------+--------+--------+
|  7499 | ALLEN | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00 |     30 |
+-------+-------+----------+------+------------+---------+--------+--------+
1 row in set (0.00 sec)
```

Q2. Solve following

1.list all employees with salary > 1250 and dept no=30

```
mysql> select  * from emp where sal>1250 and deptno=30;
+-------+--------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+--------+----------+------+------------+---------+--------+--------+
|  7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00 |     30 |
|  7698 | BLAKE  | MANAGER  | 7839 | 1981-05-01 | 2850.00 |   NULL |     30 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00 |   0.00 |     30 |
+-------+--------+----------+------+------------+---------+--------+--------+
3 rows in set (0.00 sec)
```

2. list all employees with salary >=1250 and <= 3000

```
mysql> select  * from emp where sal between 1250 and 3000;
+-------+--------+----------+------+------------+---------+---------+--------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
+-------+--------+----------+------+------------+---------+---------+--------+
|  7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1600.00 |  300.00 |     30 |
|  7521 | WARD   | SALESMAN | 7698 | 1981-02-22 | 1250.00 |  500.00 |     30 |
|  7566 | JONES  | MANAGER  | 7839 | 1981-04-02 | 2975.00 |    NULL |     20 |
|  7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250.00 | 1400.00 |     30 |
|  7698 | BLAKE  | MANAGER  | 7839 | 1981-05-01 | 2850.00 |    NULL |     30 |
|  7782 | CLARK  | MANAGER  | 7839 | 1981-06-09 | 2450.00 |    NULL |     10 |
|  7788 | SCOTT  | ANALYST  | 7566 | 1982-12-09 | 3000.00 |    NULL |     20 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00 |    0.00 |     30 |
|  7902 | FORD   | ANALYST  | 7566 | 1981-12-03 | 3000.00 |    NULL |     20 |
|  7934 | MILLER | CLERK    | 7782 | 1982-01-23 | 1300.00 |    NULL |     10 |
+-------+--------+----------+------+------------+---------+---------+--------+
10 rows in set (0.00 sec)
```

3. list all employees with salary >1250 and < 3000

```
mysql> select  * from emp where sal between 1251 and 2999;
+-------+--------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+--------+----------+------+------------+---------+--------+--------+
|  7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00 |     30 |
|  7566 | JONES  | MANAGER  | 7839 | 1981-04-02 | 2975.00 |   NULL |     20 |
|  7698 | BLAKE  | MANAGER  | 7839 | 1981-05-01 | 2850.00 |   NULL |     30 |
|  7782 | CLARK  | MANAGER  | 7839 | 1981-06-09 | 2450.00 |   NULL |     10 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00 |   0.00 |     30 |
|  7934 | MILLER | CLERK    | 7782 | 1982-01-23 | 1300.00 |   NULL |     10 |
+-------+--------+----------+------+------------+---------+--------+--------+
6 rows in set (0.00 sec)
```

4. list all employees with salary either equal to 3000 or 1250 or 2500

```
mysql> select * from emp where sal in(3000,1250,2500);
+-------+--------+----------+------+------------+---------+---------+--------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
+-------+--------+----------+------+------------+---------+---------+--------+
|  7521 | WARD   | SALESMAN | 7698 | 1981-02-22 | 1250.00 |  500.00 |     30 |
|  7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250.00 | 1400.00 |     30 |
|  7788 | SCOTT  | ANALYST  | 7566 | 1982-12-09 | 3000.00 |    NULL |     20 |
|  7902 | FORD   | ANALYST  | 7566 | 1981-12-03 | 3000.00 |    NULL |     20 |
+-------+--------+----------+------+------------+---------+---------+--------+
4 rows in set (0.00 sec)
```

5. list all employee with name=SMITH

```
mysql> select * from emp where ename='smith';
+-------+-------+-------+------+------------+--------+------+--------+
| EMPNO | ENAME | JOB   | MGR  | HIREDATE   | SAL    | COMM | DEPTNO |
+-------+-------+-------+------+------------+--------+------+--------+
|  7369 | SMITH | CLERK | 7902 | 1980-12-17 | 800.00 | NULL |     20 |
+-------+-------+-------+------+------------+--------+------+--------+
1 row in set (0.00 sec)
```

6. list all employees with name starting with S

```
mysql> select * from emp where ename regexp'^[Ss]';
+-------+-------+---------+------+------------+---------+------+--------+
| EMPNO | ENAME | JOB     | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+-------+---------+------+------------+---------+------+--------+
|  7369 | SMITH | CLERK   | 7902 | 1980-12-17 |  800.00 | NULL |     20 |
|  7788 | SCOTT | ANALYST | 7566 | 1982-12-09 | 3000.00 | NULL |     20 |
+-------+-------+---------+------+------------+---------+------+--------+
2 rows in set (0.00 sec)
```

7. list all employees with name ending with S

```
mysql> select * from emp where ename regexp'[Ss]$';
+-------+-------+---------+------+------------+---------+------+--------+
| EMPNO | ENAME | JOB     | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+-------+---------+------+------------+---------+------+--------+
|  7566 | JONES | MANAGER | 7839 | 1981-04-02 | 2975.00 | NULL |     20 |
|  7876 | ADAMS | CLERK   | 7788 | 1983-01-12 | 1100.00 | NULL |     20 |
|  7900 | JAMES | CLERK   | 7698 | 1981-12-03 |  950.00 | NULL |     30 |
+-------+-------+---------+------+------------+---------+------+--------+
3 rows in set (0.00 sec)
```

8. list all employees with name contains I at 2nd position

```
mysql> select * from emp where ename regexp'^.[I]';
+-------+--------+-----------+------+------------+---------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+--------+-----------+------+------------+---------+------+--------+
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL |     10 |
|  7934 | MILLER | CLERK     | 7782 | 1982-01-23 | 1300.00 | NULL |     10 |
+-------+--------+-----------+------+------------+---------+------+--------+
2 rows in set (0.00 sec)
```

```
mysql> select * from emp where ename regexp'^.[L]';
+-------+-------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME | JOB      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+-------+----------+------+------------+---------+--------+--------+
|  7499 | ALLEN | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00 |     30 |
|  7698 | BLAKE | MANAGER  | 7839 | 1981-05-01 | 2850.00 |   NULL |     30 |
|  7782 | CLARK | MANAGER  | 7839 | 1981-06-09 | 2450.00 |   NULL |     10 |
+-------+-------+----------+------+------------+---------+--------+--------+
3 rows in set (0.00 sec)
```

9. list all employees with name starts with A ends with N and somewhere in between L is there

```
mysql> select * from emp where ename regexp'^[A].*[L].*[N]$';
+-------+-------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME | JOB      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+-------+----------+------+------------+---------+--------+--------+
|  7499 | ALLEN | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00 |     30 |
+-------+-------+----------+------+------------+---------+--------+--------+
1 row in set (0.00 sec)
```

10. list all employees with name starts with A and B at 3 rd position and P at second last position

```
mysql> select * from emp where ename regexp'^[A].[B].*[P].$';
Empty set (0.00 sec)
```

11. List all employees with name starts with either A or starts with S or starts with W

```
mysql> select * from emp where ename regexp'^[ASW]';
+-------+-------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME | JOB      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+-------+----------+------+------------+---------+--------+--------+
|  7369 | SMITH | CLERK    | 7902 | 1980-12-17 |  800.00 |   NULL |     20 |
|  7499 | ALLEN | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00 |     30 |
|  7521 | WARD  | SALESMAN | 7698 | 1981-02-22 | 1250.00 | 500.00 |     30 |
|  7788 | SCOTT | ANALYST  | 7566 | 1982-12-09 | 3000.00 |   NULL |     20 |
|  7876 | ADAMS | CLERK    | 7788 | 1983-01-12 | 1100.00 |   NULL |     20 |
+-------+-------+----------+------+------------+---------+--------+--------+
5 rows in set (0.00 sec)
```

Practice Aggregate function:

12. find max sal and min sal for each job

```
mysql> select job,max(sal) MAX,min(sal) MIN
    -> from emp
    -> group by job;
+-----------+---------+---------+
| job       | MAX     | MIN     |
+-----------+---------+---------+
| CLERK     | 1300.00 |  800.00 |
| SALESMAN  | 1600.00 | 1250.00 |
| MANAGER   | 2975.00 | 2450.00 |
| ANALYST   | 3000.00 | 3000.00 |
| PRESIDENT | 5000.00 | 5000.00 |
+-----------+---------+---------+
5 rows in set (0.00 sec)
```

13. find how many employess have not received commission

```
mysql> select count(*)
    -> from emp
    -> where comm is null;
+----------+
| count(*) |
+----------+
|       10 |
+----------+
1 row in set (0.00 sec)
```

14. find sum of sal of all employees working in dept no 10

```
+----------+
| sum(sal) |
+----------+
|  8750.00 |
+----------+
1 row in set (0.00 sec)
```

15. find maximum salary,average sal for each job in every department

```
mysql> select deptno,job, sum(sal),avg(sal),max(sal)
    -> from emp
    -> group by job,deptno
    -> order by job,deptno;
+--------+-----------+----------+-------------+----------+
| deptno | job       | sum(sal) | avg(sal)    | max(sal) |
+--------+-----------+----------+-------------+----------+
|     20 | ANALYST   |  6000.00 | 3000.000000 |  3000.00 |
|     10 | CLERK     |  1300.00 | 1300.000000 |  1300.00 |
|     20 | CLERK     |  1900.00 |  950.000000 |  1100.00 |
|     30 | CLERK     |   950.00 |  950.000000 |   950.00 |
|     10 | MANAGER   |  2450.00 | 2450.000000 |  2450.00 |
|     20 | MANAGER   |  2975.00 | 2975.000000 |  2975.00 |
|     30 | MANAGER   |  2850.00 | 2850.000000 |  2850.00 |
|     10 | PRESIDENT |  5000.00 | 5000.000000 |  5000.00 |
|     30 | SALESMAN  |  5600.00 | 1400.000000 |  1600.00 |
+--------+-----------+----------+-------------+----------+
9 rows in set (0.00 sec)
```

16. find max salary for every department if deptno is > 15 and arrange data in deptno order.

```
mysql> select deptno,max(sal)
    -> from emp
    -> where deptno>15
    -> group by deptno
    -> order by deptno;
+--------+----------+
| deptno | max(sal) |
+--------+----------+
|     20 |  3000.00 |
|     30 |  2850.00 |
+--------+----------+
```

17. find sum salary for every department if sum is > 2000

```
mysql> select deptno,sum(sal)
    -> from emp
    -> group by deptno
    -> having sum(sal)>2000;
+--------+----------+
| deptno | sum(sal) |
+--------+----------+
|     20 | 10875.00 |
|     30 |  9400.00 |
|     10 |  8750.00 |
+--------+----------+
3 rows in set (0.00 sec)
```

18. list all department which has minimum 5 employees

```
mysql> select deptno,count(*)
    -> from emp
    -> group by deptno
    -> having count(*)>4
    -> order by deptno;
+--------+----------+
| deptno | count(*) |
+--------+----------+
|     20 |        5 |
|     30 |        6 |
+--------+----------+
2 rows in set (0.00 sec)
```

19. count how many employees earn salary more than 2000 in each job

```
mysql> select job,count(*)
    -> from emp
    -> where sal>2000
    -> group by job
    -> order by job;
+-----------+----------+
| job       | count(*) |
+-----------+----------+
| ANALYST   |        2 |
| MANAGER   |        3 |
| PRESIDENT |        1 |
+-----------+----------+
3 rows in set (0.00 sec)
```

20. list all enames and jobs in small case letter

21. list all names and jobs so that the length of name should be 15 if it is smaller then add spaces to left

```
mysql> select  lpad(ename,15,' '),job,sal
    -> from emp;
+--------------------+-----------+---------+
| lpad(ename,15,' ') | job       | sal     |
+--------------------+-----------+---------+
|           SMITH    | CLERK     |  800.00 |
|           ALLEN    | SALESMAN  | 1600.00 |
|            WARD    | SALESMAN  | 1250.00 |
|           JONES    | MANAGER   | 2975.00 |
|          MARTIN    | SALESMAN  | 1250.00 |
|           BLAKE    | MANAGER   | 2850.00 |
|           CLARK    | MANAGER   | 2450.00 |
|           SCOTT    | ANALYST   | 3000.00 |
|            KING    | PRESIDENT | 5000.00 |
|          TURNER    | SALESMAN  | 1500.00 |
|           ADAMS    | CLERK     | 1100.00 |
|           JAMES    | CLERK     |  950.00 |
|            FORD    | ANALYST   | 3000.00 |
|          MILLER    | CLERK     | 1300.00 |
+--------------------+-----------+---------+
14 rows in set (0.00 sec)
```

22. display min sal,max sal, average sal for all employees working under same manager

```
mysql> select mgr, min(sal),max(sal), avg(sal)
    -> from emp
    -> group by mgr
    -> order by mgr;
+------+----------+----------+-------------+
| mgr  | min(sal) | max(sal) | avg(sal)    |
+------+----------+----------+-------------+
| NULL |  5000.00 |  5000.00 | 5000.000000 |
| 7566 |  3000.00 |  3000.00 | 3000.000000 |
| 7698 |   950.00 |  1600.00 | 1310.000000 |
| 7782 |  1300.00 |  1300.00 | 1300.000000 |
| 7788 |  1100.00 |  1100.00 | 1100.000000 |
| 7839 |  2450.00 |  2975.00 | 2758.333333 |
| 7902 |   800.00 |   800.00 |  800.000000 |
+------+----------+----------+-------------+
7 rows in set (0.00 sec)
```

23. find sum of total earnings(sal+comm), average of sal+comm,
for all employees who earn sal > 2000 and work in either dept no 10 or 20

```
mysql> select sum(sal + ifnull(comm,0)),avg(sal+ ifnull(comm,0))
    -> from emp
    -> where sal>2000 and deptno in (10,20);
+---------------------------+--------------------------+
| sum(sal + ifnull(comm,0)) | avg(sal+ ifnull(comm,0)) |
+---------------------------+--------------------------+
|                  16425.00 |             3285.000000 |
+---------------------------+--------------------------+
1 row in set (0.00 sec)
```

24. list all employees who joined in Aug 1980 and salary is >1500 and < 2500

```
mysql> select *
    -> from emp
    -> where month(hiredate)=08 and year(hiredate)=1980 and sal between 1501 and 2499;
+-------+-------+-------+------+------------+---------+-------+--------+
| EMPNO | ENAME | JOB   | MGR  | HIREDATE   | SAL     | COMM  | DEPTNO |
+-------+-------+-------+------+------------+---------+-------+--------+
|  7936 | SIYA  | CLERK | 7782 | 1980-08-02 | 2000.00 | 10.00 |     30 |
+-------+-------+-------+------+------------+---------+-------+--------+
1 row in set (0.00 sec)
```

25. list all employees joined in either aug or may or dec

```
mysql> select * from emp where month(hiredate) in (08,05,12);
+-------+-------+---------+------+------------+---------+-------+--------+
| EMPNO | ENAME | JOB     | MGR  | HIREDATE   | SAL     | COMM  | DEPTNO |
+-------+-------+---------+------+------------+---------+-------+--------+
|  7369 | SMITH | CLERK   | 7902 | 1980-12-17 |  800.00 |  NULL |     20 |
|  7698 | BLAKE | MANAGER | 7839 | 1981-05-01 | 2850.00 |  NULL |     30 |
|  7788 | SCOTT | ANALYST | 7566 | 1982-12-09 | 3000.00 |  NULL |     20 |
|  7900 | JAMES | CLERK   | 7698 | 1981-12-03 |  950.00 |  NULL |     30 |
|  7902 | FORD  | ANALYST | 7566 | 1981-12-03 | 3000.00 |  NULL |     20 |
|  7935 | RIYA  | CLERK   | 7782 | 1980-08-02 | 1000.00 | 10.00 |     20 |
|  7936 | SIYA  | CLERK   | 7782 | 1980-08-02 | 2000.00 | 10.00 |     30 |
+-------+-------+---------+------+------------+---------+-------+--------+
7 rows in set (0.00 sec)
```

26. display name and hiredate in dd/mm/yy format for all employees whose job is clerk and they earn some commission

```
mysql> select ename, date_format(hiredate,'%d/%m/%y') date from emp where job='CLERK' and comm is not
null;
+-------+----------+
| ename | date     |
+-------+----------+
| RIYA  | 02/08/80 |
| SIYA  | 02/08/80 |
+-------+----------+
2 rows in set (0.00 sec)
```

27. list empcode,empno,name and job for each employee. (note :empcode is 3 to 5 characters from name and last 2 characters of job)

```
mysql> select concat(left(ename,4),right(job,2)) empcode,empno,ename,job from emp;
+---------+-------+--------+-----------+
| empcode | empno | ename  | job       |
+---------+-------+--------+-----------+
| SMITRK  |  7369 | SMITH  | CLERK     |
| ALLEAN  |  7499 | ALLEN  | SALESMAN  |
| WARDAN  |  7521 | WARD   | SALESMAN  |
| JONEER  |  7566 | JONES  | MANAGER   |
| MARTAN  |  7654 | MARTIN | SALESMAN  |
| BLAKER  |  7698 | BLAKE  | MANAGER   |
| CLARER  |  7782 | CLARK  | MANAGER   |
| SCOTST  |  7788 | SCOTT  | ANALYST   |
| KINGNT  |  7839 | KING   | PRESIDENT |
| TURNAN  |  7844 | TURNER | SALESMAN  |
| ADAMRK  |  7876 | ADAMS  | CLERK     |
| JAMERK  |  7900 | JAMES  | CLERK     |
| FORDST  |  7902 | FORD   | ANALYST   |
| MILLRK  |  7934 | MILLER | CLERK     |
| RIYARK  |  7935 | RIYA   | CLERK     |
| SIYARK  |  7936 | SIYA   | CLERK     |
+---------+-------+--------+-----------+
16 rows in set (0.00 sec)
```

28. display thousand separator and $ symbol for sal and commission if comm is null then display it as 0, for all employees whose name starts with A and ends with N

```
mysql> select empno,ename ,concat('$',' ',format(sal,2)) Salary, ifnull(comm,0)
    -> from emp
    -> where ename regexp '^[A].*[N]$';
+-------+-------+------------+----------------+
| empno | ename | Salary     | ifnull(comm,0) |
+-------+-------+------------+----------------+
|  7499 | ALLEN | $ 1,600.00 |         300.00 |
+-------+-------+------------+----------------+
1 row in set (0.00 sec)
```

29. Display empid,name,sal,comm,remark Remark should base on following conditions
comm >= 600 "excellent Keep it up"
if it < 600 or not null "good"
otherwise "Need improvement"

```
mysql> select empno,ename,sal,comm, case when comm>=600 then 'Excellent Keep it up'
    -> when comm is not null or comm<600 then 'Good'
    -> else 'Need improvement'
    -> end remark
    -> from emp;
+-------+--------+---------+---------+----------------------+
| empno | ename  | sal     | comm    | remark               |
+-------+--------+---------+---------+----------------------+
|  7369 | SMITH  |  800.00 |    NULL | Need improvement     |
|  7499 | ALLEN  | 1600.00 |  300.00 | Good                 |
|  7521 | WARD   | 1250.00 |  500.00 | Good                 |
|  7566 | JONES  | 2975.00 |    NULL | Need improvement     |
|  7654 | MARTIN | 1250.00 | 1400.00 | Excellent Keep it up |
|  7698 | BLAKE  | 2850.00 |    NULL | Need improvement     |
|  7782 | CLARK  | 2450.00 |    NULL | Need improvement     |
|  7788 | SCOTT  | 3000.00 |    NULL | Need improvement     |
|  7839 | KING   | 5000.00 |    NULL | Need improvement     |
|  7844 | TURNER | 1500.00 |    0.00 | Good                 |
|  7876 | ADAMS  | 1100.00 |    NULL | Need improvement     |
|  7900 | JAMES  |  950.00 |    NULL | Need improvement     |
|  7902 | FORD   | 3000.00 |    NULL | Need improvement     |
|  7934 | MILLER | 1300.00 |    NULL | Need improvement     |
|  7935 | RIYA   | 1000.00 |   10.00 | Good                 |
|  7936 | SIYA   | 2000.00 |   10.00 | Good                 |
+-------+--------+---------+---------+----------------------+
16 rows in set (0.00 sec)
```

30. Display empid, name, deptno and department name by using following conditions.
dept 10 then "Hr"
if 20 then "Admin"
if 30 then "accounts"
otherwise purchase

```
mysql> select empno,ename,deptno,case when deptno=10 then 'HR'
    -> when deptno=20 then 'Admin'
    -> when deptno=30 then 'Accounts'
    -> else 'Purchase'
    -> end 'Department name'
    -> from emp;
+-------+--------+--------+-----------------+
| empno | ename  | deptno | Department name |
+-------+--------+--------+-----------------+
|  7369 | SMITH  |     20 | Admin           |
|  7499 | ALLEN  |     30 | Accounts        |
|  7521 | WARD   |     30 | Accounts        |
|  7566 | JONES  |     20 | Admin           |
|  7654 | MARTIN |     30 | Accounts        |
|  7698 | BLAKE  |     30 | Accounts        |
|  7782 | CLARK  |     10 | HR              |
|  7788 | SCOTT  |     20 | Admin           |
|  7839 | KING   |     10 | HR              |
|  7844 | TURNER |     30 | Accounts        |
|  7876 | ADAMS  |     20 | Admin           |
|  7900 | JAMES  |     30 | Accounts        |
|  7902 | FORD   |     20 | Admin           |
|  7934 | MILLER |     10 | HR              |
|  7935 | RIYA   |     20 | Admin           |
|  7936 | SIYA   |     30 | Accounts        |
+-------+--------+--------+-----------------+
16 rows in set (0.00 sec)
```

31. Practice creating following tables

a. create table mydept_DBDA and insert 5 rows into it

```
mysql> select * from mydept_DBDA ;
+--------+-------------+-----------+
| deptid | dname       | dloc      |
+--------+-------------+-----------+
|      1 | finance     | 1st floor |
|      2 | account     | 2nd floor |
|      3 | sales       | 3rd floor |
|      4 | research    | 2nd floor |
|      5 | deveploment | 1st floor |
+--------+-------------+-----------+
5 rows in set (0.00 sec)
```

b. create myemployee table

```
mysql> desc myemployee;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| empno  | int         | NO   | PRI | NULL    |       |
| fname  | varchar(50) | NO   |     | NULL    |       |
| lname  | varchar(50) | YES  |     | NULL    |       |
| mname  | varchar(50) | YES  |     | NULL    |       |
| sal    | float       | YES  |     | NULL    |       |
| doj    | date        | YES  |     | NULL    |       |
| passno | varchar(50) | YES  | UNI | NULL    |       |
| deptno | int         | YES  | MUL | NULL    |       |
+--------+-------------+------+-----+---------+-------+
8 rows in set (0.00 sec)
```

32. Create following tables Student, Course

Create table student

```
mysql> create table student(sid int primary key, sname varchar(50) not null,email varchar(50) unique);

Query OK, 0 rows affected (0.03 sec)

mysql> desc student;
+-------+-------------+------+-----+---------+-------+
| Field | Type        | Null | Key | Default | Extra |
+-------+-------------+------+-----+---------+-------+
| sid   | int         | NO   | PRI | NULL    |       |
| sname | varchar(50) | NO   |     | NULL    |       |
| email | varchar(50) | YES  | UNI | NULL    |       |
+-------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

Create course table

```
mysql> create table course(cid int primary key, cname varchar(50) unique,duration int);
Query OK, 0 rows affected (0.03 sec)

mysql> desc course;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| cid      | int         | NO   | PRI | NULL    |       |
| cname    | varchar(50) | YES  | UNI | NULL    |       |
| duration | int         | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

Create marks table(the table has composite primary key (studid+courseid)

```
mysql> create table mark(sid int , cid int, marks int,primary key(sid,cid),constraint fk_sc foreign ke
y(sid) references student(sid) on delete cascade on update cascade, constraint fk_sc1 foreign key(cid)
 references course(cid) on delete cascade on update cascade);
Query OK, 0 rows affected (0.03 sec)

mysql> desc mark
    -> ;
+-------+------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+-------+------+------+-----+---------+-------+
| sid   | int  | NO   | PRI | NULL    |       |
| cid   | int  | NO   | PRI | NULL    |       |
| marks | int  | YES  |     | NULL    |       |
+-------+------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

Add 5 records in each student, course and marks table

```
mysql> insert into student values(1,'Sanjana','sanjana123@gmail.com')
    -> ,(2,'Vaishnavi','vaishnavi04@gmail.com'),(3,'ash','ashxxx@gmail.com'),(4,'ishwari','ishwari@gma
il.com'),(5,'Shravani','shravani@gmail.com');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from student;
+-----+-----------+-----------------------+
| sid | sname     | email                 |
+-----+-----------+-----------------------+
|   1 | Sanjana   | sanjana123@gmail.com  |
|   2 | Vaishnavi | vaishnavi04@gmail.com |
|   3 | ash       | ashxxx@gmail.com      |
|   4 | ishwari   | ishwari@gmail.com     |
|   5 | Shravani  | shravani@gmail.com    |
+-----+-----------+-----------------------+
5 rows in set (0.00 sec)
```

```
mysql> select * from course;
+-----+-------+----------+
| cid | cname | duration |
+-----+-------+----------+
|   1 | JAVA  |       24 |
|   2 | C++   |       12 |
|   3 | DBMS  |        9 |
|   4 | C     |        8 |
|   5 | OS    |        7 |
+-----+-------+----------+
5 rows in set (0.00 sec)

mysql> select * from mark;
+-----+-----+-------+
| sid | cid | marks |
+-----+-----+-------+
|   1 |   1 |    99 |
|   1 |   3 |    98 |
|   2 |   1 |    95 |
|   2 |   2 |    97 |
+-----+-----+-------+
4 rows in set (0.00 sec)
```

Create empty table emp10 with table structure same as emp table.

```
mysql> create table emp10
    -> as
    -> (select * from emp where 7=4);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc emp10
    -> ;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
```

34. Solve following using alter table

a. add primary key constraint in emp table deptno column refernces ,dept table deptno

```
mysql> alter table emp
    -> add primary key(empno,deptno);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc emp
    -> ;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| EMPNO    | int          | NO   | PRI | NULL    |       |
| ENAME    | varchar(10)  | YES  |     | NULL    |       |
| JOB      | varchar(9)   | YES  |     | NULL    |       |
| MGR      | int          | YES  |     | NULL    |       |
| HIREDATE | date         | YES  |     | NULL    |       |
| SAL      | decimal(7,2) | YES  |     | NULL    |       |
| COMM     | decimal(7,2) | YES  |     | NULL    |       |
| DEPTNO   | int          | NO   | PRI | NULL    |       |
+----------+--------------+------+-----+---------+-------+
8 rows in set (0.00 sec)
```

b. add new column in emp table netsal with constraint default 1000

```
mysql> alter table emp
    -> add netsal float default 1000;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc emp;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| EMPNO    | int          | NO   | PRI | NULL    |       |
| ENAME    | varchar(10)  | YES  |     | NULL    |       |
| JOB      | varchar(9)   | YES  |     | NULL    |       |
| MGR      | int          | YES  |     | NULL    |       |
| HIREDATE | date         | YES  |     | NULL    |       |
| SAL      | decimal(7,2) | YES  |     | NULL    |       |
| COMM     | decimal(7,2) | YES  |     | NULL    |       |
| DEPTNO   | int          | NO   | PRI | NULL    |       |
| netsal   | float        | YES  |     | 1000    |       |
+----------+--------------+------+-----+---------+-------+
9 rows in set (0.00 sec)
```

35. increase sal of each employee by 15 % sal +comm, change the job to manager and mgr to 7777 for all employees in deptno 10.

```
mysql> select * from emp;
+-------+--------+----------+------+------------+---------+---------+--------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
+-------+--------+----------+------+------------+---------+---------+--------+
|  7369 | SMITH  | CLERK    | 7902 | 1980-12-17 |  920.00 |    NULL |     20 |
|  7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1885.00 |  300.00 |     30 |
|  7521 | WARD   | SALESMAN | 7698 | 1981-02-22 | 1512.50 |  500.00 |     30 |
|  7566 | JONES  | MANAGER  | 7839 | 1981-04-02 | 3421.25 |    NULL |     20 |
|  7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1647.50 | 1400.00 |     30 |
|  7698 | BLAKE  | MANAGER  | 7839 | 1981-05-01 | 3277.50 |    NULL |     30 |
|  7782 | CLARK  | MANAGER  | 7777 | 1981-06-09 | 2817.50 |    NULL |     10 |
|  7788 | SCOTT  | ANALYST  | 7566 | 1982-12-09 | 3450.00 |    NULL |     20 |
|  7839 | KING   | MANAGER  | 7777 | 1981-11-17 | 5750.00 |    NULL |     10 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1725.00 |    0.00 |     30 |
|  7876 | ADAMS  | CLERK    | 7788 | 1983-01-12 | 1265.00 |    NULL |     20 |
|  7900 | JAMES  | CLERK    | 7698 | 1981-12-03 | 1092.50 |    NULL |     30 |
|  7902 | FORD   | ANALYST  | 7566 | 1981-12-03 | 3450.00 |    NULL |     20 |
|  7934 | MILLER | MANAGER  | 7777 | 1982-01-23 | 1495.00 |    NULL |     10 |
+-------+--------+----------+------+------------+---------+---------+--------+
14 rows in set (0.00 sec)
```

36. change job of smith to senior clerk

```
mysql> alter table emp
    -> modify job varchar(20);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> update emp set job='SENIOR CLERK' where ename='SMITH';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from emp;
+-------+--------+--------------+------+------------+---------+---------+--------+
| EMPNO | ENAME  | job          | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
+-------+--------+--------------+------+------------+---------+---------+--------+
|  7369 | SMITH  | SENIOR CLERK | 7902 | 1980-12-17 |  920.00 |    NULL |     20 |
|  7499 | ALLEN  | SALESMAN     | 7698 | 1981-02-20 | 1885.00 |  300.00 |     30 |
|  7521 | WARD   | SALESMAN     | 7698 | 1981-02-22 | 1512.50 |  500.00 |     30 |
|  7566 | JONES  | MANAGER      | 7839 | 1981-04-02 | 3421.25 |    NULL |     20 |
|  7654 | MARTIN | SALESMAN     | 7698 | 1981-09-28 | 1647.50 | 1400.00 |     30 |
|  7698 | BLAKE  | MANAGER      | 7839 | 1981-05-01 | 3277.50 |    NULL |     30 |
|  7782 | CLARK  | MANAGER      | 7777 | 1981-06-09 | 2817.50 |    NULL |     10 |
|  7788 | SCOTT  | ANALYST      | 7566 | 1982-12-09 | 3450.00 |    NULL |     20 |
|  7839 | KING   | MANAGER      | 7777 | 1981-11-17 | 5750.00 |    NULL |     10 |
|  7844 | TURNER | SALESMAN     | 7698 | 1981-09-08 | 1725.00 |    0.00 |     30 |
|  7876 | ADAMS  | CLERK        | 7788 | 1983-01-12 | 1265.00 |    NULL |     20 |
|  7900 | JAMES  | CLERK        | 7698 | 1981-12-03 | 1092.50 |    NULL |     30 |
|  7902 | FORD   | ANALYST      | 7566 | 1981-12-03 | 3450.00 |    NULL |     20 |
|  7934 | MILLER | MANAGER      | 7777 | 1982-01-23 | 1495.00 |    NULL |     10 |
+-------+--------+--------------+------+------------+---------+---------+--------+
14 rows in set (0.00 sec)
```

37. increase salary of all employees by 15% if they are earning some commission

```
mysql> update emp set sal= sal+(0.15*(sal)) where comm is not null;
Query OK, 4 rows affected, 2 warnings (0.01 sec)
Rows matched: 4  Changed: 4  Warnings: 2

mysql> select * from emp;
+-------+--------+--------------+------+------------+---------+---------+--------+
| EMPNO | ENAME  | job          | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
+-------+--------+--------------+------+------------+---------+---------+--------+
|  7369 | SMITH  | SENIOR CLERK | 7902 | 1980-12-17 |  920.00 |    NULL |     20 |
|  7499 | ALLEN  | SALESMAN     | 7698 | 1981-02-20 | 2167.75 |  300.00 |     30 |
|  7521 | WARD   | SALESMAN     | 7698 | 1981-02-22 | 1739.38 |  500.00 |     30 |
|  7566 | JONES  | MANAGER      | 7839 | 1981-04-02 | 3421.25 |    NULL |     20 |
|  7654 | MARTIN | SALESMAN     | 7698 | 1981-09-28 | 1894.63 | 1400.00 |     30 |
|  7698 | BLAKE  | MANAGER      | 7839 | 1981-05-01 | 3277.50 |    NULL |     30 |
|  7782 | CLARK  | MANAGER      | 7777 | 1981-06-09 | 2817.50 |    NULL |     10 |
|  7788 | SCOTT  | ANALYST      | 7566 | 1982-12-09 | 3450.00 |    NULL |     20 |
|  7839 | KING   | MANAGER      | 7777 | 1981-11-17 | 5750.00 |    NULL |     10 |
|  7844 | TURNER | SALESMAN     | 7698 | 1981-09-08 | 1983.75 |    0.00 |     30 |
|  7876 | ADAMS  | CLERK        | 7788 | 1983-01-12 | 1265.00 |    NULL |     20 |
|  7900 | JAMES  | CLERK        | 7698 | 1981-12-03 | 1092.50 |    NULL |     30 |
|  7902 | FORD   | ANALYST      | 7566 | 1981-12-03 | 3450.00 |    NULL |     20 |
|  7934 | MILLER | MANAGER      | 7777 | 1982-01-23 | 1495.00 |    NULL |     10 |
+-------+--------+--------------+------+------------+---------+---------+--------+
14 rows in set (0.00 sec)
```

38. list all employees with sal>smith's sal

```
mysql> select * from emp where sal > (select sal from emp e where e.ename= 'SMITH');
+-------+--------+----------+------+------------+---------+---------+--------+
| EMPNO | ENAME  | job      | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
+-------+--------+----------+------+------------+---------+---------+--------+
|  7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 2167.75 |  300.00 |     30 |
|  7521 | WARD   | SALESMAN | 7698 | 1981-02-22 | 1739.38 |  500.00 |     30 |
|  7566 | JONES  | MANAGER  | 7839 | 1981-04-02 | 3421.25 |    NULL |     20 |
|  7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1894.63 | 1400.00 |     30 |
|  7698 | BLAKE  | MANAGER  | 7839 | 1981-05-01 | 3277.50 |    NULL |     30 |
|  7782 | CLARK  | MANAGER  | 7777 | 1981-06-09 | 2817.50 |    NULL |     10 |
|  7788 | SCOTT  | ANALYST  | 7566 | 1982-12-09 | 3450.00 |    NULL |     20 |
|  7839 | KING   | MANAGER  | 7777 | 1981-11-17 | 5750.00 |    NULL |     10 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1983.75 |    0.00 |     30 |
|  7876 | ADAMS  | CLERK    | 7788 | 1983-01-12 | 1265.00 |    NULL |     20 |
|  7900 | JAMES  | CLERK    | 7698 | 1981-12-03 | 1092.50 |    NULL |     30 |
|  7902 | FORD   | ANALYST  | 7566 | 1981-12-03 | 3450.00 |    NULL |     20 |
|  7934 | MILLER | MANAGER  | 7777 | 1982-01-23 | 1495.00 |    NULL |     10 |
+-------+--------+----------+------+------------+---------+---------+--------+
13 rows in set (0.01 sec)
```

39. list all employees who are working in smith's department

```
mysql> select * from emp where deptno = (select deptno from emp e where e.ename= 'SMITH');
+-------+-------+--------------+------+------------+---------+------+--------+
| EMPNO | ENAME | job          | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+-------+--------------+------+------------+---------+------+--------+
|  7369 | SMITH | SENIOR CLERK | 7902 | 1980-12-17 |  920.00 | NULL |     20 |
|  7566 | JONES | MANAGER      | 7839 | 1981-04-02 | 3421.25 | NULL |     20 |
|  7788 | SCOTT | ANALYST      | 7566 | 1982-12-09 | 3450.00 | NULL |     20 |
|  7876 | ADAMS | CLERK        | 7788 | 1983-01-12 | 1265.00 | NULL |     20 |
|  7902 | FORD  | ANALYST      | 7566 | 1981-12-03 | 3450.00 | NULL |     20 |
+-------+-------+--------------+------+------------+---------+------+--------+
5 rows in set (0.00 sec)
```

40. list all employees with sal < smith’s sal and salary > blake’s sal

```
mysql> select * from emp where sal between (select sal from emp e where e.ename= 'SMITH') and  (select
 sal from emp e where e.ename= 'BLAKE');
+-------+--------+--------------+------+------------+---------+---------+--------+
| EMPNO | ENAME  | job          | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
+-------+--------+--------------+------+------------+---------+---------+--------+
|  7369 | SMITH  | SENIOR CLERK | 7902 | 1980-12-17 |  920.00 |    NULL |     20 |
|  7499 | ALLEN  | SALESMAN     | 7698 | 1981-02-20 | 2167.75 |  300.00 |     30 |
|  7521 | WARD   | SALESMAN     | 7698 | 1981-02-22 | 1739.38 |  500.00 |     30 |
|  7654 | MARTIN | SALESMAN     | 7698 | 1981-09-28 | 1894.63 | 1400.00 |     30 |
|  7698 | BLAKE  | MANAGER      | 7839 | 1981-05-01 | 3277.50 |    NULL |     30 |
|  7782 | CLARK  | MANAGER      | 7777 | 1981-06-09 | 2817.50 |    NULL |     10 |
|  7844 | TURNER | SALESMAN     | 7698 | 1981-09-08 | 1983.75 |    0.00 |     30 |
|  7876 | ADAMS  | CLERK        | 7788 | 1983-01-12 | 1265.00 |    NULL |     20 |
|  7900 | JAMES  | CLERK        | 7698 | 1981-12-03 | 1092.50 |    NULL |     30 |
|  7934 | MILLER | MANAGER      | 7777 | 1982-01-23 | 1495.00 |    NULL |     10 |
+-------+--------+--------------+------+------------+---------+---------+--------+
10 rows in set (0.00 sec)
```

46. list all employees who earn more than average sal of Alan's department

```
mysql> select * from emp where sal > (select avg(sal) from emp e where deptno=(select deptno from emp
where ename='ALLEN') group by deptno);
+-------+-------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME | job      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+-------+----------+------+------------+---------+--------+--------+
|  7499 | ALLEN | SALESMAN | 7698 | 1981-02-20 | 2167.75 | 300.00 |     30 |
|  7566 | JONES | MANAGER  | 7839 | 1981-04-02 | 3421.25 |   NULL |     20 |
|  7698 | BLAKE | MANAGER  | 7839 | 1981-05-01 | 3277.50 |   NULL |     30 |
|  7782 | CLARK | MANAGER  | 7777 | 1981-06-09 | 2817.50 |   NULL |     10 |
|  7788 | SCOTT | ANALYST  | 7566 | 1982-12-09 | 3450.00 |   NULL |     20 |
|  7839 | KING  | MANAGER  | 7777 | 1981-11-17 | 5750.00 |   NULL |     10 |
|  7902 | FORD  | ANALYST  | 7566 | 1981-12-03 | 3450.00 |   NULL |     20 |
+-------+-------+----------+------+------------+---------+--------+--------+
7 rows in set (0.00 sec)
```

47. list all employees who are working in purchase department

```
mysql> select * from emp where deptno = (select deptno from dept d where d.dname = 'SALES');
+-------+--------+----------+------+------------+---------+---------+--------+
| EMPNO | ENAME  | job      | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
+-------+--------+----------+------+------------+---------+---------+--------+
|  7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 2167.75 |  300.00 |     30 |
|  7521 | WARD   | SALESMAN | 7698 | 1981-02-22 | 1739.38 |  500.00 |     30 |
|  7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1894.63 | 1400.00 |     30 |
|  7698 | BLAKE  | MANAGER  | 7839 | 1981-05-01 | 3277.50 |    NULL |     30 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1983.75 |    0.00 |     30 |
|  7900 | JAMES  | CLERK    | 7698 | 1981-12-03 | 1092.50 |    NULL |     30 |
+-------+--------+----------+------+------------+---------+---------+--------+
6 rows in set (0.00 sec)
```

48. list all employees who earn more than average salary of their own department

```
mysql> select * from emp e where sal > (select avg(sal) from emp m where e.deptno=m.deptno group by de
ptno);
+-------+-------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME | job      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+-------+----------+------+------------+---------+--------+--------+
|  7499 | ALLEN | SALESMAN | 7698 | 1981-02-20 | 2167.75 | 300.00 |     30 |
|  7566 | JONES | MANAGER  | 7839 | 1981-04-02 | 3421.25 |   NULL |     20 |
|  7698 | BLAKE | MANAGER  | 7839 | 1981-05-01 | 3277.50 |   NULL |     30 |
|  7788 | SCOTT | ANALYST  | 7566 | 1982-12-09 | 3450.00 |   NULL |     20 |
|  7839 | KING  | MANAGER  | 7777 | 1981-11-17 | 5750.00 |   NULL |     10 |
|  7902 | FORD  | ANALYST  | 7566 | 1981-12-03 | 3450.00 |   NULL |     20 |
+-------+-------+----------+------+------------+---------+--------+--------+
6 rows in set (0.01 sec)
```

49. list all employees who earn sal > than their managers salary

```
mysql> select * from emp e where sal > (select sal from emp m where e.mgr=m.empno);
+-------+-------+---------+------+------------+---------+------+--------+
| EMPNO | ENAME | job     | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+-------+---------+------+------------+---------+------+--------+
|  7788 | SCOTT | ANALYST | 7566 | 1982-12-09 | 3450.00 | NULL |     20 |
|  7902 | FORD  | ANALYST | 7566 | 1981-12-03 | 3450.00 | NULL |     20 |
+-------+-------+---------+------+------------+---------+------+--------+
2 rows in set (0.00 sec)
```

50. list all employees who are earning more than average salary of their job

```
mysql> select * from emp e where sal > (select avg(sal) from emp m where e.job=m.job group by job);
+-------+--------+----------+------+------------+---------+--------+--------+
| EMPNO | ENAME  | job      | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
+-------+--------+----------+------+------------+---------+--------+--------+
|  7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 2167.75 | 300.00 |     30 |
|  7566 | JONES  | MANAGER  | 7839 | 1981-04-02 | 3421.25 |   NULL |     20 |
|  7839 | KING   | MANAGER  | 7777 | 1981-11-17 | 5750.00 |   NULL |     10 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1983.75 |   0.00 |     30 |
|  7876 | ADAMS  | CLERK    | 7788 | 1983-01-12 | 1265.00 |   NULL |     20 |
+-------+--------+----------+------+------------+---------+--------+--------+
5 rows in set (0.00 sec)
```

51. display employee name and department name
52. display empno,name,department name and grade (use emp,dept and salgrade table)
53. list all employees number,name, mgrno and manager name
54. create following tables and solve following questions(primary keys are marked in yellow) foreign keys are marked in green
product(pid,pname,price,qty,cid,sid)
salesman (sid,sname,address)
category(cid,cnam,descritpion)

```
mysql> select * from product;
+-----+------------+--------+------+------+------+
| pid | pname      | price  | qty  | cid  | sid  |
+-----+------------+--------+------+------+------+
| 501 | Smartphone |  699.99 |   50 |    1 |  101 |
| 502 | Laptop     | 1199.99 |   20 |    1 |  102 |
| 503 | T-Shirt    |   19.99 |  200 |    2 |  101 |
| 504 | Jeans      |   49.99 |  120 |    2 |  103 |
| 505 | Desk Lamp  |   29.99 |   35 |    3 |  102 |
+-----+------------+--------+------+------+------+
5 rows in set (0.00 sec)
```

```
mysql> select * from salesman;
+-----+-------------+---------+
| sid | sname       | address |
+-----+-------------+---------+
| 101 | Alice Smith | Pune    |
| 102 | Bob         | Mumbai  |
| 103 | Charlie     | Beed    |
+-----+-------------+---------+
3 rows in set (0.00 sec)
```

```
mysql> select * from category;
+-----+-------------+-------------------------+
| cid | cname       | descr                   |
+-----+-------------+-------------------------+
|   1 | Electronics | Gadgets and accessories |
|   2 | Clothing    | Apparel and garments    |
|   3 | Home Decor  | Furniture and ornaments |
+-----+-------------+-------------------------+
3 rows in set (0.00 sec)
```

1. list all product name,their category name and name of a person, who sold that product

```
mysql> select pname,cname,sname from product p,category c,salesman s
    -> where p.cid=c.cid and p.sid=s.sid;
+------------+-------------+-------------+
| pname      | cname       | sname       |
+------------+-------------+-------------+
| Smartphone | Electronics | Alice Smith |
| Laptop     | Electronics | Bob         |
| T-Shirt    | Clothing    | Alice Smith |
| Jeans      | Clothing    | Charlie     |
| Desk Lamp  | Home Decor  | Bob         |
+------------+-------------+-------------+
5 rows in set (0.00 sec)
```

2. list all product name and salesman name for all salesman who stays in pune

```
mysql> select pname,cname,sname from product p,category c,salesman s
    -> where p.cid=c.cid and p.sid=s.sid and s.address='pune';
+------------+-------------+-------------+
| pname      | cname       | sname       |
+------------+-------------+-------------+
| Smartphone | Electronics | Alice Smith |
| T-Shirt    | Clothing    | Alice Smith |
+------------+-------------+-------------+
2 rows in set (0.00 sec)
```

3. list all product name and category name

```
mysql> select pname,cname from product p left join category c
    -> on p.cid=c.cid;
+------------+-------------+
| pname      | cname       |
+------------+-------------+
| Smartphone | Electronics |
| Laptop     | Electronics |
| T-Shirt    | Clothing    |
| Jeans      | Clothing    |
| Desk Lamp  | Home Decor  |
+------------+-------------+
5 rows in set (0.00 sec)
```

55.create following tables and solve following questions(primary keys are marked in yellow) foreign keys are marked in green
faculty(fid,fname,sp.skill1,sp.skill2)
courses(cid,cname,rid,fid)
room(roomid,rname,rloc)
1. list all courses for which no room is assigned and all rooms for which are Available

```
mysql> select cname,rid,rname from course c left join room r on r.roomid=c.r
id where rid is null
    -> union
    -> select cname,rid,rname from course c right join room r on r.roomid=c.
rid where cid is null;
+-------+------+-------+
| cname | rid  | rname |
+-------+------+-------+
| DTISS | NULL | NULL  |
| NULL  | NULL | Mogra |
+-------+------+-------+
2 rows in set (0.01 sec)
```

2. list all faculties who are not allocated to any course and rooms which are not allocated to any course

```
mysql> select fname,cname,null rname from faculty f left join course c on c.
fid=f.fid where cid is null
    -> union
    -> select null fname,cname,rname from course c right join room r on c.ri
d=r.roomid where cid is null;
+-------+-------+-------+
| fname | cname | rname |
+-------+-------+-------+
| Sid   | NULL  | NULL  |
| Bob   | NULL  | NULL  |
| NULL  | NULL  | Mogra |
+-------+-------+-------+
3 rows in set (0.00 sec)
```

3. list all rooms which are allocated or not allocated to any courses

```
mysql> select cname,rname from course c right join room r on c.rid=r.roomid;

+-------+--------+
| cname | rname  |
+-------+--------+
| DBDA  | jasmin |
| DAC   | rose   |
| NULL  | Mogra  |
| DIOT  | Lotus  |
+-------+--------+
4 rows in set (0.00 sec)
```

4. list all rooms which are not allocated to any courses

```
mysql> select rname,cname from course c right join room r on c.rid=r.roomid
where cname is null;
+-------+-------+
| rname | cname |
+-------+-------+
| Mogra | NULL  |
+-------+-------+
1 row in set (0.00 sec)
```

5. display courses and faculty assigned to those courses whose special skill is database

6. display course id, course name , faculty name and special courses and room name and room location

```
mysql> select cid,cname,fname,sp_skill1,sp_skill2,rname,rloc from course c l
eft join faculty f on c.fid=f.fid left join room r on c.rid=r.roomid ;
+-----+-------+-------+-----------+-----------+--------+-----------+
| cid | cname | fname | sp_skill1 | sp_skill2 | rname  | rloc      |
+-----+-------+-------+-----------+-----------+--------+-----------+
| 121 | DBDA  | John  | a         | b         | jasmin | 1st floor |
| 131 | DAC   | NULL  | NULL      | NULL      | rose   | 2nd floor |
| 141 | DTISS | NULL  | NULL      | NULL      | NULL   | NULL      |
| 151 | DIOT  | Kim   | a         | x         | Lotus  | 1st floor |
+-----+-------+-------+-----------+-----------+--------+-----------+
4 rows in set (0.00 sec)
```

(Day 6)

location

```
mysql> select * from location;
+--------+-------+--------------+
| loc_id | lname | address      |
+--------+-------+--------------+
|    100 | lab A | IACSD Pune   |
|    101 | lab B | Acts Pune    |
|    102 | lab C | Acts Pune    |
|    103 | lab D | IACSD Pune   |
|    104 | lab E | SunBeam Pune |
+--------+-------+--------------+
5 rows in set (0.00 sec)
```

Server

```
mysql> select * from server;
+----------+--------+--------------------+-------+---------+
| serverid | sname  | configuration      | locid | adminid |
+----------+--------+--------------------+-------+---------+
|        1 | omega1 | I5 processor 16 GB |   100 |    1001 |
|        2 | omega2 | I4 processor 32 GB |   100 |    1001 |
|        3 | omega3 | I7 processor 32 GB |   101 |    1002 |
|        4 | omega4 | I5 processor 16 GB |   101 |    1002 |
|        5 | test1  | I4 processor 16 GB |   102 |    1002 |
|        6 | test2  | I3 processor 24 GB |   102 |    1003 |
|        7 | test3  | I5 processor 8 GB  |   101 |    1003 |
|        8 | Lotus1 | I5 processor 16 GB |   103 |    1003 |
|        9 | oracle | I5 processor 256 GB |   103 |    1002 |
|       10 | omega5 | I5 processor 256 GB |   103 |    1001 |
|       11 | mysql2 | I5 processor 256 GB |  NULL |    NULL |
|       12 | mysql3 | I5 processor 256 GB |  NULL |    1001 |
|       13 | mysql4 | I5 processor 256 GB |  NULL |    1001 |
+----------+--------+--------------------+-------+---------+
13 rows in set (0.00 sec)
```

admin

```
mysql> select * from admin;
+---------+-------+----------+--------+----------+
| adminid | uname | password | mobile | name     |
+---------+-------+----------+--------+----------+
|    1001 | user1 | user1    | 11111  | Namrata  |
|    1002 | user2 | user2    | 2222   | Rohit    |
|    1003 | user3 | user3    | 3333   | Priyanka |
|    1004 | user4 | user4    | 44444  | Shilpi   |
|    1005 | user5 | user5    | 55555  | Swati    |
+---------+-------+----------+--------+----------+
5 rows in set (0.00 sec)
```

1. display all server-name, location name of the server

```
mysql> select s.sname , l.lname
    -> from server s
    -> left join location l
    -> on s.locid = l.loc_id;
+--------+-------+
| sname  | lname |
+--------+-------+
| omega1 | lab A |
| omega2 | lab A |
| omega3 | lab B |
| omega4 | lab B |
| test1  | lab C |
| test2  | lab C |
| test3  | lab B |
| Lotus1 | lab D |
| oracle | lab D |
| omega5 | lab D |
| mysql2 | NULL  |
| mysql3 | NULL  |
| mysql4 | NULL  |
+--------+-------+
13 rows in set (0.00 sec)
```

2. display all server-name, admin-name, admin mobile

```
mysql> select s.sname ,a.name,a.mobile
    -> from server s
    -> left join admin a
    -> on s.adminid=a.adminid;
+--------+----------+--------+
| sname  | name     | mobile |
+--------+----------+--------+
| omega1 | Namrata  | 11111  |
| omega2 | Namrata  | 11111  |
| omega3 | Rohit    | 2222   |
| omega4 | Rohit    | 2222   |
| test1  | Rohit    | 2222   |
| test2  | Priyanka | 3333   |
| test3  | Priyanka | 3333   |
| Lotus1 | Priyanka | 3333   |
| oracle | Rohit    | 2222   |
| omega5 | Namrata  | 11111  |
| mysql2 | NULL     | NULL   |
| mysql3 | Namrata  | 11111  |
| mysql4 | Namrata  | 11111  |
+--------+----------+--------+
13 rows in set (0.00 sec)
```

3. display all server which are managed by admin rohit

```
mysql> select s.sname,s.serverid
    -> from server s
    -> join admin a on s.adminid=a.adminid
    -> where a.name='rohit';
+--------+----------+
| sname  | serverid |
+--------+----------+
| omega3 |        3 |
| omega4 |        4 |
| test1  |        5 |
| oracle |        9 |
+--------+----------+
4 rows in set (0.00 sec)
```

4. display all server and the locations, for which admin is rohit

```
mysql> select s.serverid , s.sname , l.lname , l.address
    -> from server s
    -> left join location l
    -> on s.locid = l.loc_id
    -> join admin a
    -> on s.adminid = a.adminid
    -> where a.name = 'Rohit';
+----------+--------+-------+------------+
| serverid | sname  | lname | address    |
+----------+--------+-------+------------+
|        3 | omega3 | Lab B | Acts Pune  |
|        4 | omega4 | Lab B | Acts Pune  |
|        5 | test1  | Lab C | Acts Pune  |
|        9 | oracle | Lab D | IACSD Pune |
+----------+--------+-------+------------+
4 rows in set (0.00 sec)
```

5. display all admins, for whom no server is assigned

```
mysql> SELECT
    ->     a.Adminid,
    ->     a.Uname,
    ->     a.Name,
    ->     a.Mobile
    -> FROM Admin a
    -> LEFT JOIN Server s ON a.Adminid = s.Adminid
    -> WHERE s.Serverid IS NULL;
+----------+-------+--------+--------+
| Adminid | Uname | Name   | Mobile |
+----------+-------+--------+--------+
|    1004 | user4 | Shilpi | 44444  |
|    1005 | user5 | Swati  | 55555  |
+----------+-------+--------+--------+
2 rows in set (0.00 sec)
```

6. display all servers and admin details, and also display admins for whom no server is assigned

```
mysql> select s.serverid,s.sname,s.configuration,a.adminid,a.name,a.mobile,a.uname,a.password
    -> from server s
    -> join admin a
    -> on s.adminid=a.adminid;
+----------+--------+--------------------+---------+----------+--------+-------+----------+
| serverid | sname  | configuration      | adminid | name     | mobile | uname | password |
+----------+--------+--------------------+---------+----------+--------+-------+----------+
|        1 | omega1 | I5 processor 16 GB |    1001 | Namrata  | 11111  | user1 | user1    |
|        2 | omega2 | I4 processor 32 GB |    1001 | Namrata  | 11111  | user1 | user1    |
|       10 | omega5 | I5 processor 256 GB |    1001 | Namrata  | 11111  | user1 | user1    |
|       12 | mysql3 | I5 processor 256 GB |    1001 | Namrata  | 11111  | user1 | user1    |
|       13 | mysql4 | I5 processor 256 GB |    1001 | Namrata  | 11111  | user1 | user1    |
|        3 | omega3 | I7 processor 32 GB |    1002 | Rohit    | 2222   | user2 | user2    |
|        4 | omega4 | I5 processor 16 GB |    1002 | Rohit    | 2222   | user2 | user2    |
|        5 | test1  | I4 processor 16 GB |    1002 | Rohit    | 2222   | user2 | user2    |
|        9 | oracle | I5 processor 256 GB |    1002 | Rohit    | 2222   | user2 | user2    |
|        6 | test2  | I3 processor 24 GB |    1003 | Priyanka | 3333   | user3 | user3    |
|        7 | test3  | I5 processor 8 GB  |    1003 | Priyanka | 3333   | user3 | user3    |
|        8 | Lotus1 | I5 processor 16 GB |    1003 | Priyanka | 3333   | user3 | user3    |
+----------+--------+--------------------+---------+----------+--------+-------+----------+
12 rows in set (0.00 sec)
```

7. display servers, admins and location of the server

```
mysql> SELECT
    ->     s.Sname AS Server_Name,
    ->     a.Name AS Admin_Name,
    ->     l.Lname AS Location_Name
    -> FROM Server s
    -> JOIN Admin a ON s.Adminid = a.Adminid
    -> JOIN Location l ON s.Locid = l.Loc_id;
+-------------+------------+---------------+
| Server_Name | Admin_Name | Location_Name |
+-------------+------------+---------------+
| omega1      | Namrata    | lab A         |
| omega2      | Namrata    | lab A         |
| omega5      | Namrata    | lab D         |
| omega3      | Rohit      | lab B         |
| omega4      | Rohit      | lab B         |
| test1       | Rohit      | lab C         |
| oracle      | Rohit      | lab D         |
| test2       | Priyanka   | lab C         |
| test3       | Priyanka   | lab B         |
| Lotus1      | Priyanka   | lab D         |
+-------------+------------+---------------+
10 rows in set (0.00 sec)
```

8. to display all admins for whom no server is assigned also display locations at which no server is placed

```
mysql> SELECT a.adminid, a.name, NULL AS Lname, NULL AS Address
    -> FROM admin a
    -> LEFT JOIN server s ON a.adminid = s.adminid
    -> WHERE s.serverid IS NULL
    ->
    -> UNION
    ->
    -> SELECT NULL, NULL, l.lname, l.address
    -> FROM location l
    -> LEFT JOIN server s ON s.locid = l.loc_id
    -> WHERE s.serverid IS NULL;
+---------+--------+-------+--------------+
| adminid | name   | Lname | Address      |
+---------+--------+-------+--------------+
|    1004 | Shilpi | NULL  | NULL         |
|    1005 | Swati  | NULL  | NULL         |
|    NULL | NULL   | lab E | SunBeam Pune |
+---------+--------+-------+--------------+
3 rows in set (0.00 sec)
```

9. display all servers for which no location is assigned

```
mysql> SELECT
    ->     Serverid,
    ->     Sname,
    ->     Configuration,
    ->     Adminid
    -> FROM Server
    -> WHERE Locid IS NULL;
+----------+--------+---------------------+---------+
| Serverid | Sname  | Configuration       | Adminid |
+----------+--------+---------------------+---------+
|       11 | mysql2 | I5 processor 256 GB |    NULL |
|       12 | mysql3 | I5 processor 256 GB |    1001 |
|       13 | mysql4 | I5 processor 256 GB |    1001 |
+----------+--------+---------------------+---------+
3 rows in set (0.00 sec)
```

10. display all servers, for which no location is assigned, whose admin is namrata

```
mysql> SELECT s.*
    -> FROM server s
    -> JOIN admin a ON s.adminid = a.adminid
    -> WHERE a.name = 'namrata' AND s.locid IS NULL;
+----------+--------+---------------------+-------+---------+
| serverid | sname  | configuration       | locid | adminid |
+----------+--------+---------------------+-------+---------+
|       12 | mysql3 | I5 processor 256 GB |  NULL |    1001 |
|       13 | mysql4 | I5 processor 256 GB |  NULL |    1001 |
+----------+--------+---------------------+-------+---------+
2 rows in set (0.00 sec)
```

(Day 7)

SQL Window Functions

Design following tables and solve questions using window functions

- demand2 → day, qty
- demand → product, day, qty
- Sales data → product, location, sales, refunds

Demand2 table

```
mysql> SELECT
    ->     day,
    ->     qty,
    ->     SUM(qty) OVER (ORDER BY day) AS running_total_qty,
    ->     LAG(qty, 1, 0) OVER (ORDER BY day) AS prev_day_qty,
    ->     qty - LAG(qty, 1, 0) OVER (ORDER BY day) AS day_over_day_diff
    -> FROM demand2;
+-----+------+-------------------+--------------+--------------------+
| day | qty  | running_total_qty | prev_day_qty | day_over_day_diff |
+-----+------+-------------------+--------------+--------------------+
|   1 |   10 |                10 |            0 |                 10 |
|   2 |    6 |                16 |           10 |                 -4 |
|   3 |   21 |                37 |            6 |                 15 |
|   4 |    9 |                46 |           21 |                -12 |
|   6 |   12 |                58 |            9 |                  3 |
|   7 |   18 |                76 |           12 |                  6 |
|   8 |   15 |                91 |           18 |                 -3 |
+-----+------+-------------------+--------------+--------------------+
7 rows in set (0.00 sec)
```

Demand table

```
mysql> SELECT
    ->     product,
    ->     day,
    ->     qty,
    ->     SUM(qty) OVER (PARTITION BY product ORDER BY day) AS cumulative_qty,
    ->     AVG(qty) OVER (
    ->         PARTITION BY product
    ->         ORDER BY day
    ->         ROWS BETWEEN 1 PRECEDING AND CURRENT ROW
    ->     ) AS moving_avg_2_days
    -> FROM demand;
+---------+-----+------+----------------+-------------------+
| product | day | qty  | cumulative_qty | moving_avg_2_days |
+---------+-----+------+----------------+-------------------+
| Laptop  |   1 |   10 |             10 |           10.0000 |
| Laptop  |   2 |   15 |             25 |           12.5000 |
| Laptop  |   3 |   20 |             45 |           17.5000 |
| Phone   |   1 |    5 |              5 |            5.0000 |
| Phone   |   2 |    8 |             13 |            6.5000 |
| Phone   |   3 |   12 |             25 |           10.0000 |
+---------+-----+------+----------------+-------------------+
6 rows in set (0.00 sec)
```

Sales data

```
mysql> WITH CalculatedSales AS (
    ->     SELECT
    ->         product,
    ->         location,
    ->         sales,
    ->         refunds,
    ->         (sales - refunds) AS net_sales
    ->     FROM sales_data
    -> )
    -> SELECT
    ->     product,
    ->     location,
    ->     sales,
    ->     refunds,
    ->     net_sales,
    ->     DENSE_RANK() OVER (PARTITION BY product ORDER BY net_sales DESC) AS location_rank,
    ->     ROUND((sales / SUM(sales) OVER (PARTITION BY product)) * 100, 2) AS sales_pct_contribution
    -> FROM CalculatedSales;
+---------+----------+---------+---------+-----------+---------------+-------------------------+
| product | location | sales   | refunds | net_sales | location_rank | sales_pct_contribution |
+---------+----------+---------+---------+-----------+---------------+-------------------------+
| Laptop  | North    | 5000.00 |  200.00 |   4800.00 |             1 |                   41.67 |
| Laptop  | East     | 4000.00 |    0.00 |   4000.00 |             2 |                   33.33 |
| Laptop  | South    | 3000.00 |  100.00 |   2900.00 |             3 |                   25.00 |
| Phone   | North    | 2500.00 |  150.00 |   2350.00 |             1 |                   43.10 |
| Phone   | East     | 1800.00 |   20.00 |   1780.00 |             2 |                   31.03 |
| Phone   | South    | 1500.00 |   50.00 |   1450.00 |             3 |                   25.86 |
+---------+----------+---------+---------+-----------+---------------+-------------------------+
6 rows in set (0.00 sec)
```

```
mysql> WITH RECURSIVE DaysSequence AS (
    ->     SELECT 1 AS day
    ->     UNION ALL
    ->     SELECT day + 1
    ->     FROM DaysSequence
    ->     WHERE day < (SELECT MAX(day) FROM demand2)
    -> )
    -> SELECT
    ->     ds.day,
    ->     COALESCE(d.qty, 0) AS qty,
    ->     SUM(COALESCE(d.qty, 0)) OVER (ORDER BY ds.day) AS continuous_running_total
    -> FROM DaysSequence ds
    -> LEFT JOIN demand2 d ON ds.day = d.day;
+------+-----+---------------------------+
| day  | qty | continuous_running_total |
+------+-----+---------------------------+
|    1 |  10 |                        10 |
|    2 |   6 |                        16 |
|    3 |  21 |                        37 |
|    4 |   9 |                        46 |
|    5 |   0 |                        46 |
|    6 |  12 |                        58 |
|    7 |  18 |                        76 |
|    8 |  15 |                        91 |
+------+-----+---------------------------+
8 rows in set (0.00 sec)
```

SQL Window Functions – Questions

1. Write a query to display the day and qty from the demand2 table along with the **previous day's quantity** using a window function.

```
mysql> SELECT
    ->      day,
    ->      qty,
    ->      LAG(qty, 1) OVER (ORDER BY day) AS prev_qty
    -> FROM demand2;
+-----+------+----------+
| day | qty  | prev_qty |
+-----+------+----------+
|   1 |   10 |     NULL |
|   2 |    6 |       10 |
|   3 |   21 |        6 |
|   4 |    9 |       21 |
|   6 |   12 |        9 |
|   7 |   18 |       12 |
|   8 |   15 |       18 |
+-----+------+----------+
7 rows in set (0.00 sec)
```

2. Write a query to display the day, qty, and the **next day's quantity** using a window function.

```
mysql> select day,qty,lead(qty,1) over(order by day) as next_qty
    -> from demand2;
+-----+------+----------+
| day | qty  | next_qty |
+-----+------+----------+
|   1 |   10 |        6 |
|   2 |    6 |       21 |
|   3 |   21 |        9 |
|   4 |    9 |       12 |
|   6 |   12 |       18 |
|   7 |   18 |       15 |
|   8 |   15 |     NULL |
+-----+------+----------+
7 rows in set (0.00 sec)
```

3. Write a query to display the day, qty, and the **difference between the current day's quantity and the previous day's quantity**.

```
mysql> select day,qty,qty-lag(qty,1) over(order by day) as diff_day
    -> from demand2;
+-----+------+----------+
| day | qty  | diff_day |
+-----+------+----------+
|   1 |   10 |     NULL |
|   2 |    6 |       -4 |
|   3 |   21 |       15 |
|   4 |    9 |      -12 |
|   6 |   12 |        3 |
|   7 |   18 |        6 |
|   8 |   15 |       -3 |
+-----+------+----------+
7 rows in set (0.00 sec)
```

4. Write a query to display the product, day, qty, and the **previous day's quantity for each product** using a window function.

```
mysql> select product,day,qty,lag(qty,1) over(partition by product order by day
    -> ) as prev_day
    -> from demand;
+---------+-----+------+----------+
| product | day | qty  | prev_day |
+---------+-----+------+----------+
| Laptop  |   1 |   10 |     NULL |
| Laptop  |   2 |   15 |       10 |
| Laptop  |   3 |   20 |       15 |
| Phone   |   1 |    5 |     NULL |
| Phone   |   2 |    8 |        5 |
| Phone   |   3 |   12 |        8 |
+---------+-----+------+----------+
6 rows in set (0.00 sec)
```

5. Write a query to display the product, day, qty, and the **next day's quantity for each product**.

```
mysql>  select product,day,qty,lead(qty,1) over(partition by product order by day) as next_qty
    -> from demand;
+---------+-----+------+----------+
| product | day | qty  | next_qty |
+---------+-----+------+----------+
| Laptop  |   1 |   10 |       15 |
| Laptop  |   2 |   15 |       20 |
| Laptop  |   3 |   20 |     NULL |
| Phone   |   1 |    5 |        8 |
| Phone   |   2 |    8 |       12 |
| Phone   |   3 |   12 |     NULL |
+---------+-----+------+----------+
6 rows in set (0.00 sec)
```

6. Write a query to display the product, day, qty, and the **difference between the current day's quantity and the previous day's quantity for each product**.

```
mysql>  select product,day,qty,qty-lag(qty,1) over(partition by product order by day) as diff_day
    -> from demand;
+---------+-----+------+----------+
| product | day | qty  | diff_day |
+---------+-----+------+----------+
| Laptop  |   1 |   10 |     NULL |
| Laptop  |   2 |   15 |        5 |
| Laptop  |   3 |   20 |        5 |
| Phone   |   1 |    5 |     NULL |
| Phone   |   2 |    8 |        3 |
| Phone   |   3 |   12 |        4 |
+---------+-----+------+----------+
6 rows in set (0.00 sec)
```

7. Write a query to display the day, qty, and the **cumulative sum of quantity** using a window function

```
mysql> select day,qty,sum(qty) over (order by day)as cumulative
    -> from demand;
+-----+------+------------+
| day | qty  | cumulative |
+-----+------+------------+
|   1 |   10 |         15 |
|   1 |    5 |         15 |
|   2 |   15 |         38 |
|   2 |    8 |         38 |
|   3 |   20 |         70 |
|   3 |   12 |         70 |
+-----+------+------------+
6 rows in set (0.00 sec)
```

8. Using the sales_data table, write a query to display product, location, sales, refunds, and **net sales (sales - refunds)**.

```
mysql> select product,location,sales,refunds, (sales-refunds) as net_sales
    -> from sales_data;
+---------+----------+---------+---------+-----------+
| product | location | sales   | refunds | net_sales |
+---------+----------+---------+---------+-----------+
| Laptop  | North    | 5000.00 |  200.00 |   4800.00 |
| Laptop  | South    | 3000.00 |  100.00 |   2900.00 |
| Laptop  | East     | 4000.00 |    0.00 |   4000.00 |
| Phone   | North    | 2500.00 |  150.00 |   2350.00 |
| Phone   | South    | 1500.00 |   50.00 |   1450.00 |
| Phone   | East     | 1800.00 |   20.00 |   1780.00 |
+---------+----------+---------+---------+-----------+
6 rows in set (0.00 sec)
```

9. Write a query to **rank the products based on sales within each location** using a window function.

```
mysql> select product,location,sales,dense_rank() over(partition by location order
 by sales)drnk
    -> from sales_data;
+---------+----------+---------+------+
| product | location | sales   | drnk |
+---------+----------+---------+------+
| Phone   | East     | 1800.00 |    1 |
| Laptop  | East     | 4000.00 |    2 |
| Phone   | North    | 2500.00 |    1 |
| Laptop  | North    | 5000.00 |    2 |
| Phone   | South    | 1500.00 |    1 |
| Laptop  | South    | 3000.00 |    2 |
+---------+----------+---------+------+
6 rows in set (0.00 sec)
```

10. Write a query to **calculate the total sales from the top 3 best-selling products in each location**.

```
mysql> SELECT
    ->     location,
    ->     SUM(sales) AS total_sales
    -> FROM (
    ->     SELECT
    ->         product,
    ->         location,
    ->         sales,
    ->         DENSE_RANK() OVER (PARTITION BY location ORDER BY sales DESC) AS dr
nk_rank
    ->     FROM sales_data
    -> ) AS drnk
    -> WHERE drnk_rank <= 3
    -> GROUP BY location;
+----------+-------------+
| location | total_sales |
+----------+-------------+
| East     |     5800.00 |
| North    |     7500.00 |
| South    |     4500.00 |
+----------+-------------+
3 rows in set (0.00 sec)
```

11. Write a query to display each product's sales along with the **total sales of its location** using a window function.

```
mysql> SELECT
    ->     product,
    ->     location,
    ->     sales,
    ->     SUM(sales) OVER (PARTITION BY location) AS location_total_sales
    -> FROM sales_data;
+---------+----------+---------+----------------------+
| product | location | sales   | location_total_sales |
+---------+----------+---------+----------------------+
| Laptop  | East     | 4000.00 |              5800.00 |
| Phone   | East     | 1800.00 |              5800.00 |
| Laptop  | North    | 5000.00 |              7500.00 |
| Phone   | North    | 2500.00 |              7500.00 |
| Laptop  | South    | 3000.00 |              4500.00 |
| Phone   | South    | 1500.00 |              4500.00 |
+---------+----------+---------+----------------------+
6 rows in set (0.00 sec)
```

12. Write a query to display each product's sales along with the **percentage contribution of that product to the total sales of its location**.

```
mysql>     SELECT
    ->     product,
    ->     location,
    ->     sales,
    ->   (sales/SUM(sales) OVER(PARTITION BY location))*100.00 AS pct_total_sales
    -> FROM sales_data;
+---------+----------+---------+-----------------+
| product | location | sales   | pct_total_sales |
+---------+----------+---------+-----------------+
| Laptop  | East     | 4000.00 |     68.96551724 |
| Phone   | East     | 1800.00 |     31.03448276 |
| Laptop  | North    | 5000.00 |     66.66666667 |
| Phone   | North    | 2500.00 |     33.33333333 |
| Laptop  | South    | 3000.00 |     66.66666667 |
| Phone   | South    | 1500.00 |     33.33333333 |
+---------+----------+---------+-----------------+
6 rows in set (0.00 sec)
```

13. Write a query to display the **running total of sales for each location** using a window function.

```
mysql>     SELECT
    ->     product,
    ->     location,
    ->     sales,
    ->  sum(sales) over(partition by location order by sales) as totalsales
    -> from sales_data;
+---------+----------+---------+------------+
| product | location | sales   | totalsales |
+---------+----------+---------+------------+
| Phone   | East     | 1800.00 |    1800.00 |
| Laptop  | East     | 4000.00 |    5800.00 |
| Phone   | North    | 2500.00 |    2500.00 |
| Laptop  | North    | 5000.00 |    7500.00 |
| Phone   | South    | 1500.00 |    1500.00 |
| Laptop  | South    | 3000.00 |    4500.00 |
+---------+----------+---------+------------+
6 rows in set (0.00 sec)
```

14. Write a query to display each product's sales along with the **previous product's sales within the same location**.

```
mysql> SELECT
    ->     product,
    ->     location,
    ->     sales,
    ->     lag(sales,1) OVER (PARTITION BY location ORDER BY sales DESC) AS pri_to
tal_sales
    -> FROM sales_data;
+---------+----------+---------+-----------------+
| product | location | sales   | pri_total_sales |
+---------+----------+---------+-----------------+
| Laptop  | East     | 4000.00 |            NULL |
| Phone   | East     | 1800.00 |         4000.00 |
| Laptop  | North    | 5000.00 |            NULL |
| Phone   | North    | 2500.00 |         5000.00 |
| Laptop  | South    | 3000.00 |            NULL |
| Phone   | South    | 1500.00 |         3000.00 |
+---------+----------+---------+-----------------+
6 rows in set (0.00 sec)
```

15. Write a query to display each product's sales along with the **difference between its sales and the previous product's sales within the same location**.

```
mysql> SELECT
    ->     product,
    ->     location,
    ->     sales,
    ->    sales-LAG(sales, 1) OVER (PARTITION BY location ORDER BY sales) AS diffp
rev_product_sales
    -> FROM sales_data;
+---------+----------+---------+-----------------------+
| product | location | sales   | diffprev_product_sales |
+---------+----------+---------+-----------------------+
| Phone   | East     | 1800.00 |                  NULL |
| Laptop  | East     | 4000.00 |               2200.00 |
| Phone   | North    | 2500.00 |                  NULL |
| Laptop  | North    | 5000.00 |               2500.00 |
| Phone   | South    | 1500.00 |                  NULL |
| Laptop  | South    | 3000.00 |               1500.00 |
+---------+----------+---------+-----------------------+
6 rows in set (0.00 sec)
```

(Day 8)

Question 1. (Procedures, Functions)

1.a

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE pattern_a(IN num INT)
    -> BEGIN
    ->     DECLARE str VARCHAR(200) DEFAULT '';
    ->     DECLARE row_str VARCHAR(200) DEFAULT '';
    ->     DECLARE cnt INT DEFAULT 1;
    ->     DECLARE stars INT DEFAULT 1;
    ->     WHILE cnt <= num DO
    ->         SET row_str = REPEAT('*', stars);
    ->         SET str = CONCAT(str, row_str, '\n');
    ->         SET stars = stars + 1;
    ->         SET cnt = cnt + 1;
    ->     END WHILE;
    ->     SELECT str AS Pattern_A;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL pattern_a(4);
+----------------+
| Pattern_A      |
+----------------+
| *
**
***
****
 |
+----------------+
1 row in set (0.00 sec)
```

1.b

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE pattern_b(IN num INT)
    -> BEGIN
    ->     DECLARE str VARCHAR(500) DEFAULT '';
    ->     DECLARE cnt INT DEFAULT 1;
    ->     DECLARE stars INT DEFAULT 1;
    ->     DECLARE spaces INT;
    ->     -- Upper half including middle
    ->     WHILE cnt <= num DO
    ->         SET spaces = num - cnt;
    ->         SET str = CONCAT(str, REPEAT(' ', spaces), REPEAT('*', stars), '\n'
);
    ->         SET stars = stars + 2;
    ->         SET cnt = cnt + 1;
    ->     END WHILE;
    ->     -- Lower half (mirror, skip middle row)
    ->     SET cnt = num - 1;
    ->     SET stars = (num - 1) * 2 - 1;
    ->     WHILE cnt >= 1 DO
    ->         SET spaces = num - cnt;
    ->         SET str = CONCAT(str, REPEAT(' ', spaces), REPEAT('*', stars), '\n'
);
    ->         SET stars = stars - 2;
    ->         SET cnt = cnt - 1;
    ->     END WHILE;
    ->     SELECT str AS Pattern_B;
    -> END//
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL pattern_b(3);
+-------------------------+
| Pattern_B               |
+-------------------------+
|   *
 ***
*****
 ***
  *
 |
+-------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

1.c

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE pattern_c(IN num INT)
    -> BEGIN
    ->     DECLARE str VARCHAR(500) DEFAULT '';
    ->     DECLARE row_str VARCHAR(200);
    ->     DECLARE cnt INT;
    ->     DECLARE i INT;
    ->     SET cnt = (num * 2) - 1;  -- starting digit count (odd numbers: 7,5,3,1
)
    ->     WHILE cnt >= 1 DO
    ->         SET row_str = '';
    ->         SET i = 1;
    ->         WHILE i <= cnt DO
    ->             IF i % 2 = 1 THEN
    ->                 SET row_str = CONCAT(row_str, '1');
    ->             ELSE
    ->                 SET row_str = CONCAT(row_str, '0');
    ->             END IF;
    ->             SET i = i + 1;
    ->         END WHILE;
    ->         SET str = CONCAT(str, row_str, '\n');
    ->         SET cnt = cnt - 2;
    ->     END WHILE;
    ->     SELECT str AS Pattern_C;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL pattern_c(4);
+----------------------+
| Pattern_C            |
+----------------------+
| 1010101
10101
101
1
 |
+----------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

1.d

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE pattern_d(IN num INT)
    -> BEGIN
    ->     DECLARE str VARCHAR(500) DEFAULT '';
    ->     DECLARE row_str VARCHAR(200);
    ->     DECLARE row_cnt INT DEFAULT 1;
    ->     DECLARE col_cnt INT;
    ->     WHILE row_cnt <= num DO
    ->         SET row_str = '';
    ->         SET col_cnt = 1;
    ->         WHILE col_cnt <= row_cnt DO
    ->             IF col_cnt = 1 THEN
    ->                 SET row_str = CONCAT(row_str, col_cnt);
    ->             ELSE
    ->                 SET row_str = CONCAT(row_str, ' ', col_cnt);
    ->             END IF;
    ->             SET col_cnt = col_cnt + 1;
    ->         END WHILE;
    ->         SET str = CONCAT(str, row_str, '\n');
    ->         SET row_cnt = row_cnt + 1;
    ->     END WHILE;
    ->     SELECT str AS Pattern_D;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL pattern_d(5);
+--------------------------------+
| Pattern_D                      |
+--------------------------------+
| 1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
 |
+--------------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

2.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE insert_rec(
    ->     IN peno INT,
    ->     IN pnm VARCHAR(20),
    ->     IN psal DECIMAL(9,2),
    ->     IN pjob VARCHAR(20),
    ->     IN phiredate DATE
    -> )
    -> BEGIN
    ->     INSERT INTO emp(empno, ename, sal, job, hiredate)
    ->     VALUES(peno, pnm, psal, pjob, phiredate);
    ->     SELECT 'Record inserted successfully' AS Message;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql>
mysql> CALL insert_rec(104, 'Sneha', 55000, 'Analyst', '2018-03-20');
+-----------------------------+
| Message                     |
+-----------------------------+
| Record inserted successfully |
+-----------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

```
mysql> select * from emp;
+--------+--------+----------+------------+-------------+--------+
| empno  | ename  | sal      | job        | hiredate    | deptno |
+--------+--------+----------+------------+-------------+--------+
|    101 | Amit   | 45000.00 | Developer  | 2015-06-01  |     10 |
|    102 | Priya  | 85000.00 | Manager    | 1985-05-15  |     20 |
|    103 | Rahul  | 32000.00 | Tester     | 2020-01-10  |     10 |
|    104 | Sneha  | 55000.00 | Analyst    | 2018-03-20  |   NULL |
+--------+--------+----------+------------+-------------+--------+
4 rows in set (0.00 sec)
```

3.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE delete_emp(IN peno INT)
    -> BEGIN
    ->     DELETE FROM emp WHERE empno = peno;
    ->     SELECT ROW_COUNT() AS Rows_Deleted;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL delete_emp(104);
+--------------+
| Rows_Deleted |
+--------------+
|            1 |
+--------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM emp;
+-------+--------+----------+------------+-------------+--------+
| empno | ename  | sal      | job        | hiredate    | deptno |
+-------+--------+----------+------------+-------------+--------+
|   101 | Amit   | 45000.00 | Developer  | 2015-06-01  |     10 |
|   102 | Priya  | 85000.00 | Manager    | 1985-05-15  |     20 |
|   103 | Rahul  | 32000.00 | Tester     | 2020-01-10  |     10 |
+-------+--------+----------+------------+-------------+--------+
3 rows in set (0.00 sec)
```

4.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE emp_above_sal(IN psal DECIMAL(9,2))
    -> BEGIN
    ->     SELECT e.empno, e.ename, e.deptno, d.dname
    ->     FROM emp e
    ->     JOIN dept d ON e.deptno = d.deptno
    ->     WHERE e.sal > psal;
    -> END//
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL emp_above_sal(40000);
+-------+-------+--------+-------+
| empno | ename | deptno | dname |
+-------+-------+--------+-------+
|   101 | Amit  |     10 | IT    |
|   102 | Priya |     20 | HR    |
+-------+-------+--------+-------+
2 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

5.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE dept_stats(
    ->     IN  pdeptno INT,
    ->     OUT pmin    DECIMAL(9,2),
    ->     OUT pmax    DECIMAL(9,2),
    ->     OUT pavg    DECIMAL(9,2),
    ->     OUT pcnt    INT
    -> )
    -> BEGIN
    ->     SELECT MIN(sal), MAX(sal), AVG(sal), COUNT(*)
    ->     INTO pmin, pmax, pavg, pcnt
    ->     FROM emp
    ->     WHERE deptno = pdeptno;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL dept_stats(10, @mn, @mx, @av, @ct);
Query OK, 1 row affected (0.00 sec)

mysql> SELECT @mn AS Min_Sal, @mx AS Max_Sal, @av AS Avg_Sal, @ct AS Emp_Count;
+----------+----------+----------+-----------+
| Min_Sal  | Max_Sal  | Avg_Sal  | Emp_Count |
+----------+----------+----------+-----------+
| 32000.00 | 45000.00 | 38500.00 |         2 |
+----------+----------+----------+-----------+
1 row in set (0.00 sec)
```

6.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE display_products()
    -> BEGIN
    ->     SELECT p.pid, p.pname, c.cid, c.cname, s.sname AS Salesman_Name
    ->     FROM product p
    ->     JOIN category c ON p.cid = c.cid
    ->     JOIN salesman s ON p.sid = s.sid;
    -> END//
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL display_products();
+-----+--------+-----+-------------+---------------+
| pid | pname  | cid | cname       | Salesman_Name |
+-----+--------+-----+-------------+---------------+
|  10 | Laptop |   1 | Electronics | Rajesh        |
|  11 | Mobile |   1 | Electronics | Seema         |
+-----+--------+-----+-------------+---------------+
2 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

7.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE vehicles_by_customer(IN pcname VARCHAR(50))
    -> BEGIN
    ->     SELECT v.vid, v.vname, v.price, cv.buy_price, s.sname AS Salesman_Name
    ->     FROM customer c
    ->     JOIN cust_vehicle cv ON c.custid = cv.custid
    ->     JOIN vehicle v ON cv.vid = v.vid
    ->     JOIN salesman s ON cv.sid = s.sid
    ->     WHERE c.cname = pcname;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL vehicles_by_customer('Nilima');
+------+---------+-----------+-------------+----------------+
| vid  | vname   | price     | buy_price   | Salesman_Name  |
+------+---------+-----------+-------------+----------------+
|    1 | Activa  | 80000.00  |  75000.00   | Rajesh         |
+------+---------+-----------+-------------+----------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE emp_salary_status()
    -> BEGIN
    ->     DECLARE vempno     INT DEFAULT 0;
    ->     DECLARE vename     VARCHAR(20);
    ->     DECLARE vjob       VARCHAR(20);
    ->     DECLARE vsal       DECIMAL(9,2);
    ->     DECLARE vdeptno    INT;
    ->     DECLARE vavgsal    DECIMAL(9,2);
    ->     DECLARE vstatus    VARCHAR(10);
    ->     DECLARE vset       INT DEFAULT 0;
    ->     DECLARE vtotalcnt INT DEFAULT 0;
    ->
    ->     DECLARE empcur CURSOR FOR
    ->         SELECT e.empno, e.ename, e.job, e.sal, e.deptno,
    ->                AVG(e2.sal) OVER (PARTITION BY e.deptno)
    ->         FROM emp e
    ->         JOIN emp e2 ON e.deptno = e2.deptno;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET vset = 1;
    ->
    ->     SELECT COUNT(*) INTO vtotalcnt FROM emp;
    ->     IF vtotalcnt = 0 THEN
    ->         SELECT 'Emp table is empty' AS ErrorMessage;
    ->     ELSE
    ->         OPEN empcur;
    ->         label1: LOOP
    ->             FETCH empcur INTO vempno, vename, vjob, vsal, vdeptno, vavgsal;
    ->             IF vset = 1 THEN
    ->                 LEAVE label1;
    ->             END IF;
    ->
    ->             IF vsal > vavgsal THEN
    ->                 SET vstatus = 'Greater';
    ->             ELSEIF vsal < vavgsal THEN
    ->                 SET vstatus = 'Lesser';
    ->             ELSE
    ->                 SET vstatus = 'Equal';
    ->             END IF;
    ->
    ->             SELECT vempno AS Empno, vename AS Name, vjob AS Job,
    ->                    vsal AS Salary, vstatus AS Status, vdeptno AS Deptno;
    ->         END LOOP;
    ->         CLOSE empcur;
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql>
```

8.

```
mysql> CALL emp_salary_status();
+-------+-------+--------+----------+--------+--------+
| Empno | Name  | Job    | Salary   | Status | Deptno |
+-------+-------+--------+----------+--------+--------+
|   103 | Rahul | Tester | 32000.00 | Lesser |     10 |
+-------+-------+--------+----------+--------+--------+
1 row in set (0.01 sec)

+-------+------+-----------+----------+---------+--------+
| Empno | Name | Job       | Salary   | Status  | Deptno |
+-------+------+-----------+----------+---------+--------+
|   101 | Amit | Developer | 45000.00 | Greater |     10 |
+-------+------+-----------+----------+---------+--------+
1 row in set (0.01 sec)

+-------+-------+--------+----------+--------+--------+
| Empno | Name  | Job    | Salary   | Status | Deptno |
+-------+-------+--------+----------+--------+--------+
|   103 | Rahul | Tester | 32000.00 | Lesser |     10 |
+-------+-------+--------+----------+--------+--------+
1 row in set (0.01 sec)

+-------+------+-----------+----------+---------+--------+
| Empno | Name | Job       | Salary   | Status  | Deptno |
+-------+------+-----------+----------+---------+--------+
|   101 | Amit | Developer | 45000.00 | Greater |     10 |
+-------+------+-----------+----------+---------+--------+
1 row in set (0.01 sec)

+-------+-------+---------+----------+--------+--------+
| Empno | Name  | Job     | Salary   | Status | Deptno |
+-------+-------+---------+----------+--------+--------+
|   102 | Priya | Manager | 85000.00 | Equal  |     20 |
+-------+-------+---------+----------+--------+--------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)
```

9.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE update_sal_by_exp()
    -> BEGIN
    ->     DECLARE vempno  INT DEFAULT 0;
    ->     DECLARE vsal    DECIMAL(9,2);
    ->     DECLARE vhire   DATE;
    ->     DECLARE vexp    INT;
    ->     DECLARE vnewsal DECIMAL(9,2);
    ->     DECLARE vset    INT DEFAULT 0;
    ->
    ->     DECLARE expcur CURSOR FOR
    ->         SELECT empno, sal, hiredate FROM emp;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET vset = 1;
    ->
    ->     OPEN expcur;
    ->     label1: LOOP
    ->         FETCH expcur INTO vempno, vsal, vhire;
    ->         IF vset = 1 THEN
    ->             LEAVE label1;
    ->         END IF;
    ->
    ->         SET vexp = TIMESTAMPDIFF(YEAR, vhire, CURDATE());
    ->
    ->         IF vexp <= 35 THEN
    ->             SET vnewsal = vsal;  -- No update
    ->         ELSEIF vexp > 35 AND vexp <= 38 THEN
    ->             SET vnewsal = vsal + (vsal * 0.20);
    ->         ELSE
    ->             SET vnewsal = vsal + (vsal * 0.25);
    ->         END IF;
    ->
    ->         UPDATE emp SET sal = vnewsal WHERE empno = vempno;
    ->     END LOOP;
    ->     CLOSE expcur;
    ->
    ->     SELECT 'Salary updated based on experience rules' AS Message;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL update_sal_by_exp();
+---------------------------------------+
| Message                               |
+---------------------------------------+
| Salary updated based on experience rules |
+---------------------------------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM emp;
+--------+--------+-----------+-----------+------------+--------+
| empno  | ename  | sal       | job       | hiredate   | deptno |
+--------+--------+-----------+-----------+------------+--------+
|    101 | Amit   |  45000.00 | Developer | 2015-06-01 |     10 |
|    102 | Priya  | 106250.00 | Manager   | 1985-05-15 |     20 |
|    103 | Rahul  |  32000.00 | Tester    | 2020-01-10 |     10 |
+--------+--------+-----------+-----------+------------+--------+
3 rows in set (0.00 sec)
```

10.

```
mysql> DELIMITER //
mysql> CREATE FUNCTION calc_experience(phiredate DATE)
    -> RETURNS INT
    -> BEGIN
    ->     DECLARE vexp INT;
    ->     SET vexp = TIMESTAMPDIFF(YEAR, phiredate, CURDATE());
    ->     RETURN vexp;
    -> END//
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql>
mysql> DELIMITER //
mysql> CREATE PROCEDURE calc_allowance()
    -> BEGIN
    ->     DECLARE vempno      INT DEFAULT 0;
    ->     DECLARE vename      VARCHAR(20);
    ->     DECLARE vhiredate   DATE;
    ->     DECLARE vexp        INT;
    ->     DECLARE vallowance DECIMAL(9,2);
    ->     DECLARE vset        INT DEFAULT 0;
    ->
    ->     DECLARE empcur CURSOR FOR
    ->         SELECT empno, ename, hiredate FROM emp;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET vset = 1;
    ->
    ->     DELETE FROM emp_allowance;
    ->
    ->     OPEN empcur;
    ->     label1: LOOP
    ->         FETCH empcur INTO vempno, vename, vhiredate;
    ->         IF vset = 1 THEN
    ->             LEAVE label1;
    ->         END IF;
    ->
    ->         SET vexp = calc_experience(vhiredate);
    ->         SET vallowance = vexp * 3000;
    ->
    ->         INSERT INTO emp_allowance(empno, ename, hiredate, experience, allowance)
    ->         VALUES(vempno, vename, vhiredate, vexp, vallowance);
    ->     END LOOP;
    ->     CLOSE empcur;
    ->
    ->     SELECT * FROM emp_allowance;
    -> END//
Query OK, 0 rows affected (0.00 sec)
mysql> DELIMITER ;
```

```
mysql> CALL calc_allowance();
+-------+-------+------------+------------+-----------+
| empno | ename | hiredate   | experience | allowance |
+-------+-------+------------+------------+-----------+
|   101 | Amit  | 2015-06-01 |         10 |  30000.00 |
|   102 | Priya | 1985-05-15 |         40 | 120000.00 |
|   103 | Rahul | 2020-01-10 |          6 |  18000.00 |
+-------+-------+------------+------------+-----------+
3 rows in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)
```

11.

```
mysql> DELIMITER //
mysql> CREATE FUNCTION calc_ctc(psal DECIMAL(9,2), phiredate DATE)
    -> RETURNS DECIMAL(9,2)
    -> BEGIN
    ->     DECLARE vda         DECIMAL(9,2);
    ->     DECLARE vhra        DECIMAL(9,2);
    ->     DECLARE vta         DECIMAL(9,2);
    ->     DECLARE vspecial    DECIMAL(9,2) DEFAULT 0;
    ->     DECLARE vservice    INT;
    ->     DECLARE vctc        DECIMAL(9,2);
    ->
    ->     SET vda      = psal * 0.15;
    ->     SET vhra     = psal * 0.20;
    ->     SET vta      = psal * 0.08;
    ->     SET vservice = TIMESTAMPDIFF(YEAR, phiredate, CURDATE());
    ->
    ->     IF vservice < 1 THEN
    ->         SET vspecial = 0;
    ->     ELSEIF vservice >= 1 AND vservice < 2 THEN
    ->         SET vspecial = psal * 0.10;
    ->     ELSEIF vservice >= 2 AND vservice < 4 THEN
    ->         SET vspecial = psal * 0.20;
    ->     ELSE
    ->         SET vspecial = psal * 0.30;
    ->     END IF;
    ->
    ->     SET vctc = psal + vda + vhra + vta + vspecial;
    ->     RETURN vctc;
    -> END//
Query OK, 0 rows affected (0.00 sec)

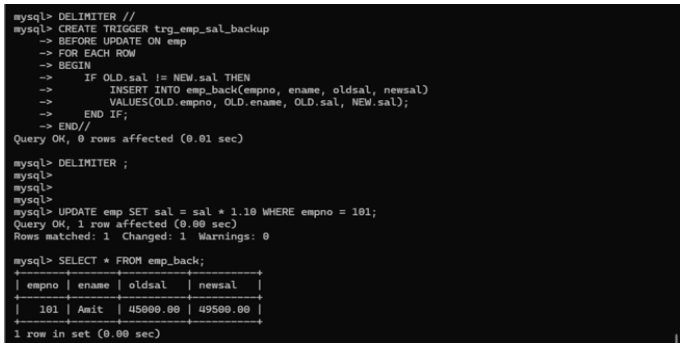
mysql> DELIMITER ;
```

12.

```
mysql> SELECT empno, ename, sal,
    ->        calc_ctc(sal, hiredate) AS Cost_To_Company
    -> FROM emp;
+-------+-------+-----------+-----------------+
| empno | ename | sal       | Cost_To_Company |
+-------+-------+-----------+-----------------+
|   101 | Amit  |  45000.00 |        77850.00 |
|   102 | Priya | 106250.00 |       183812.50 |
|   103 | Rahul |  32000.00 |        55360.00 |
+-------+-------+-----------+-----------------+
3 rows in set (0.00 sec)
```

QUESTION 2. (Triggers)

1.

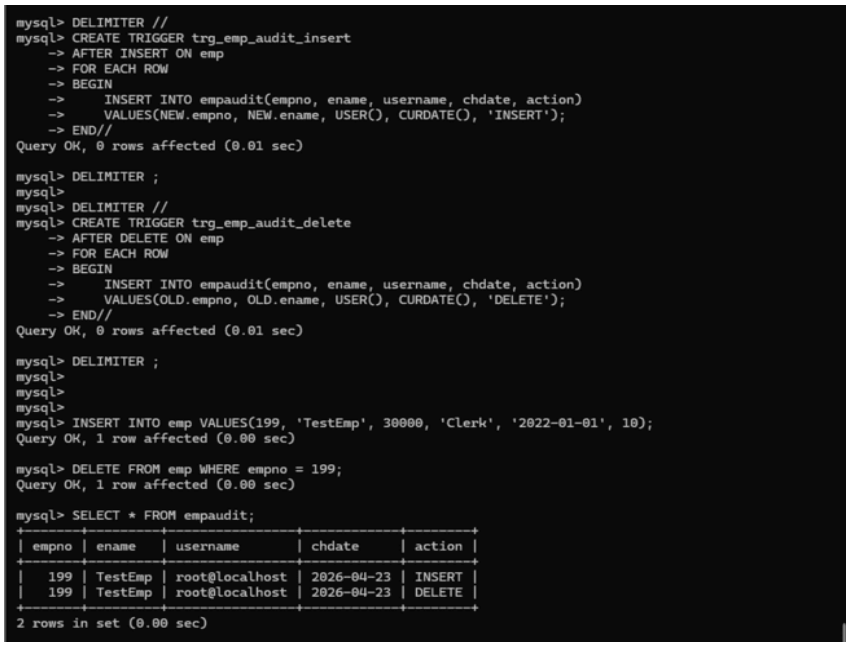


```
mysql> DELIMITER //
mysql> CREATE TRIGGER trg_emp_sal_backup
    -> BEFORE UPDATE ON emp
    -> FOR EACH ROW
    -> BEGIN
    ->     IF OLD.sal != NEW.sal THEN
    ->         INSERT INTO emp_back(empno, ename, oldsal, newsal)
    ->         VALUES(OLD.empno, OLD.ename, OLD.sal, NEW.sal);
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql>
mysql>
mysql> UPDATE emp SET sal = sal * 1.10 WHERE empno = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM emp_back;
+-------+-------+----------+----------+
| empno | ename | oldsal   | newsal   |
+-------+-------+----------+----------+
|   101 | Amit  | 45000.00 | 49500.00 |
+-------+-------+----------+----------+
1 row in set (0.00 sec)
```

2.



```
mysql> DELIMITER //
mysql> CREATE TRIGGER trg_emp_audit_insert
    -> AFTER INSERT ON emp
    -> FOR EACH ROW
    -> BEGIN
    ->     INSERT INTO empaudit(empno, ename, username, chdate, action)
    ->     VALUES(NEW.empno, NEW.ename, USER(), CURDATE(), 'INSERT');
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql> DELIMITER //
mysql> CREATE TRIGGER trg_emp_audit_delete
    -> AFTER DELETE ON emp
    -> FOR EACH ROW
    -> BEGIN
    ->     INSERT INTO empaudit(empno, ename, username, chdate, action)
    ->     VALUES(OLD.empno, OLD.ename, USER(), CURDATE(), 'DELETE');
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql>
mysql>
mysql> INSERT INTO emp VALUES(199, 'TestEmp', 30000, 'Clerk', '2022-01-01', 10);
Query OK, 1 row affected (0.00 sec)

mysql> DELETE FROM emp WHERE empno = 199;
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM empaudit;
+-------+---------+----------------+------------+--------+
| empno | ename   | username       | chdate     | action |
+-------+---------+----------------+------------+--------+
|   199 | TestEmp | root@localhost | 2026-04-23 | INSERT |
|   199 | TestEmp | root@localhost | 2026-04-23 | DELETE |
+-------+---------+----------------+------------+--------+
2 rows in set (0.00 sec)
```

3.



```
mysql> DELIMITER //
mysql> CREATE TRIGGER trg_vehicle_price_history
    -> BEFORE UPDATE ON vehicle
    -> FOR EACH ROW
    -> BEGIN
    ->     IF OLD.price != NEW.price THEN
    ->         INSERT INTO vehicle_history(vno, vname, oldprice, newprice, chdate, username)
    ->         VALUES(OLD.vid, OLD.vname, OLD.price, NEW.price, CURDATE(), USER());
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql>
mysql>
mysql> UPDATE vehicle SET price = 90000 WHERE vid = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM vehicle_history;
+------+--------+----------+----------+------------+----------------+
| vno  | vname  | oldprice | newprice | chdate     | username       |
+------+--------+----------+----------+------------+----------------+
|    1 | Activa | 80000.00 | 90000.00 | 2026-04-23 | root@localhost |
+------+--------+----------+----------+------------+----------------+
1 row in set (0.00 sec)
```

(Day 11)

Q1.

```
test> db.restaurent.find()
[
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c66'),
    address: {
      building: '2780',
      coord: [ -73.98241999999999, 40.579505 ],
      street: 'Stillwell Avenue',
      zipcode: '11224'
    },
    borough: 'Brooklyn',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-06-10T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2013-06-05T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2012-04-13T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2011-10-12T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: 'Riviera Caterer',
    restaurant_id: '40356018'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c67'),
    address: {
      building: '1007',
      coord: [ -73.856077, 40.848447 ],
      street: 'Morris Park Ave',
      zipcode: '10462'
    },
    borough: 'Bronx',
    cuisine: 'Bakery',
    grades: [
      { date: ISODate('2014-03-03T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-09-11T00:00:00.000Z'), grade: 'A', score: 6 },
```

Q2.

```
test> db.restaurent.find(
...   {},
...   { restaurant_id: 1, name: 1, borough: 1, cuisine: 1 }
... )
[
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c66'),
    borough: 'Brooklyn',
    cuisine: 'American',
    name: 'Riviera Caterer',
    restaurant_id: '40356018'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c67'),
    borough: 'Bronx',
    cuisine: 'Bakery',
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c68'),
    borough: 'Queens',
    cuisine: 'Jewish/Kosher',
    name: 'Tov Kosher Kitchen',
    restaurant_id: '40356068'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c69'),
    borough: 'Brooklyn',
    cuisine: 'Chinese',
    name: 'May May Kitchen',
    restaurant_id: '40358429'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c6a'),
    borough: 'Manhattan',
    cuisine: 'American',
    name: '1 East 66Th Street Kitchen',
    restaurant_id: '40359480'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c6b'),
    borough: 'Brooklyn',
    cuisine: 'Hamburgers',
    name: "Wendy'S",
    restaurant_id: '30112340'
```

Q3.

```
test> db.restaurent.find(
|   {},
|   { restaurant_id: 1, name: 1, borough: 1, cuisine: 1, _id: 0 }
| )
|
[
  {
    borough: 'Brooklyn',
    cuisine: 'American',
    name: 'Riviera Caterer',
    restaurant_id: '40356018'
  },
  {
    borough: 'Bronx',
    cuisine: 'Bakery',
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    borough: 'Queens',
    cuisine: 'Jewish/Kosher',
    name: 'Tov Kosher Kitchen',
    restaurant_id: '40356068'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Chinese',
    name: 'May May Kitchen',
    restaurant_id: '40358429'
  },
  {
    borough: 'Manhattan',
    cuisine: 'American',
    name: '1 East 66Th Street Kitchen',
    restaurant_id: '40359480'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Hamburgers',
    name: "Wendy'S",
    restaurant_id: '30112340'
  },
```

Q4.

```
test> db.restaurent.find(
|   {},
|   { restaurant_id: 1, name: 1, borough: 1, "address.zipcode": 1, _id: 0 }
| )
|
[
  {
    address: { zipcode: '11224' },
    borough: 'Brooklyn',
    name: 'Riviera Caterer',
    restaurant_id: '40356018'
  },
  {
    address: { zipcode: '10462' },
    borough: 'Bronx',
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    address: { zipcode: '11374' },
    borough: 'Queens',
    name: 'Tov Kosher Kitchen',
    restaurant_id: '40356068'
  },
  {
    address: { zipcode: '11208' },
    borough: 'Brooklyn',
    name: 'May May Kitchen',
    restaurant_id: '40358429'
  },
  {
    address: { zipcode: '10065' },
    borough: 'Manhattan',
    name: '1 East 66Th Street Kitchen',
    restaurant_id: '40359480'
  },
  {
    address: { zipcode: '11225' },
    borough: 'Brooklyn',
    name: "Wendy'S",
    restaurant_id: '30112340'
  },
  {
    address: { zipcode: '11223' },
    borough: 'Brooklyn',
```

Q5.

```
test> db.restaurent.find(
| {borough:"Bronx"})
[
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c67'),
    address: {
      building: '1007',
      coord: [ -73.856077, 40.848447 ],
      street: 'Morris Park Ave',
      zipcode: '10462'
    },
    borough: 'Bronx',
    cuisine: 'Bakery',
    grades: [
      { date: ISODate('2014-03-03T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-09-11T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2013-01-24T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-11-23T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2011-03-10T00:00:00.000Z'), grade: 'B', score: 14 }
    ],
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c81'),
    address: {
      building: '2300',
      coord: [ -73.8786113, 40.8502883 ],
      street: 'Southern Boulevard',
      zipcode: '10460'
    },
    borough: 'Bronx',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-05-28T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2013-06-19T00:00:00.000Z'), grade: 'A', score: 4 },
      { date: ISODate('2012-06-15T00:00:00.000Z'), grade: 'A', score: 3 }
    ],
    name: 'Wild Asia',
    restaurant_id: '40357217'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c88'),
    address: {
      building: '1236',
```

Q6.

```
test> db.restaurent.find( { borough: "Bronx" }).limit(5)
[
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c67'),
    address: {
      building: '1007',
      coord: [ -73.856077, 40.848447 ],
      street: 'Morris Park Ave',
      zipcode: '10462'
    },
    borough: 'Bronx',
    cuisine: 'Bakery',
    grades: [
      { date: ISODate('2014-03-03T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-09-11T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2013-01-24T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-11-23T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2011-03-10T00:00:00.000Z'), grade: 'B', score: 14 }
    ],
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c81'),
    address: {
      building: '2300',
      coord: [ -73.8786113, 40.8502883 ],
      street: 'Southern Boulevard',
      zipcode: '10460'
    },
    borough: 'Bronx',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-05-28T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2013-06-19T00:00:00.000Z'), grade: 'A', score: 4 },
      { date: ISODate('2012-06-15T00:00:00.000Z'), grade: 'A', score: 3 }
    ],
    name: 'Wild Asia',
    restaurant_id: '40357217'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c88'),
    address: {
```

Q7.

```
test> db.restaurent.find( { borough: "Bronx" }).skip(5).limit(5)
[
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034ca3'),
    address: {
      building: '658',
      coord: [ -73.81363999999999, 40.82941100000001 ],
      street: 'Clarence Ave',
      zipcode: '10465'
    },
    borough: 'Bronx',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-06-21T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2012-07-11T00:00:00.000Z'), grade: 'A', score: 10 }
    ],
    name: 'Manhem Club',
    restaurant_id: '40364363'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034cbc'),
    address: {
      building: '2222',
      coord: [ -73.84971759999999, 40.8304811 ],
      street: 'Haviland Avenue',
      zipcode: '10462'
    },
    borough: 'Bronx',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-12-18T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2014-05-01T00:00:00.000Z'), grade: 'B', score: 17 },
      { date: ISODate('2013-03-14T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-09-20T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2012-02-08T00:00:00.000Z'), grade: 'B', score: 19 }
    ],
    name: 'The New Starling Athletic Club Of The Bronx',
    restaurant_id: '40364956'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034cce'),
    address: {
      building: '72',
      coord: [ -73.92506, 40.8275556 ],
      street: 'East  161 Street',
      zipcode: '10451'
    },
```

Q8.

```
test> db.restaurent.find(
|   { "grades.score": { $gt: 90 } }
| )
[
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034dc5'),
    address: {
      building: '65',
      coord: [ -73.9782725, 40.7624022 ],
      street: 'West   54 Street',
      zipcode: '10019'
    },
    borough: 'Manhattan',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-08-22T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2014-03-28T00:00:00.000Z'), grade: 'C', score: 99 },
      { date: ISODate('2013-09-25T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2013-04-08T00:00:00.000Z'), grade: 'B', score: 25 },
      { date: ISODate('2012-10-15T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2011-10-19T00:00:00.000Z'), grade: 'A', score: 13 }
    ],
    name: "Murals On 54/Randolphs'S",
    restaurant_id: '40372466'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034e60'),
    address: {
      building: '345',
      coord: [ -73.9864626, 40.7266739 ],
      street: 'East 6 Street',
      zipcode: '10003'
    },
    borough: 'Manhattan',
    cuisine: 'Indian',
    grades: [
      { date: ISODate('2014-09-15T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2014-01-14T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2013-05-30T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2013-04-24T00:00:00.000Z'), grade: 'P', score: 2 },
      { date: ISODate('2012-10-01T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2012-04-06T00:00:00.000Z'), grade: 'C', score: 92 },
      { date: ISODate('2011-11-03T00:00:00.000Z'), grade: 'C', score: 41 }
    ],
    name: 'Gandhi',
```

Q9.

```
test> db.restaurent.find(
|   { "grades.score": { $gt: 80, $lt: 100 } }
| )
[
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034dc5'),
    address: {
      building: '65',
      coord: [ -73.9782725, 40.7624022 ],
      street: 'West   54 Street',
      zipcode: '10019'
    },
    borough: 'Manhattan',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-08-22T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2014-03-28T00:00:00.000Z'), grade: 'C', score: 99 },
      { date: ISODate('2013-09-25T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2013-04-08T00:00:00.000Z'), grade: 'B', score: 25 },
      { date: ISODate('2012-10-15T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2011-10-19T00:00:00.000Z'), grade: 'A', score: 13 }
    ],
    name: "Murals On 54/Randolphs'S",
    restaurant_id: '40372466'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034e60'),
    address: {
      building: '345',
      coord: [ -73.9864626, 40.7266739 ],
      street: 'East 6 Street',
      zipcode: '10003'
    },
    borough: 'Manhattan',
    cuisine: 'Indian',
    grades: [
      { date: ISODate('2014-09-15T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2014-01-14T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2013-05-30T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2013-04-24T00:00:00.000Z'), grade: 'P', score: 2 },
      { date: ISODate('2012-10-01T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2012-04-06T00:00:00.000Z'), grade: 'C', score: 92 },
      { date: ISODate('2011-11-03T00:00:00.000Z'), grade: 'C', score: 41 }
    ],
    name: 'Gandhi',
    restaurant_id: '40381295'
```

Q10.

```
test> db.restaurent.find(
|    { "address.coord.1": { $lt: -95.754168 } }
| )
```

Q11.

```
test> db.restaurent.find(
|    {
|      cuisine: { $ne: "American" },
|      "grades.score": { $gt: 70 },
|      "address.coord.1": { $lt: -65.754168 }
|    }
| )
```

Q12.

```
test> db.restaurent.find(
|   {
|     cuisine: { $ne: "American" },
|     "grades.score": { $gt: 70 },
|     "address.coord.0": { $lt: -65.754168 }
|   }
| )
[
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034e60'),
    address: {
      building: '345',
      coord: [ -73.9864626, 40.7266739 ],
      street: 'East 6 Street',
      zipcode: '10003'
    },
    borough: 'Manhattan',
    cuisine: 'Indian',
    grades: [
      { date: ISODate('2014-09-15T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2014-01-14T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2013-05-30T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2013-04-24T00:00:00.000Z'), grade: 'P', score: 2 },
      { date: ISODate('2012-10-01T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2012-04-06T00:00:00.000Z'), grade: 'C', score: 92 },
      { date: ISODate('2011-11-03T00:00:00.000Z'), grade: 'C', score: 41 }
    ],
    name: 'Gandhi',
    restaurant_id: '40381295'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034fc6'),
    address: {
      building: '130',
      coord: [ -73.984758, 40.7457939 ],
      street: 'Madison Avenue',
      zipcode: '10016'
    },
    borough: 'Manhattan',
    cuisine: 'Pizza/Italian',
    grades: [
      { date: ISODate('2014-12-24T00:00:00.000Z'), grade: 'Z', score: 31 },
      { date: ISODate('2014-06-17T00:00:00.000Z'), grade: 'C', score: 98 },
```

Q13.

```
test> db.restaurent.find(
|   {
|     cuisine: { $ne: "American" },
|     "grades.grade": "A",
|     borough: { $ne: "Brooklyn" }
|   }
| ).sort({ cuisine: -1 })
[
  {
    _id: ObjectId('69e1fc89c3cf3ffef8038c84'),
    address: {
      building: '217',
      coord: [ -73.9961458, 40.7186459 ],
      street: 'Grand Street',
      zipcode: '10013'
    },
    borough: 'Manhattan',
    cuisine: 'Vietnamese/Cambodian/Malaysia',
    grades: [
      { date: ISODate('2014-04-04T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2013-10-29T00:00:00.000Z'), grade: 'B', score: 20 },
      { date: ISODate('2013-03-25T00:00:00.000Z'), grade: 'B', score: 22 },
      { date: ISODate('2012-07-13T00:00:00.000Z'), grade: 'A', score: 11 }
    ],
    name: 'Paris Sandwich',
    restaurant_id: '41674140'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8035f29'),
    address: {
      building: '156-32',
      coord: [ -73.8087286, 40.7639199 ],
      street: 'Northern Boulevard',
      zipcode: '11354'
    },
    borough: 'Queens',
    cuisine: 'Vietnamese/Cambodian/Malaysia',
    grades: [
      { date: ISODate('2014-09-11T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2014-04-08T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2013-10-07T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-09-13T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2012-04-26T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2011-11-09T00:00:00.000Z'), grade: 'A', score: 10 }
    ],
```

Q14.

```
test> db.restaurent.find(
|   { name: { $regex: /^Wil/ } },
|   { restaurant_id: 1, name: 1, borough: 1, cuisine: 1, _id: 0 }
| )
[
  {
    borough: 'Brooklyn',
    cuisine: 'Delicatessen',
    name: "Wilken'S Fine Food",
    restaurant_id: '40356483'
  },
  {
    borough: 'Bronx',
    cuisine: 'American',
    name: 'Wild Asia',
    restaurant_id: '40357217'
  },
  {
    borough: 'Bronx',
    cuisine: 'Pizza',
    name: 'Wilbel Pizza',
    restaurant_id: '40871979'
  },
  {
    borough: 'Manhattan',
    cuisine: 'Seafood',
    name: 'Wild Edibles',
    restaurant_id: '40928482'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Bagels/Pretzels',
    name: 'Wild Bagels',
    restaurant_id: '41225826'
  },
  {
    borough: 'Bronx',
    cuisine: 'Latin (Cuban, Dominican, Puerto Rican, South & Central American)',
    name: "Willie'S Steak House",
    restaurant_id: '41239497'
  },
  {
    borough: 'Manhattan',
    cuisine: 'American',
    name: 'Willis North America',
    restaurant_id: '41255267'
  },
```

Q15.

```
Type "it" for more
test> db.restaurent.find(
|   { name: { $regex: /ces$/ } },
|   { restaurant_id: 1, name: 1, borough: 1, cuisine: 1, _id: 0 }
| )
[
  {
    borough: 'Manhattan',
    cuisine: 'American',
    name: 'Pieces',
    restaurant_id: '40399910'
  },
  {
    borough: 'Queens',
    cuisine: 'American',
    name: 'S.M.R Restaurant Services',
    restaurant_id: '40403857'
  },
  {
    borough: 'Manhattan',
    cuisine: 'American',
    name: 'Good Shepherd Services',
    restaurant_id: '40403989'
  },
  {
    borough: 'Queens',
    cuisine: 'Ice Cream, Gelato, Yogurt, Ices',
    name: "The Ice Box-Ralph'S Famous Italian Ices",
    restaurant_id: '40698899'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Jewish/Kosher',
    name: 'Alices',
    restaurant_id: '40782042'
  },
  {
    borough: 'Manhattan',
    cuisine: 'American',
    name: 'Re: Sources',
    restaurant_id: '40876068'
  },
  {
    borough: 'Staten Island',
    cuisine: 'Ice Cream, Gelato, Yogurt, Ices',
```

Q16.

```
test> db.restaurent.find(
|   { name: { $regex: /Reg/ } },
|   { restaurant_id: 1, name: 1, borough: 1, cuisine: 1, _id: 0 }
| )
[
  {
    borough: 'Brooklyn',
    cuisine: 'American',
    name: 'Regina Caterers',
    restaurant_id: '40356649'
  },
  {
    borough: 'Manhattan',
    cuisine: 'Café/Coffee/Tea',
    name: 'Caffe Reggio',
    restaurant_id: '40369418'
  },
  {
    borough: 'Manhattan',
    cuisine: 'American',
    name: 'Regency Hotel',
    restaurant_id: '40382679'
  },
  {
    borough: 'Manhattan',
    cuisine: 'American',
    name: 'Regency Whist Club',
    restaurant_id: '40402377'
  },
  {
    borough: 'Queens',
    cuisine: 'American',
    name: 'Rego Park Cafe',
    restaurant_id: '40523342'
  },
  {
    borough: 'Queens',
    cuisine: 'Pizza',
    name: 'Regina Pizza',
    restaurant_id: '40801325'
  },
  {
    borough: 'Manhattan',
    cuisine: 'American',
```

Q17.

```
test> db.restaurent.find(
|   {
|     borough: "Bronx",
|     cuisine: { $in: ["American", "Chinese"] }
|   }
| )
[
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c81'),
    address: {
      building: '2300',
      coord: [ -73.8786113, 40.8502883 ],
      street: 'Southern Boulevard',
      zipcode: '10460'
    },
    borough: 'Bronx',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-05-28T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2013-06-19T00:00:00.000Z'), grade: 'A', score: 4 },
      { date: ISODate('2012-06-15T00:00:00.000Z'), grade: 'A', score: 3 }
    ],
    name: 'Wild Asia',
    restaurant_id: '40357217'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c88'),
    address: {
      building: '1236',
      coord: [ -73.8893654, 40.81376179999999 ],
      street: '238 Spofford Ave',
      zipcode: '10474'
    },
    borough: 'Bronx',
    cuisine: 'Chinese',
    grades: [
      { date: ISODate('2013-12-30T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2013-01-08T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2012-06-12T00:00:00.000Z'), grade: 'B', score: 15 }
    ],
    name: 'Happy Garden',
    restaurant_id: '40363289'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c9c'),
    address: {
```

Q18.

```
test> db.restaurent.find(
|   {
|     borough: { $in: ["Staten Island", "Queens", "Bronx", "Brooklyn"] }
|   },
|   { restaurant_id: 1, name: 1, borough: 1, cuisine: 1, _id: 0 }
| )
[
  {
    borough: 'Brooklyn',
    cuisine: 'American',
    name: 'Riviera Caterer',
    restaurant_id: '40356018'
  },
  {
    borough: 'Bronx',
    cuisine: 'Bakery',
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    borough: 'Queens',
    cuisine: 'Jewish/Kosher',
    name: 'Tov Kosher Kitchen',
    restaurant_id: '40356068'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Chinese',
    name: 'May May Kitchen',
    restaurant_id: '40358429'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Hamburgers',
    name: "Wendy'S",
    restaurant_id: '30112340'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Jewish/Kosher',
    name: 'Seuda Foods',
    restaurant_id: '40360045'
  },
  {
    borough: 'Queens',
    cuisine: 'Ice Cream, Gelato, Yogurt, Ices',
```

Q19.

```
test> db.restaurent.find(
|   {
|     borough: { $nin: ["Staten Island", "Queens", "Bronx", "Brooklyn"] }
|   },
|   { restaurant_id: 1, name: 1, borough: 1, cuisine: 1, _id: 0 }
| )
[
  {
    borough: 'Manhattan',
    cuisine: 'American',
    name: '1 East 66Th Street Kitchen',
    restaurant_id: '40359480'
  },
  {
    borough: 'Manhattan',
    cuisine: 'American',
    name: 'Glorious Food',
    restaurant_id: '40361521'
  },
  {
    borough: 'Manhattan',
    cuisine: 'Delicatessen',
    name: "Bully'S Deli",
    restaurant_id: '40361708'
  },
  {
    borough: 'Manhattan',
    cuisine: 'American',
    name: 'P & S Deli Grocery',
    restaurant_id: '40362264'
  },
  {
    borough: 'Manhattan',
    cuisine: 'Chicken',
    name: "Harriet'S Kitchen",
    restaurant_id: '40362098'
  },
  {
    borough: 'Manhattan',
    cuisine: 'American',
    name: 'Angelika Film Center',
    restaurant_id: '40362274'
  },
```

Q20.

```
test> db.restaurent.find(
|   { "grades.score": { $lte: 10 } },
|   { restaurant_id: 1, name: 1, borough: 1, cuisine: 1, _id: 0 }
| )
[
  {
    borough: 'Brooklyn',
    cuisine: 'American',
    name: 'Riviera Caterer',
    restaurant_id: '40356018'
  },
  {
    borough: 'Bronx',
    cuisine: 'Bakery',
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Chinese',
    name: 'May May Kitchen',
    restaurant_id: '40358429'
  },
  {
    borough: 'Manhattan',
    cuisine: 'American',
    name: '1 East 66Th Street Kitchen',
    restaurant_id: '40359480'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Hamburgers',
    name: "Wendy'S",
    restaurant_id: '30112340'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Jewish/Kosher',
    name: 'Seuda Foods',
    restaurant_id: '40360045'
  },
```

Q21.

```
test> db.restaurent.find(
|   {
|     $or: [
|       { cuisine: { $nin: ["American", "Chinese"] } },
|       { name: { $regex: /^Wil/ } }
|     ]
|   },
|   { restaurant_id: 1, name: 1, borough: 1, cuisine: 1, _id: 0 }
| )
[
  {
    borough: 'Bronx',
    cuisine: 'Bakery',
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    borough: 'Queens',
    cuisine: 'Jewish/Kosher',
    name: 'Tov Kosher Kitchen',
    restaurant_id: '40356068'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Hamburgers',
    name: "Wendy'S",
    restaurant_id: '30112340'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Jewish/Kosher',
    name: 'Seuda Foods',
    restaurant_id: '40360045'
  },
  {
    borough: 'Queens',
    cuisine: 'Ice Cream, Gelato, Yogurt, Ices',
    name: 'Carvel Ice Cream',
    restaurant_id: '40361322'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Delicatessen',
    name: 'Nordic Delicacies',
    restaurant_id: '40361390'
  },
```

Q22.

```
test> db.restaurent.find(
|   {
|     grades: {
|       $elemMatch: {
|         grade: "A",
|         score: 11,
|         date: ISODate("2014-08-11T00:00:00Z")
|       }
|     }
|   },
|   { restaurant_id: 1, name: 1, grades: 1, _id: 0 }
| )
[
  {
    grades: [
      { date: ISODate('2014-08-11T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2013-12-10T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2013-06-10T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-06-08T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2012-01-25T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2011-09-13T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: 'Don Filippo Restaurant',
    restaurant_id: '40372417'
  },
  {
    grades: [
      { date: ISODate('2014-08-11T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2014-02-04T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2013-06-19T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2012-06-25T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-02-08T00:00:00.000Z'), grade: 'A', score: 5 }
    ],
    name: 'Cosi',
    restaurant_id: '40922983'
  },
  {
    grades: [
      { date: ISODate('2014-08-11T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2014-02-07T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-12-05T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2012-07-13T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-02-08T00:00:00.000Z'), grade: 'A', score: 9 }
    ],
    name: 'Subway',
```

Q23.

```
test> db.restaurent.find(
|   {
|     "grades.1.grade": "A",
|     "grades.1.score": 9,
|     "grades.1.date": ISODate("2014-08-11T00:00:00Z")
|   },
|   { restaurant_id: 1, name: 1, grades: 1, _id: 0 }
| )
[
  {
    grades: [
      { date: ISODate('2015-01-12T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2014-08-11T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2014-01-14T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2013-02-07T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2012-04-30T00:00:00.000Z'), grade: 'A', score: 11 }
    ],
    name: 'Club Macanudo (Cigar Bar)',
    restaurant_id: '40526406'
  },
  {
    grades: [
      { date: ISODate('2015-01-15T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2014-08-11T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2013-06-18T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2013-01-24T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2012-06-11T00:00:00.000Z'), grade: 'A', score: 11 }
    ],
    name: "Shoolbred'S",
    restaurant_id: '41302014'
  },
  {
    grades: [
      { date: ISODate('2014-12-27T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2014-08-11T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2014-02-24T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2013-01-08T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-06-27T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2012-01-26T00:00:00.000Z'), grade: 'A', score: 5 }
    ],
    name: 'Cabana',
    restaurant_id: '41309951'
  },
  {
    grades: [
```

Q24.

```
test> db.restaurent.find(
|   { "address.coord.1": { $gt: 42, $lte: 52 } },
|   { restaurant_id: 1, name: 1, address: 1, _id: 0 }
| )
[
  {
    address: {
      building: '47',
      coord: [ -78.877224, 42.89546199999999 ],
      street: 'Broadway @ Trinity Pl',
      zipcode: '10006'
    },
    name: "T.G.I. Friday'S",
    restaurant_id: '40387990'
  },
  {
    address: {
      building: '1',
      coord: [ -0.7119979, 51.6514664 ],
      street: 'Pennplaza E, Penn Sta',
      zipcode: '10001'
    },
    name: 'T.G.I. Fridays',
    restaurant_id: '40388936'
  },
  {
    address: {
      building: '3000',
      coord: [ -87.86567699999999, 42.61150920000001 ],
      street: '47 Avenue',
      zipcode: '11101'
    },
    name: "Di Luvio'S Deli",
    restaurant_id: '40402284'
  },
  {
    address: {
      building: '21972199',
      coord: [ -78.589606, 42.8912372 ],
      street: 'Broadway',
      zipcode: '10024'
    },
    name: 'La Caridad 78',
    restaurant_id: '40568285'
  },
  {
```

Q25.

```
test> db.restaurent.find().sort({ name: 1 })
[
  {
    _id: ObjectId('69e1fc89c3cf3ffef803af14'),
    address: {
      building: '535',
      coord: [ -73.8958518, 40.8468137 ],
      street: 'E Tremont Ave',
      zipcode: '10457'
    },
    borough: 'Bronx',
    cuisine: 'Other',
    grades: [],
    name: '',
    restaurant_id: '50018749'
  },
  {
    _id: ObjectId('69e1fc89c3cf3ffef803aeef'),
    address: {
      building: '4307',
      coord: [ -73.86765059999999, 40.8986752 ],
      street: 'Katonah Ave',
      zipcode: '10470'
    },
    borough: 'Bronx',
    cuisine: 'Other',
    grades: [],
    name: '',
    restaurant_id: '50018680'
  },
  {
    _id: ObjectId('69e1fc89c3cf3ffef803aef2'),
    address: {
      building: '10318',
      coord: [ -73.8620663, 40.7495891 ],
      street: 'Roosevelt Ave',
      zipcode: '11368'
    },
    borough: 'Queens',
    cuisine: 'Other',
    grades: [],
    name: '',
    restaurant_id: '50018693'
  },
  {
```

Q26.

```
test> db.restaurent.find().sort({ name: -1 })
|
[
  {
    _id: ObjectId('69e1fc89c3cf3ffef80390d6'),
    address: {
      building: '1',
      coord: [ -74.073156, 40.6457369 ],
      street: 'Richmond Terrace',
      zipcode: '10301'
    },
    borough: 'Staten Island',
    cuisine: 'Pizza',
    grades: [
      { date: ISODate('2015-01-13T00:00:00.000Z'), grade: 'Z', score: 18 },
      { date: ISODate('2014-07-24T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2013-11-08T00:00:00.000Z'), grade: 'B', score: 21 },
      { date: ISODate('2013-04-17T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: "Zz'S Pizza & Grill",
    restaurant_id: '41702705'
  },
  {
    _id: ObjectId('69e1fc89c3cf3ffef8039768'),
    address: {
      building: '169',
      coord: [ -74.000466, 40.727688 ],
      street: 'Thompson Street',
      zipcode: '10012'
    },
    borough: 'Manhattan',
    cuisine: 'Other',
    grades: [
      { date: ISODate('2014-05-15T00:00:00.000Z'), grade: 'A', score: 10 }
    ],
    name: 'Zz Clam Bar',
    restaurant_id: '50001062'
  },
  {
    _id: ObjectId('69e1fc89c3cf3ffef8039331'),
    address: {
      building: '726',
      coord: [ -73.8662887, 40.8594314 ],
      street: 'Astor Avenue',
      zipcode: '10467'
```

Q27.

```
test> db.restaurent.find().sort({ cuisine: 1, borough: -1 })
[
  {
    _id: ObjectId('69e1fc89c3cf3ffef8038b25'),
    address: {
      building: '6729',
      coord: [ -73.8204147, 40.724243 ],
      street: 'Main St',
      zipcode: '11367'
    },
    borough: 'Queens',
    cuisine: 'Afghan',
    grades: [
      { date: ISODate('2014-08-20T00:00:00.000Z'), grade: 'A', score: 4 },
      { date: ISODate('2013-07-31T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2012-08-04T00:00:00.000Z'), grade: 'A', score: 5 }
    ],
    name: 'Bakhtar Kabab',
    restaurant_id: '41661199'
  },
  {
    _id: ObjectId('69e1fc89c3cf3ffef8037cbe'),
    address: {
      building: '259-11',
      coord: [ -73.708831, 40.73748399999999 ],
      street: 'Hillside Avenue',
      zipcode: '11004'
    },
    borough: 'Queens',
    cuisine: 'Afghan',
    grades: [
      { date: ISODate('2014-09-15T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2013-09-18T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2013-04-18T00:00:00.000Z'), grade: 'B', score: 21 },
      { date: ISODate('2012-08-29T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2012-02-09T00:00:00.000Z'), grade: 'A', score: 11 }
    ],
    name: 'Afghan Kebob House',
    restaurant_id: '41535706'
  },
  {
    _id: ObjectId('69e1fc89c3cf3ffef803a238'),
    address: {
      building: '33-35',
      coord: [ -73.7896131, 40.7666136 ],
      street: 'Francis Lewis Boulevard',
      zipcode: '11358'
```

Q28.

```
test> db.restaurent.find(
|   { "address.street": { $exists: false } }
| )
```

Q29.

```
test> db.restaurent.find(
|   { "address.coord": { $type: 1 } }
| )
[
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c66'),
    address: {
      building: '2780',
      coord: [ -73.98241999999999, 40.579505 ],
      street: 'Stillwell Avenue',
      zipcode: '11224'
    },
    borough: 'Brooklyn',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-06-10T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2013-06-05T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2012-04-13T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2011-10-12T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: 'Riviera Caterer',
    restaurant_id: '40356018'
  },
  {
    _id: ObjectId('69e1fc88c3cf3ffef8034c67'),
    address: {
      building: '1007',
      coord: [ -73.856077, 40.848447 ],
      street: 'Morris Park Ave',
      zipcode: '10462'
    },
    borough: 'Bronx',
    cuisine: 'Bakery',
    grades: [
      { date: ISODate('2014-03-03T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-09-11T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2013-01-24T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-11-23T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2011-03-10T00:00:00.000Z'), grade: 'B', score: 14 }
    ],
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
```

Q30.

```
test> db.restaurent.find(
|   { "grades.score": { $mod: [7, 0] } },
|   { restaurant_id: 1, name: 1, grades: 1, _id: 0 }
| )
|
[
  {
    grades: [
      { date: ISODate('2014-06-10T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2013-06-05T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2012-04-13T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2011-10-12T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: 'Riviera Caterer',
    restaurant_id: '40356018'
  },
  {
    grades: [
      { date: ISODate('2014-03-03T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-09-11T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2013-01-24T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-11-23T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2011-03-10T00:00:00.000Z'), grade: 'B', score: 14 }
    ],
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    grades: [
      { date: ISODate('2014-09-16T00:00:00.000Z'), grade: 'B', score: 21 },
      { date: ISODate('2013-08-28T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2013-04-02T00:00:00.000Z'), grade: 'C', score: 56 },
      { date: ISODate('2012-08-15T00:00:00.000Z'), grade: 'B', score: 27 },
      { date: ISODate('2012-03-28T00:00:00.000Z'), grade: 'B', score: 27 }
    ],
    name: 'May May Kitchen',
    restaurant_id: '40358429'
  },
  {
    grades: [
      { date: ISODate('2014-05-07T00:00:00.000Z'), grade: 'A', score: 3 },
      { date: ISODate('2013-05-03T00:00:00.000Z'), grade: 'A', score: 4 },
      { date: ISODate('2012-04-30T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2011-12-27T00:00:00.000Z'), grade: 'A', score: 0 }
    ],
```

Q31.

```
test> db.restaurent.find(
|   { name: { $regex: /mon/i } },
|   {
|     name: 1,
|     borough: 1,
|     "address.coord": 1,
|     cuisine: 1,
|     _id: 0
|   }
| )
[
  {
    address: { coord: [ -73.8221418, 40.7272376 ] },
    borough: 'Queens',
    cuisine: 'Jewish/Kosher',
    name: 'Shimons Kosher Pizza'
  },
  {
    address: { coord: [ -74.10465599999999, 40.58834 ] },
    borough: 'Staten Island',
    cuisine: 'American',
    name: 'Richmond County Country Club'
  },
  {
    address: { coord: [ -73.98306099999999, 40.7441419 ] },
    borough: 'Manhattan',
    cuisine: 'American',
    name: "Desmond'S Tavern"
  },
  {
    address: { coord: [ -73.9812843, 40.5947365 ] },
    borough: 'Brooklyn',
    cuisine: 'Pizza/Italian',
    name: 'Lb Spumoni Gardens'
  },
  {
    address: { coord: [ -73.951199, 40.7166026 ] },
    borough: 'Brooklyn',
    cuisine: 'Italian',
    name: "Bamonte'S Restaurant"
  },
  {
    address: { coord: [ -73.924072, 40.76108900000001 ] },
    borough: 'Queens',
    cuisine: 'Greek',
    name: 'Omonia Cafe'
  },
```

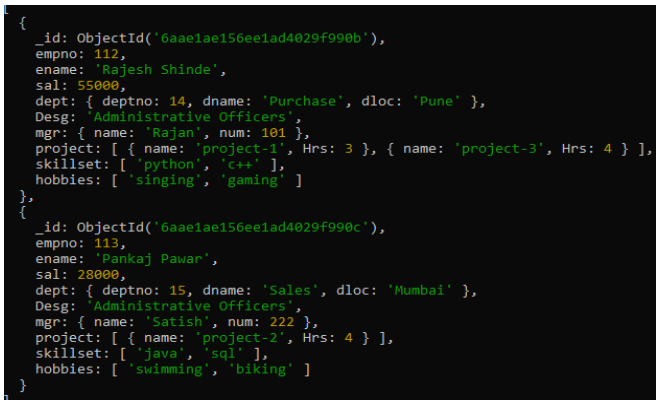
Q32.

```
test> db.restaurent.find(
|   { name: { $regex: /^Mad/ } },
|   {
|     name: 1,
|     borough: 1,
|     "address.coord": 1,
|     cuisine: 1,
|     _id: 0
|   }
| )
[
  {
    address: { coord: [ -73.9860597, 40.7431194 ] },
    borough: 'Manhattan',
    cuisine: 'American',
    name: 'Madison Square'
  },
  {
    address: { coord: [ -73.98302199999999, 40.742313 ] },
    borough: 'Manhattan',
    cuisine: 'Indian',
    name: 'Madras Mahal'
  },
  {
    address: { coord: [ -74.000002, 40.72735 ] },
    borough: 'Manhattan',
    cuisine: 'American',
    name: 'Madame X'
  },
  {
    address: { coord: [ -73.98171959999999, 40.7499406 ] },
    borough: 'Manhattan',
    cuisine: 'French',
    name: 'Madison Bistro'
  },
  {
    address: { coord: [ -73.9717845, 40.6897199 ] },
    borough: 'Brooklyn',
    cuisine: 'African',
    name: 'Madiba'
  },
  {
    address: { coord: [ -73.9040753, 40.9069011 ] },
    borough: 'Bronx',
    cuisine: 'Italian',
    name: "Madison'S"
  },
```

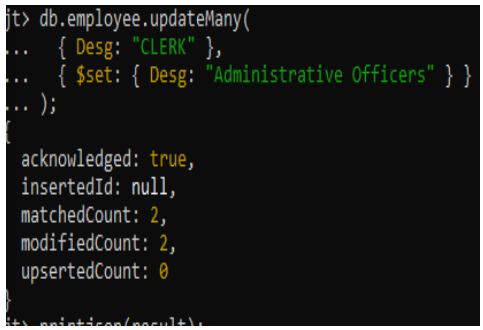
(DAY 12)

Create a Employee Collection add 5 documents:
Example:
{empno:111,
ename:"Deepali Vaidya",
sal:40000.00,
dept:{deptno:12,dname:,"Hr",dloc:"Mumbai"},
Desg:"Analyst",
mgr:{name:"Satish",num:222},
project:[{name:"project-1",Hrs:4},
{name:"project-2",Hrs:4}],
skillset:['perl','python','java'],
hobbies:['reading','biking']}

1. All Employee's with the desg as 'CLERK' are now called as (AO) Administrative Officers. Update the Employee collection for this.

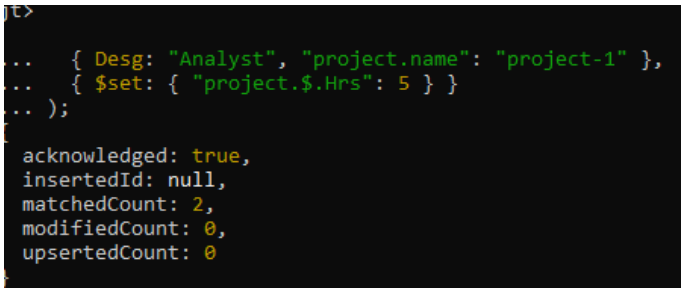


```
{
  _id: ObjectId('6aae1ae156ee1ad4029f990b'),
  empno: 112,
  ename: 'Rajesh Shinde',
  sal: 55000,
  dept: { deptno: 14, dname: 'Purchase', dloc: 'Pune' },
  Desg: 'Administrative Officers',
  mgr: { name: 'Rajan', num: 101 },
  project: [ { name: 'project-1', Hrs: 3 }, { name: 'project-3', Hrs: 4 } ],
  skillset: [ 'python', 'c++' ],
  hobbies: [ 'singing', 'gaming' ]
},
{
  _id: ObjectId('6aae1ae156ee1ad4029f990c'),
  empno: 113,
  ename: 'Pankaj Pawar',
  sal: 28000,
  dept: { deptno: 15, dname: 'Sales', dloc: 'Mumbai' },
  Desg: 'Administrative Officers',
  mgr: { name: 'Satish', num: 222 },
  project: [ { name: 'project-2', Hrs: 4 } ],
  skillset: [ 'java', 'sql' ],
  hobbies: [ 'swimming', 'biking' ]
}
```



```
t> db.employee.updateMany(
...   { Desg: "CLERK" },
...   { $set: { Desg: "Administrative Officers" } }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

2. Change the number of hours for project-1 to 5 for all employees with designation analyst.



```
t>
...   { Desg: "Analyst", "project.name": "project-1" },
...   { $set: { "project.$.Hrs": 5 } }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 0,
  upsertedCount: 0
}
```

3. Add 2 projects project-3 and project-4 for employee whose name starts with "Deep" with 2 hrs

```
}
...   { ename: /^Deep/i },
...   {
...     $push: {
...       project: {
...         $each: [
...           { name: "project-3", Hrs: 2 },
...           { name: "project-4", Hrs: 2 }
...         ]
...       }
...     }
...   }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

4. Add bonus rs 2000 for all employees with salary > 50000

```
...   { sal: { $gt: 50000 } },
...   { $set: { bonus: 2000 } }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

5. Add bonus rs 1500 if salary <50000 and > 30000
db.Employee.updateMany({sal:{$gt:30000,$lt:50000}},{$set:{bonus:1500}});

```
...   { sal: { $gt: 30000, $lt: 50000 } },
...   { $set: { bonus: 1500 } }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
jt>
```

6. increment bounus by 1000 for all employees if salary <=30000

```
jt> db.employee.updateMany({sal:{$lte:30000}}, {$inc:{bonus:1000}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

7. Change manager name to Tushar for all employees whose manager is currently “satish” And manager number to 3333

```
jt> db.employee.updateMany({"mgr.name":/^satish$/i},{$set:{"mgr.name":"Tushar","
mgr.name":3333}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

8. Increase salary of all employees from “purchase department” by 15000

```
jt> db.employee.updateMany({"dept.dname":/^purchase$/i},{$inc:{sal:15000}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

9. Decrease number of hrs by 2 for all employees who are working on project-2

```
jt> db.employee.updateMany({"project.name":"project-2"},{$inc:{"project.$.Hrs":-
2}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
jt>
```

10. Delete project-2 from all employee document if they are working on the project for 4

```
jt> db.employee.updateMany(
...   { project: { $elemMatch: { name: "project-2", Hrs: 4 } } },
...   { $pull: { project: { name: "project-2", Hrs: 4 } } }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
jt> db.employee.find(
...   { "project.name": "project-2" },
...   { ename: 1, project: 1 }
... );
[
  {
    _id: ObjectId('6aae1ae156ee1ad4029f990e'),
    ename: 'Rutuja Kulkarni',
    project: [ { name: 'project-2', Hrs: 2 }, { name: 'project-3', Hrs: 2 } ]
  }
```

11. Change the salary of employees to 10000 only if their salary is < 10000

```
jt> db.employee.updateMany({sal:{$lt:10000}},{$set:{sal:10000}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
jt> db.employee.find({}, { ename: 1, sal: 1 });
[
  {
    _id: ObjectId('6aae1ae156ee1ad4029f990a'),
    ename: 'Deepali Vaidya',
    sal: 40000
  },
  {
    _id: ObjectId('6aae1ae156ee1ad4029f990b'),
    ename: 'Rajesh Shinde',
    sal: 55000
  },
  {
    _id: ObjectId('6aae1ae156ee1ad4029f990c'),
    ename: 'Pankaj Pawar',
    sal: 28000
  },
  {
    _id: ObjectId('6aae1ae156ee1ad4029f990d'),
    ename: 'Deepak Joshi',
    sal: 10000
  },
  {
    _id: ObjectId('6aae1ae156ee1ad4029f990e'),
    ename: 'Rutuja Kulkarni',
    sal: 33000
  }
]
```

12. Increase bonus of all employees by 500 if the bonus is <2000 or their salary is < 20000 or if employee belong to sales department

```
jt> db.employee.updateMany(
...   {
...     $or: [
...       { bonus: { $lt: 2000 } },
...       { sal: { $lt: 20000 } },
...       { "dept.dname": /^sales$/i }
...     ]
...   },
...   { $inc: { bonus: 500 } }
jt>
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
```

13. Add 2 new project at position 2 for all employees with designation analyst or salary is equal to either 30000 or 33000 or 35000

```
jt> db.employee.updateMany(
...   {
...     $or: [
...       { Desg: "Analyst" },
...       { sal: { $in: [30000, 33000, 35000] } }
...     ]
...   },
...   {
...     $push: {
...       project: {
...         $each: [
...           { name: "project-5", Hrs: 4 },
...           { name: "project-6", Hrs: 4 }
...         ],
...         $position: 1
...       }
...     }
...   }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

14. Delete last project of all employees with department name is “HR” and if the location is Mumbai

```
...   {
...     "dept.dname": /^hr$/i,
...     "dept.dloc": /^mumbai$/i
...   },
...   { $pop: { project: 1 } }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
...   {},
...   { ename: 1, "dept.dname": 1, "dept.dloc": 1, project: 1 }
... ).pretty()
[
  {
    _id: ObjectId('6aae1ae156ee1ad4029f990a'),
    ename: 'Deepali Vaidya',
    dept: { dname: 'Hr', dloc: 'Mumbai' },
    project: [
      { name: 'project-1', Hrs: 5 },
      { name: 'project-5', Hrs: 4 },
      { name: 'project-6', Hrs: 4 },
      { name: 'project-3', Hrs: 2 }
    ]
  },
```

15. Change designation of all employees to senior programmer if they are working on name:”Project-1” for 4 hrs

```
...   { project: { $elemMatch: { name: /^project-1$/i, Hrs: 4 } } },
...   { $set: { Desg: "Senior Programmer" } }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
...   { "project.name": /^project-1$/i },
...   { ename: 1, Desg: 1, project: 1 }
... ).pretty();
[
  {
    _id: ObjectId('6aae1ae156ee1ad4029f990a'),
    ename: 'Deepali Vaidya',
    Desg: 'Senior Programmer',
    project: [
      { name: 'project-1', Hrs: 4 },
      { name: 'project-5', Hrs: 4 },
      { name: 'project-6', Hrs: 4 },
      { name: 'project-3', Hrs: 2 }
    ]
  },
  {
    _id: ObjectId('6aae1ae156ee1ad4029f990b'),
    ename: 'Rajesh Shinde',
    Desg: 'Administrative Officers',
    project: [ { name: 'project-1', Hrs: 3 }, { name: 'project-3', Hrs: 4 } ]
  },
  {
```

16. Add list of hobbies in all employees document whose manager is Rajan or Revati

```
jt> db.employee.updateMany( { "mgr.name": { $in: ["Rajan", "Revati"] } }, { $pus
h: { hobbies: { $each: ["photography", "gardening"] } } } );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
...   { "mgr.name": { $in: ["Rajan", "Revati"] } },
...   { ename: 1, "mgr.name": 1, hobbies: 1 }
... ).pretty();
[
  {
    _id: ObjectId('6aae1ae156ee1ad4029f990b'),
    ename: 'Rajesh Shinde',
    mgr: { name: 'Rajan' },
    hobbies: [
      'singing',
      'gaming',
      'photography',
      'gardening',
      'photography',
      'gardening'
    ]
  },
  {
    _id: ObjectId('6aae1ae156ee1ad4029f990d'),
    ename: 'Deepak Joshi',
    mgr: { name: 'Revati' },
    hobbies: [
      'reading',
      'trekking',
      'photography',
      'gardening',
      'photography',
      'gardening'
```

17. Add list of skillset in all employee documents who are working on project-4 for 3 hrs or on project-3 for 4 hrs

```
jt> db.employee.updateMany(
...   {
...     $or: [
...       { project: { $elemMatch: { name: "project-4", Hrs: 3 } } },
...       { project: { $elemMatch: { name: "project-3", Hrs: 4 } } }
...     ]
...   },
...   {
...     $push: {
...       skillset: { $each: ["MongoDB", "NodeJS"] }
...     }
...   }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

18. Add a new hobby as blogging at 3 position in hobbies array for all employess whose name starts with R or p and ends with j or s

```
jt> db.employee.updateMany(
...   {
...     $or: [
...       { ename: { $regex: "^[rp]", $options: "i" } },
...       { ename: { $regex: "[js]$", $options: "i" } }
...     ]
...   },
...   {
...     $push: {
...       hobbies: {
...         $each: ["blogging"],
...         $position: 2
...       }
...     }
...   }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

19. Increase salary by 10000 for all employees who are working on project-2 or project-3 or project-1

Decrease bonus by 1000 rs And increase salary by 1000rs for all employees whose department location is Mumbai

```
jt> db.employee.bulkWrite([
...   {
...     updateMany: {
...       filter: { "project.name": { $in: ["project-1", "project-2", "project-3
"] } },
...       update: { $inc: { sal: 10000 } }
...     }
...   },
...   {
...     updateMany: {
...       filter: { "dept.dloc": /^mumbai$/i },
...       update: { $inc: { bonus: -1000, sal: 1000 } }
...     }
...   }
... ]);
{
  acknowledged: true,
  insertedCount: 0,
  insertedIds: {},
  matchedCount: 6,
  modifiedCount: 6,
  deletedCount: 0,
  upsertedCount: 0,
  upsertedIds: {}
}
```

20. Remove all employees working on project-1

```
...   { "project.name": /^project-1$/i }
... );
{ acknowledged: true, deletedCount: 3 }
jt>
```

21. Replace document of employee with name “Deepak to some new document

```
...   { ename: { $regex: "deepak", $options: "i" } },
...   {
...     empno: 104,
...     ename: "Deepak",
...     sal: 55000,
...     bonus: 2000,
...     dept: { deptno: 10, dname: "Testing", dloc: "Pune" },
...     Desg: "Senior QA Analyst",
...     mgr: { name: "Rajan", num: 101 },
...     project: [{ name: "project-4", Hrs: 5 }],
...     skillset: ["Automation", "Selenium", "Java"],
...     hobbies: ["reading", "travelling"]
...   }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

22. Change skill python to python 3.8 for all employees if python is there in the skillset

```
tanvi> db.Employee.updateMany(
|   { skillset: "python" },
|   { $set: { "skillset.$": "python 3.8" } }
| );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
tanvi> db.Employee.find({skillset:"python 3.8"});
[
  {
    _id: ObjectId('6ab0f3e7705c00ac924a1a21'),
    empno: 111,
    ename: 'Deepali Vaidya',
    sal: 40000,
    bonus: 1000,
    dept: { deptno: 12, dname: 'Hr', dloc: 'Mumbai' },
    Desg: 'Analyst',
    mgr: { name: 'Satish', num: 222 },
    project: [ { name: 'project-1', Hrs: 4 }, { name: 'project-2', Hrs: 4 }
],
    skillset: [ 'perl', 'python 3.8', 'java' ],
    hobbies: [ 'reading', 'biking' ]
  },
  {
    _id: ObjectId('6ab0f3e7705c00ac924a1a22'),
    empno: 112,
    ename: 'Ramesh Raj',
    sal: 28000,
    bonus: 500,
    dept: { deptno: 10, dname: 'Purchase', dloc: 'Pune' },
    Desg: 'CLERK',
    mgr: { name: 'Rajan', num: 101 },
    project: [ { name: 'project-1', Hrs: 3 }, { name: 'project-3', Hrs: 4 }
],
    skillset: [ 'python 3.8', 'c++' ],
    hobbies: [ 'cooking', 'music' ]
```

23. Add 2 skills MongoDb and Perl at the end of skillset array for all employees who are working at Pune location

```
db.Employee.updateMany(
  { "dept.dloc": { $regex: "^pune$", $options: "i" } },
  { $set: { skillset: ["mongoDb", "Perl"] } }
);
db.Employee.find({dept.dloc:pune});
```

```
tanvi> db.Employee.find({ "dept.dloc": { $regex: /^pune$/i } });
[
  {
    _id: ObjectId('6ab0f3e7705c00ac924a1a22'),
    empno: 112,
    ename: 'Ramesh Raj',
    sal: 28000,
    bonus: 500,
    dept: { deptno: 10, dname: 'Purchase', dloc: 'Pune' },
    Desg: 'CLERK',
    mgr: { name: 'Rajan', num: 101 },
    project: [ { name: 'project-1', Hrs: 3 }, { name: 'project-3', Hrs: 4 }
],
    skillset: [ 'mongoDb', 'Perl' ],
    hobbies: [ 'cooking', 'music' ]
  },
  {
    _id: ObjectId('6ab0f3e7705c00ac924a1a25'),
    empno: 115,
    ename: 'Priya Nair',
    sal: 30000,
    bonus: 1800,
    dept: { deptno: 15, dname: 'Engineering', dloc: 'Pune' },
    Desg: 'Developer',
    mgr: { name: 'Anish', num: 333 },
    project: [ { name: 'project-3', Hrs: 2 }, { name: 'project-2', Hrs: 4 }
],
    skillset: [ 'mongoDb', 'Perl' ],
    hobbies: [ 'painting', 'gardening' ]
  }
]
```

24. Delete first hobby from hobby array for all employees who are working on project-1 or project-2

```
tanvi> db.Employee.updateMany({"project.name":{$in:["project-1","project-2"]
}},{$pop:{hobbies: -1}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
tanvi> db.Employee.find().pretty();
[
  {
    _id: ObjectId('6ab0f3e7705c00ac924a1a21'),
    empno: 111,
    ename: 'Deepali Vaidya',
    sal: 40000,
    bonus: 1000,
    dept: { deptno: 12, dname: 'Hr', dloc: 'Mumbai' },
    Desg: 'Analyst',
    mgr: { name: 'Satish', num: 222 },
    project: [ { name: 'project-1', Hrs: 4 }, { name: 'project-2', Hrs: 4 }
],
    skillset: [ 'perl', 'python 3.8', 'java' ],
    hobbies: [ 'biking' ]
  },
  {
    _id: ObjectId('6ab0f3e7705c00ac924a1a22'),
    empno: 112,
    ename: 'Ramesh Raj',
    sal: 28000,
    bonus: 500,
    dept: { deptno: 10, dname: 'Purchase', dloc: 'Pune' },
    Desg: 'CLERK',
    mgr: { name: 'Rajan', num: 101 },
    project: [ { name: 'project-1', Hrs: 3 }, { name: 'project-3', Hrs: 4 }
],
    skillset: [ 'mongoDb', 'Perl' ],
    hobbies: [ 'music' ]
  },
  {
```

25. Delete last hobby from hobbies array for all employees who are working on project which is at 2 nd position in projects array for 4 hrs

```
tanvi> db.Employee.updateMany({"project.1.Hrs":4},{$pop:{hobbies:1}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
tanvi> db.Employee.find().pretty();
[
  {
    _id: ObjectId('6ab0f3e7705c00ac924a1a21'),
    empno: 111,
    ename: 'Deepali Vaidya',
    sal: 40000,
    bonus: 1000,
    dept: { deptno: 12, dname: 'Hr', dloc: 'Mumbai' },
    Desg: 'Analyst',
    mgr: { name: 'Satish', num: 222 },
    project: [ { name: 'project-1', Hrs: 4 }, { name: 'project-2', Hrs: 4 }
],
    skillset: [ 'perl', 'python 3.8', 'java' ],
    hobbies: []
  },
  {
    _id: ObjectId('6ab0f3e7705c00ac924a1a22'),
    empno: 112,
    ename: 'Ramesh Raj',
    sal: 28000,
    bonus: 500,
    dept: { deptno: 10, dname: 'Purchase', dloc: 'Pune' },
    Desg: 'CLERK',
    mgr: { name: 'Rajan', num: 101 },
    project: [ { name: 'project-1', Hrs: 3 }, { name: 'project-3', Hrs: 4 }
],
    skillset: [ 'mongoDb', 'Perl' ],
    hobbies: []
  },
```

26. Add 2 new projects at the end of array for all employees whose skillset contains Perl or python

```
},
{
  _id: ObjectId('6ab0f3e7705c00ac924a1a22'),
  empno: 112,
  ename: 'Ramesh Raj',
  sal: 28000,
  bonus: 500,
  dept: { deptno: 10, dname: 'Purchase', dloc: 'Pune' },
  Desg: 'CLERK',
  mgr: { name: 'Rajan', num: 101 },
  project: [
    { name: 'project-1', Hrs: 3 },
    { name: 'project-3', Hrs: 4 },
    { name: 'project-5', Hrs: 4 },
    { name: 'project-6', Hrs: 4 }
  ],
  skillset: [ 'mongoDb', 'Perl' ],
  hobbies: []
},
```

```
tanvi> db.Employee.updateMany(
|   { skillset: { $in: ["Perl", "python"] } },
|   {
|     $push: {
|       project: {
|         $each: [
|           { name: "project-5", Hrs: 4 },
|           { name: "project-6", Hrs: 4 }
|         ]
|       }
|     }
|   }
| );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
```

27. Change hrs to 6 for project-1 for all employees if they working on the project-1 for < 6 hrs. otherwise keep the existing value.

```
tanvi> db.Employee.updateMany({project:{$elemMatch:{name:"project-1",Hrs:{$l
t:6}}}},{$set:{"project.$.Hrs":6}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
tanvi> db.Employee.find({"project.Hrs":6});
[
  {
    _id: ObjectId('6ab0f3e7705c00ac924a1a21'),
    empno: 111,
    ename: 'Deepali Vaidya',
    sal: 40000,
    bonus: 1000,
    dept: { deptno: 12, dname: 'Hr', dloc: 'Mumbai' },
    Desg: 'Analyst',
    mgr: { name: 'Satish', num: 222 },
    project: [ { name: 'project-1', Hrs: 6 }, { name: 'project-2', Hrs: 4 }
],
    skillset: [ 'perl', 'python 3.8', 'java' ],
    hobbies: []
  },
```